

LESSON 11

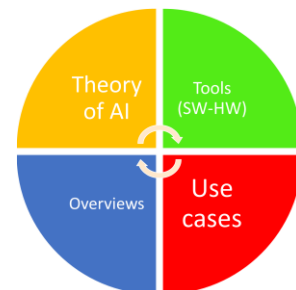
Data Representation and Visualization for Machine Learning





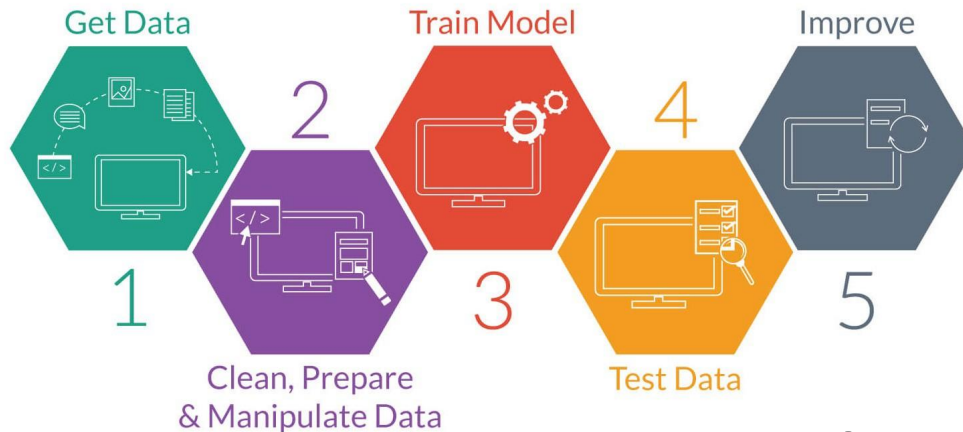
THEORY

Data Visualization for ML



Step 2 of the ML workflow.

Substep: Data visualization



2a) Data Munging/Wrangling

2b) Exploratory Data Analysis

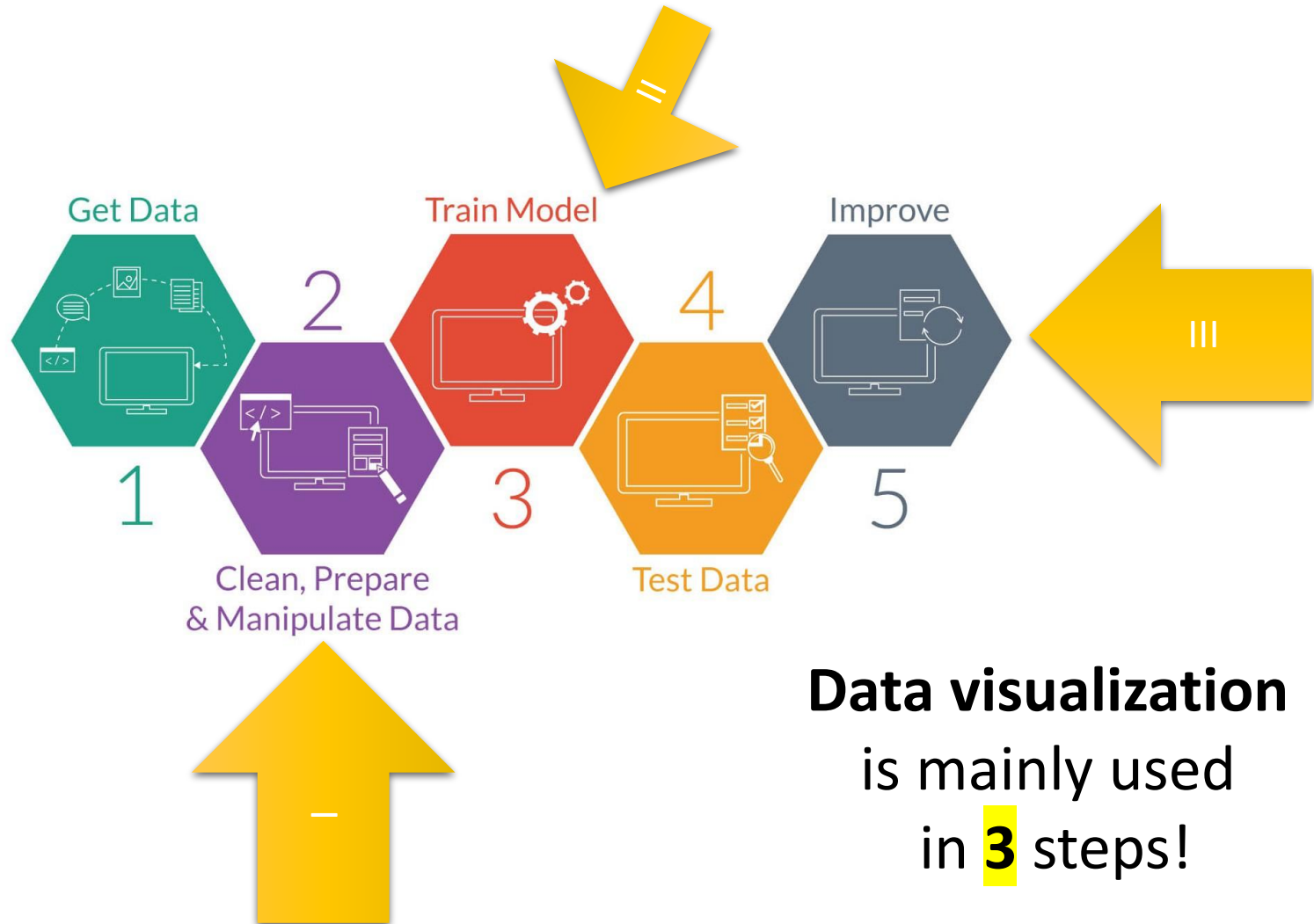
...

Data visualization

....

2c) Feature engineering

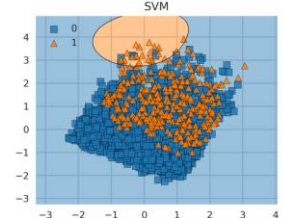
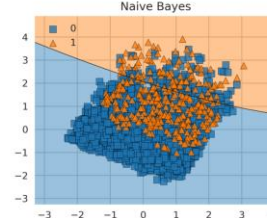
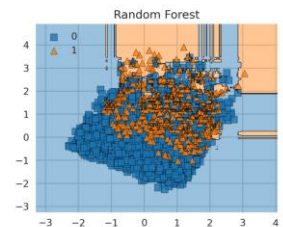
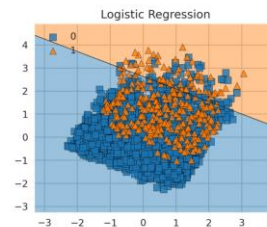
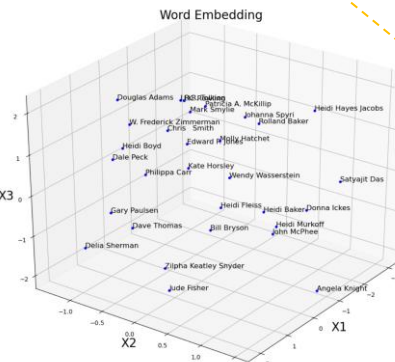
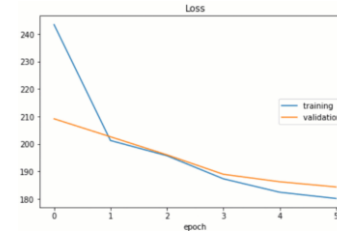
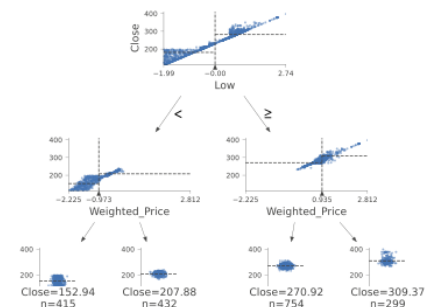
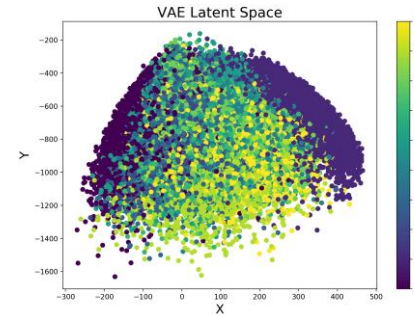
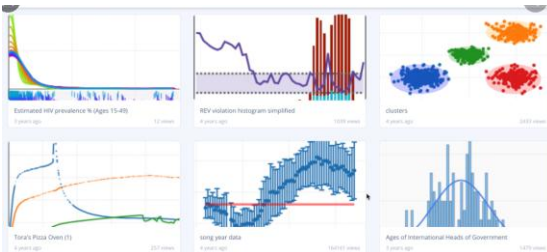
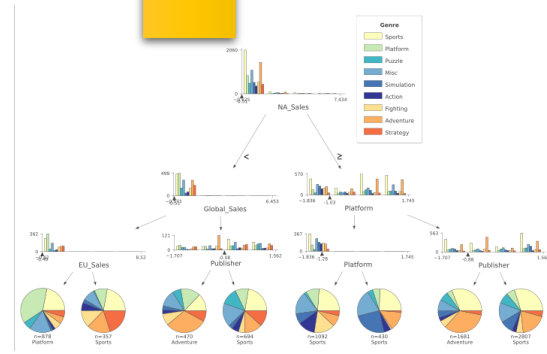
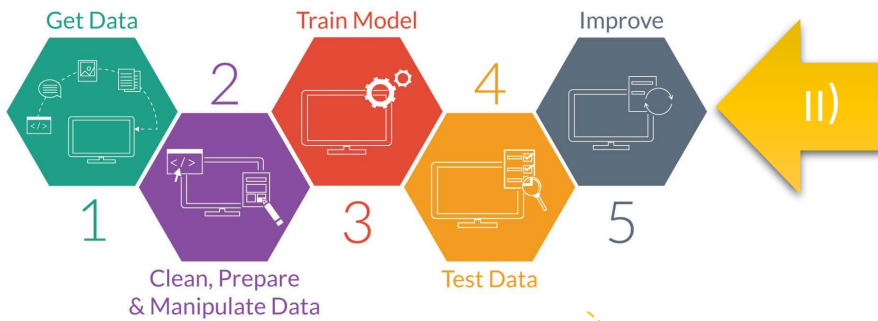
Steps 2-5 of the ML workflow



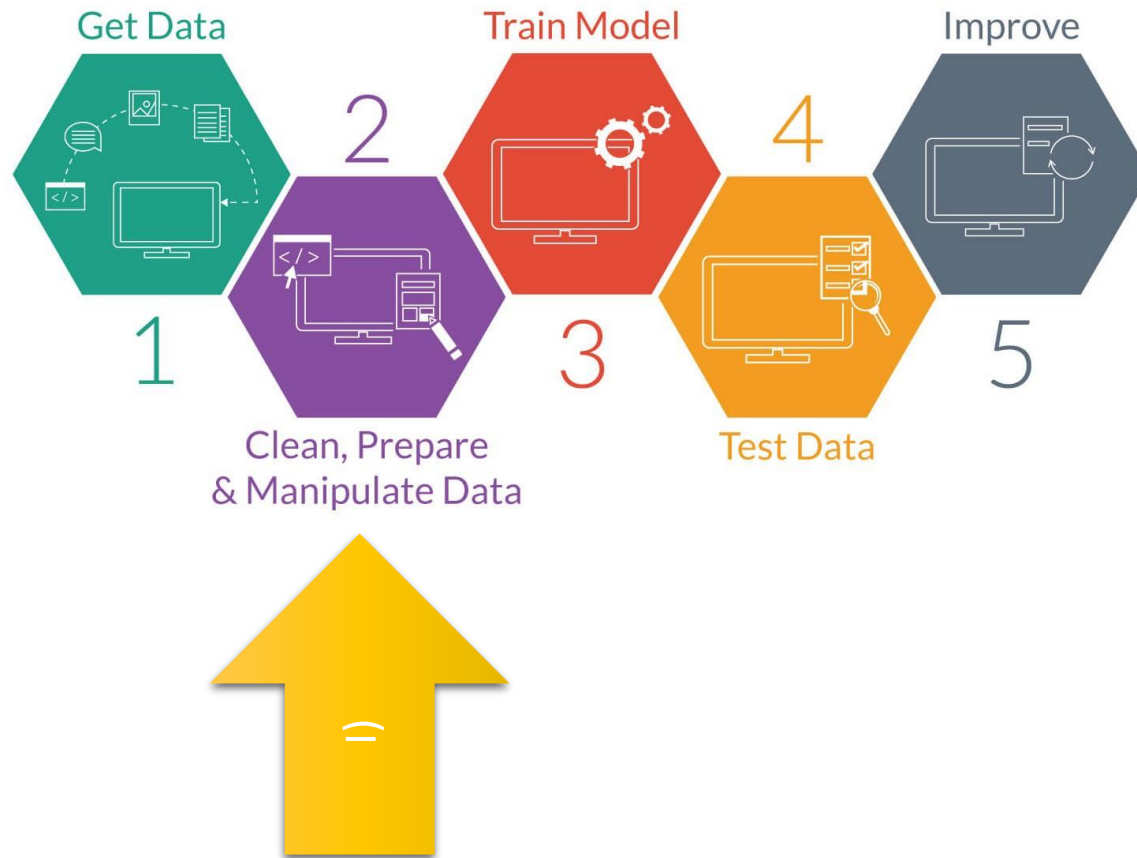
Data visualization
is mainly used
in **3** steps!

Step 2-5 of the ML workflow

Focus on



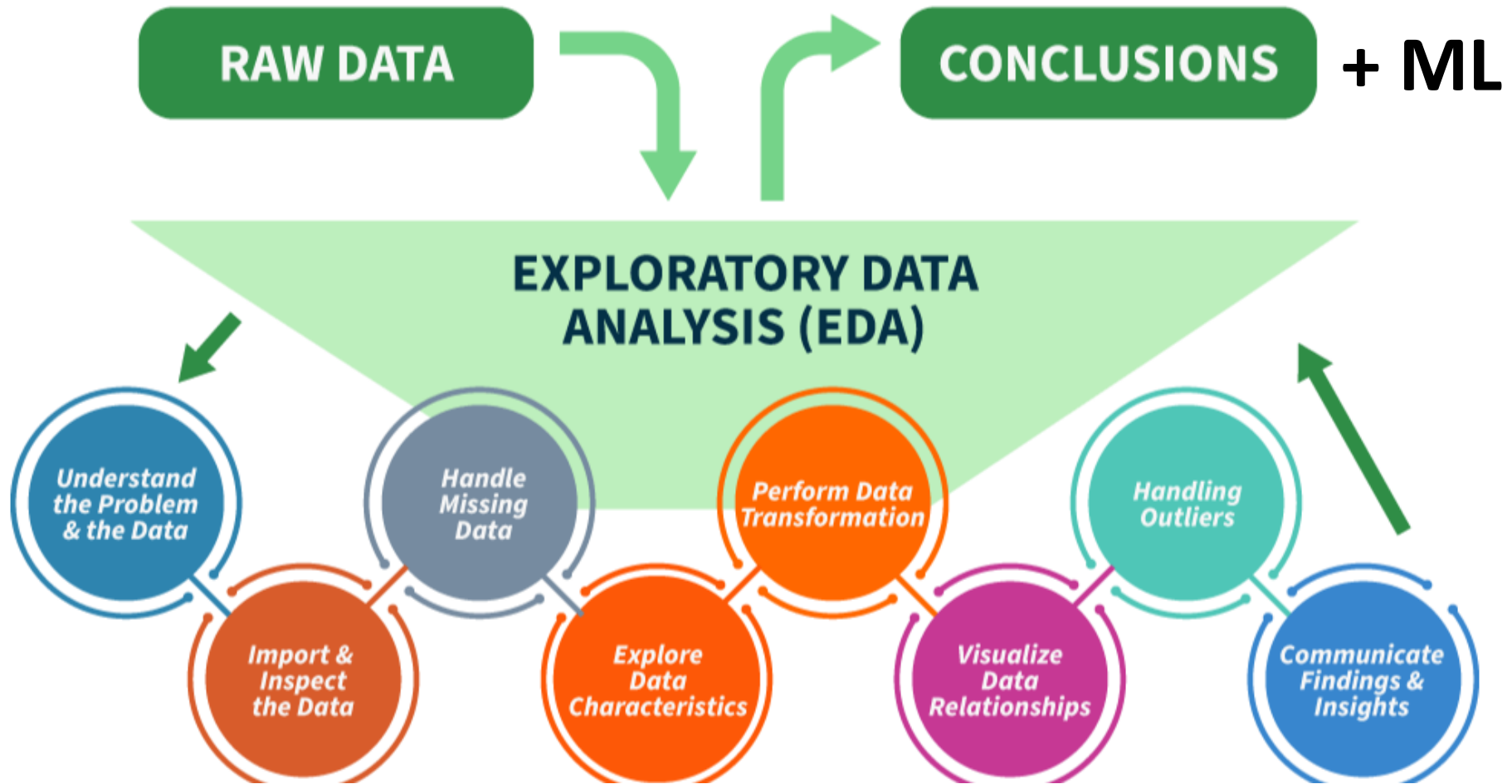
Step 2 of the ML workflow



Why EDA before ML?

- Verify the number of samples and features
- Detect missing values, outliers, and wrong ranges
- Understand distributions and feature scales
- Feature engineering and dimensionality reduction
- Identify visible clusters or biases before training
- Avoid training a model on misunderstood data

Why EDA before ML?

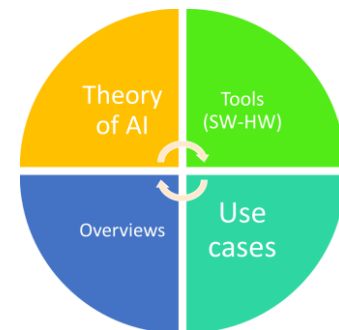




Toolboxes

EDA in Python

A simple example
to review the techniques
we just studied



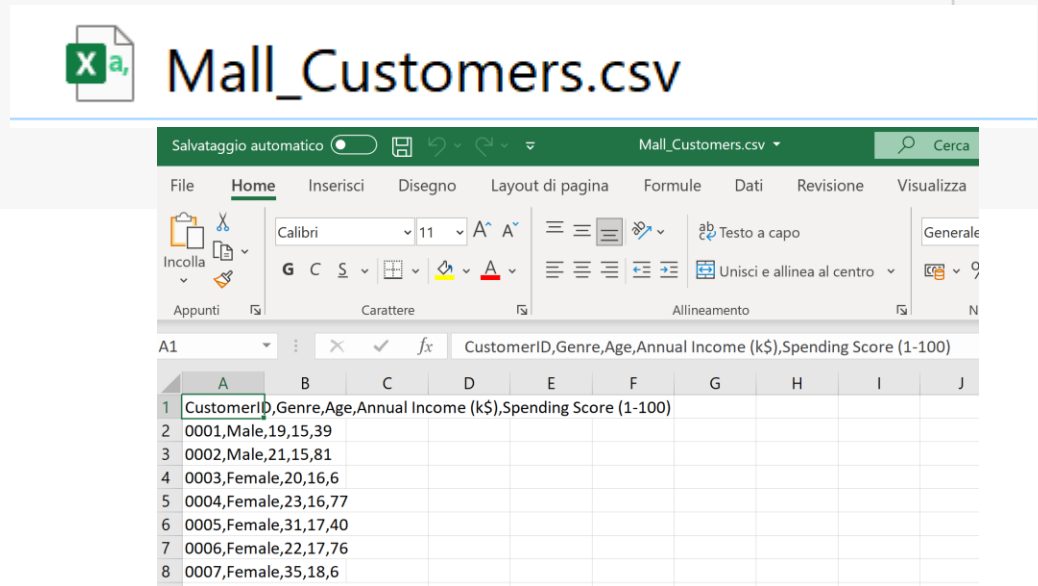
Loading files in Google Colab

```
[1] from google.colab import files
    uploaded = files.upload()
```

Scegli file Mall_Customers.csv

- **Mall_Customers.csv**(application/vnd.ms-excel) - 4286 bytes, last modified: 25/9/2019 - 100% done
Saving Mall_Customers.csv to Mall_Customers.csv

```
#import libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```



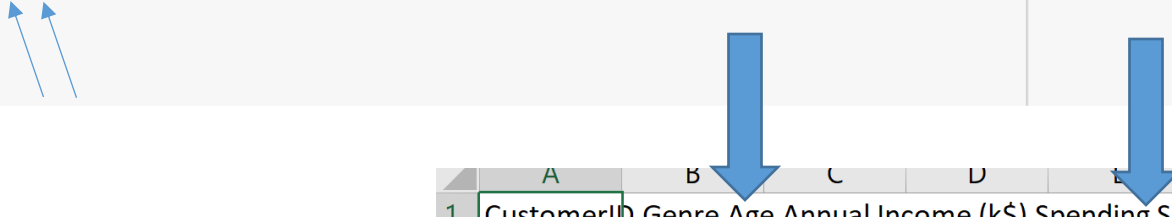
	A	B	C	D	E	F	G	H	I	J
1	CustomerID,Genre,Age,Annual Income (k\$),Spending Score (1-100)									
2	0001,Male,19,15,39									
3	0002,Male,21,15,81									
4	0003,Female,20,16,6									
5	0004,Female,23,16,77									
6	0005,Female,31,17,40									
7	0006,Female,22,17,76									
8	0007,Female,35,18,6									

pandas - Python data analysis library

Import data and feature selection

```
#import data set
dataset = pd.read_csv('Mall_Customers.csv') # CustomerID,Genre,Age,Annual Income (k$),Spending Score (1-100)
print(dataset.shape)
#cut the dabase: only Age, Spending Score (1-100)
X = dataset.iloc[:,[2,4]].values
X[0:9]
```

```
(200, 5)
array([[19, 39],
       [21, 81],
       [20,  6],
       [23, 77],
       [31, 40],
       [22, 76],
       [35,  6],
       [23, 94],
       [64,  3]])
```



	A	B	C	D	E	F
1	CustomerID,Genre,Age,Annual Income (k\$),Spending Score (1-100)					
2	0001,Male,	19,15,	39			
3	0002,Male,	21,15,	81			
4	0003,Female,	20,16,	6			
5	0004,Female,	23,16,	77			
6	0005,Female,	31,17,	40			
7	0006,Female,	22,17,	76			
8	0007,Female,	35,18,	6			

Data exploration 1 using `head()`



#DATA EXPLORATION

```
dataset.head(10)
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
5	6	Female	22	17	76
6	7	Female	35	18	6
7	8	Female	23	18	94
8	9	Male	64	19	3
9	10	Female	30	19	72

Data exploration 2 using .shape and .describe()



```
dataset.shape
```

(200, 5)



check



	A	B	C	D	E	F
1	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	
2	0001	Male	19	15	39	
3	0002	Male	21	15	81	
4	0003	Female	20	16	6	
5	0004	Female	23	16	77	
6	0005	Female	31	17	40	
7	0006	Female	22	17	76	
8	0007	Female	35	18	6	



```
dataset.describe().T
```

	count	mean	std	min	25%	50%	75%	max
CustomerID	200.0	100.50	57.879185	1.0	50.75	100.5	150.25	200.0
Age	200.0	38.85	13.969007	18.0	28.75	36.0	49.00	70.0
Annual Income (k\$)	200.0	60.56	26.264721	15.0	41.50	61.5	78.00	137.0
Spending Score (1-100)	200.0	50.20	25.823522	1.0	34.75	50.0	73.00	99.0

Data exploration 3 .info



```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 200 entries, 0 to 199
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	CustomerID	200 non-null	int64
1	Genre	200 non-null	object
2	Age	200 non-null	int64
3	Annual Income (k\$)	200 non-null	int64
4	Spending Score (1-100)	200 non-null	int64

```
dtypes: int64(4), object(1)
```

```
memory usage: 7.9+ KB
```

Data exploration 4 .isnull

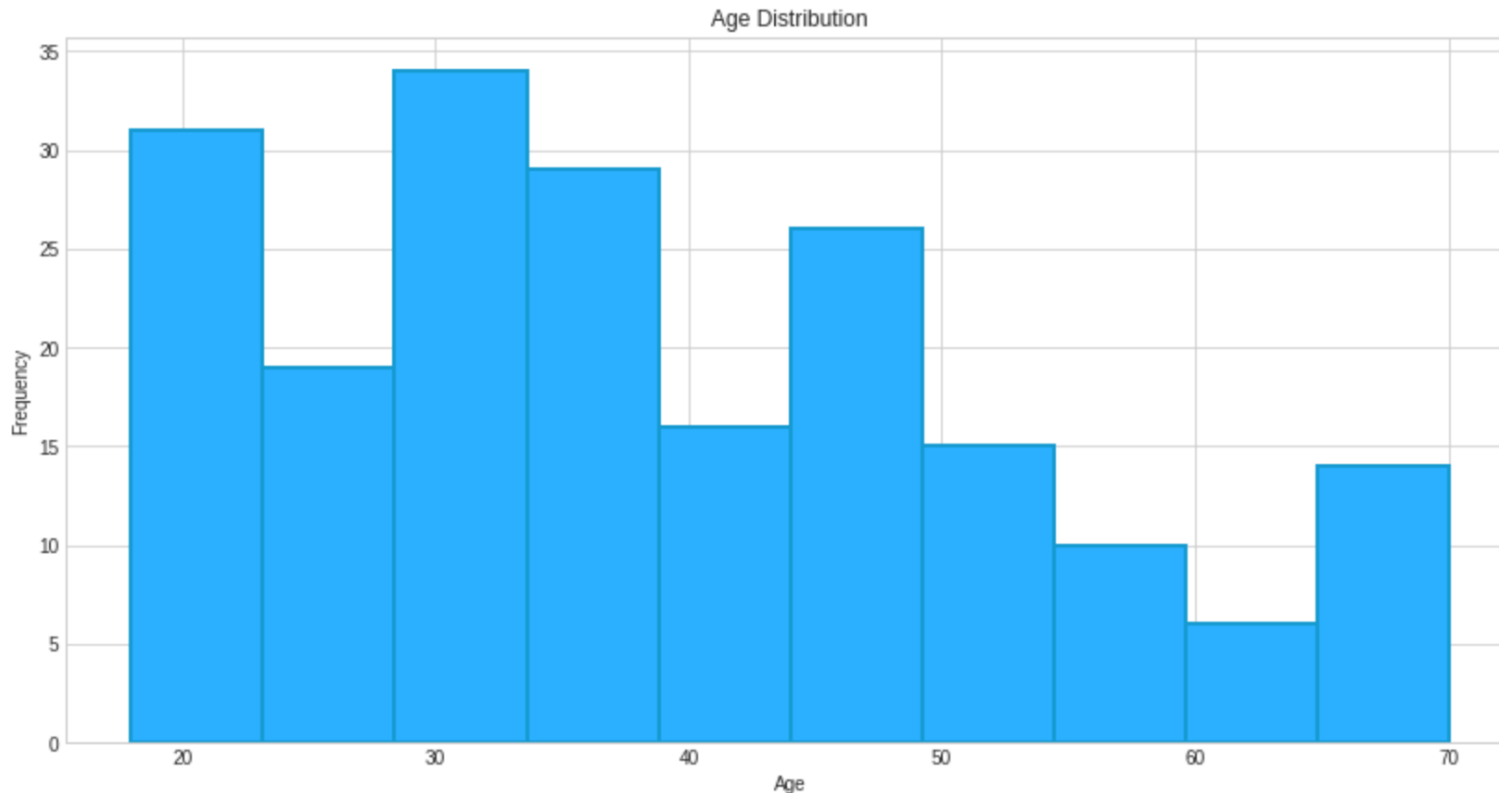


```
dataset.isnull().sum()
```

```
↳ CustomerID      0
   Genre           0
   Age             0
   Annual Income (k$)  0
   Spending Score (1-100)  0
   dtype: int64
```

Data exploration 5 plt.hist

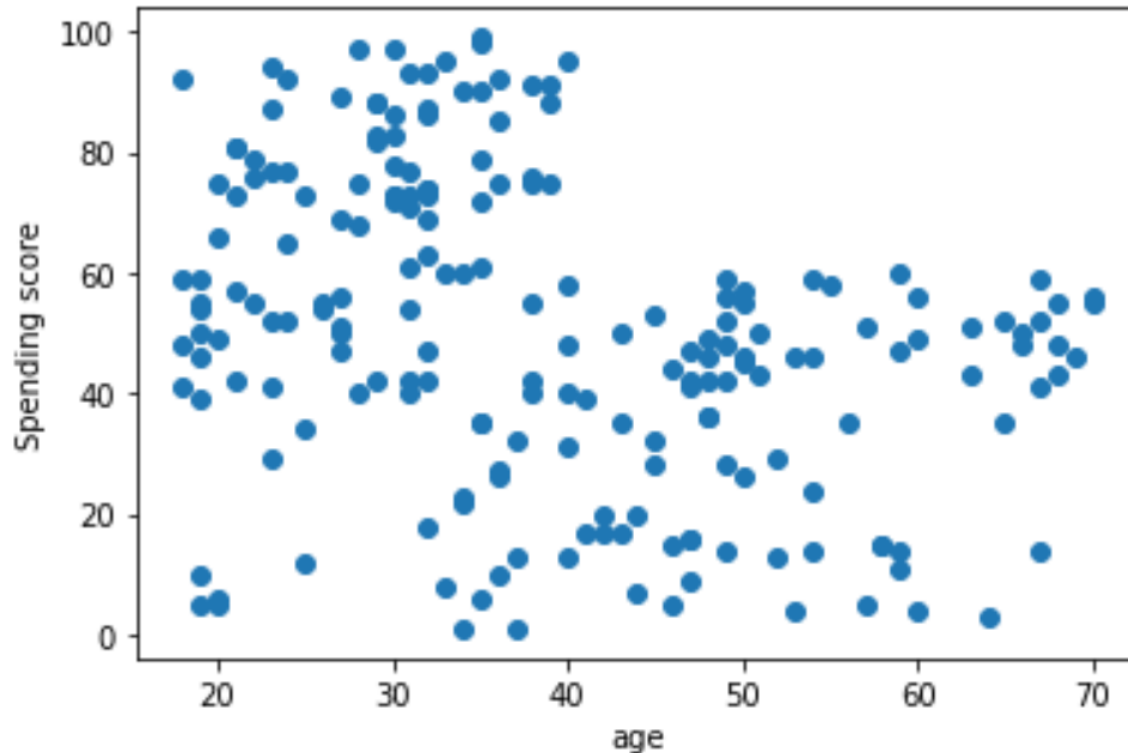
```
# age distribution
plt.style.use('seaborn-whitegrid')
plt.figure(figsize=(14,7))
plt.hist(dataset['Age'], bins=10, facecolor = '#2ab0ff', edgecolor='#169acf', linewidth=2)
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



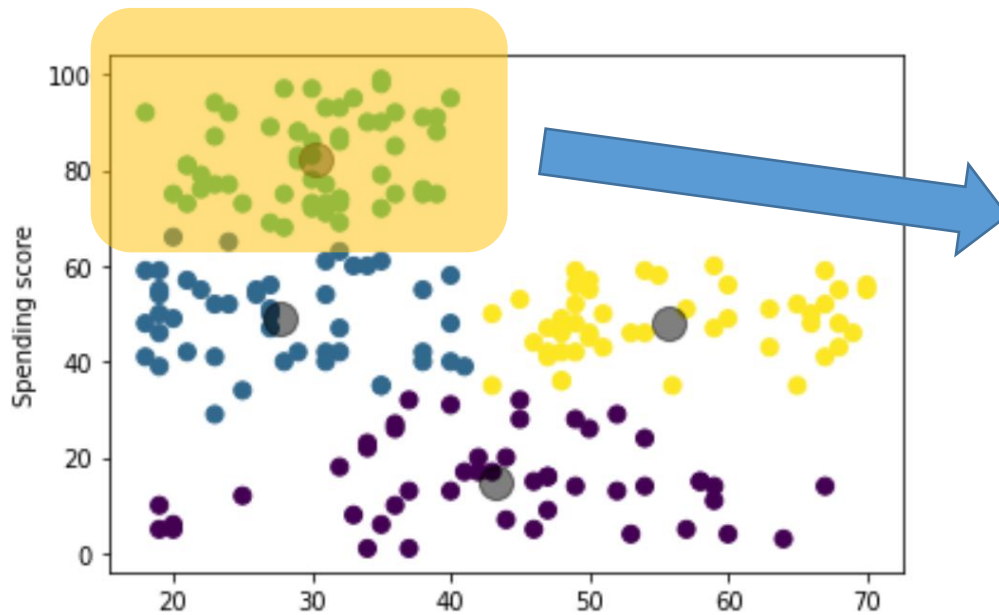
Plotting the data

```
plt.xlabel('age')  
plt.ylabel('Spending score')  
plt.scatter(X[:, 0], X[:, 1])
```

<matplotlib.collections.PathCollection at 0x7f5bb4909710>



Clustering can be used for insights!

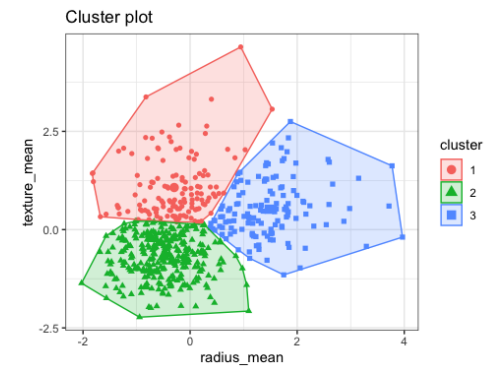


We identified a cluster characterized by

- young age
- high spending score

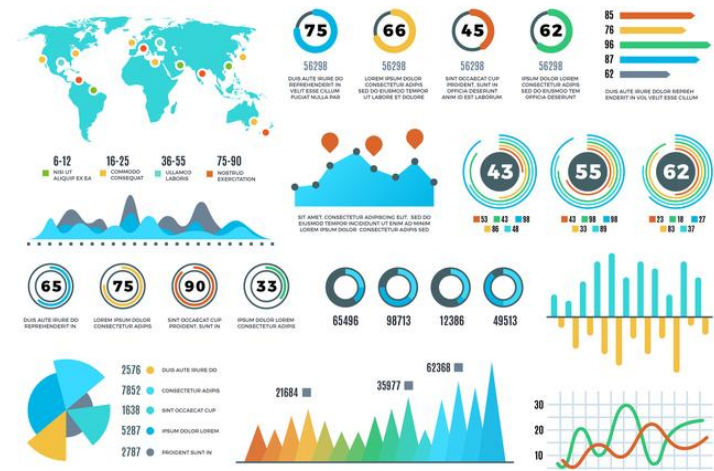
Send coupons and dedicated offers frequently

About clusters



- In the next lessons we will review main clustering methods
- **A cluster is not yet a causal explanation**
 - It suggests a segment with similar behavior
 - Actions should be tested with testing or controlled evaluation
 - individuals in the cluster to be compared with a random selection to compare the outputs.
 - Example action: personalized offers, with privacy and fairness checks

Visualization in ML

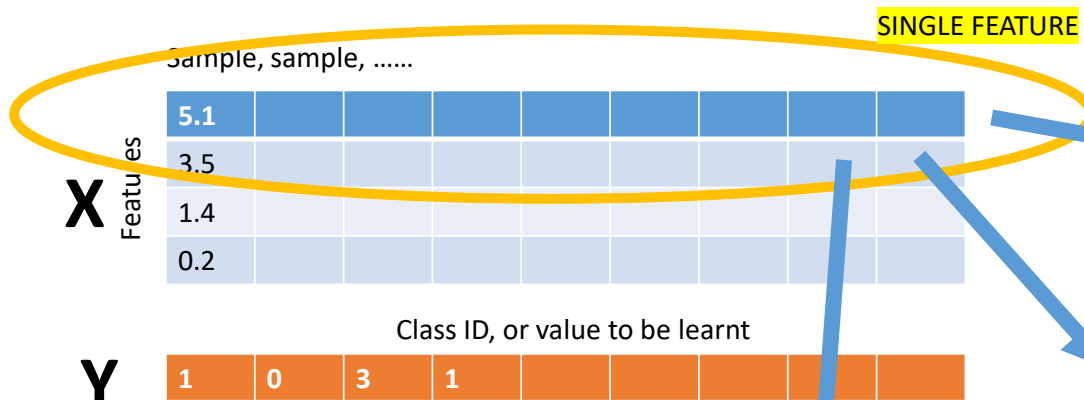


- Training Data
- Model Performance
- Interpretability + model inspection
- High-dimensional data
- Education and communication

Y = FUNC(X)

Exploratory Data Analysis / Feature engineering

- Optimizing one single feature at a time



Descriptive statistics

```
data = [1 2 3 4 50];

% The arithmetic mean of the data:
mean(data) → → ans: 12

% The median of the data:
median(data) → → ans: 3

% The standard deviation:
std(data) → → ans: 21.27

% The smallest value in the data:
min(data) → → ans: 1

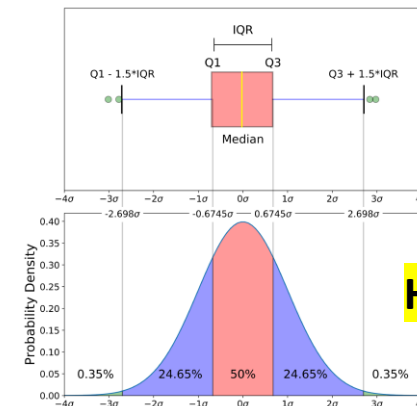
% The largest value in the data:
max(data) → → ans: 50
```

$$x' = (x - x_{min}) / (x_{max} - x_{min})$$

Simple data processing and normalization

Boxplots

Outliers/Errors

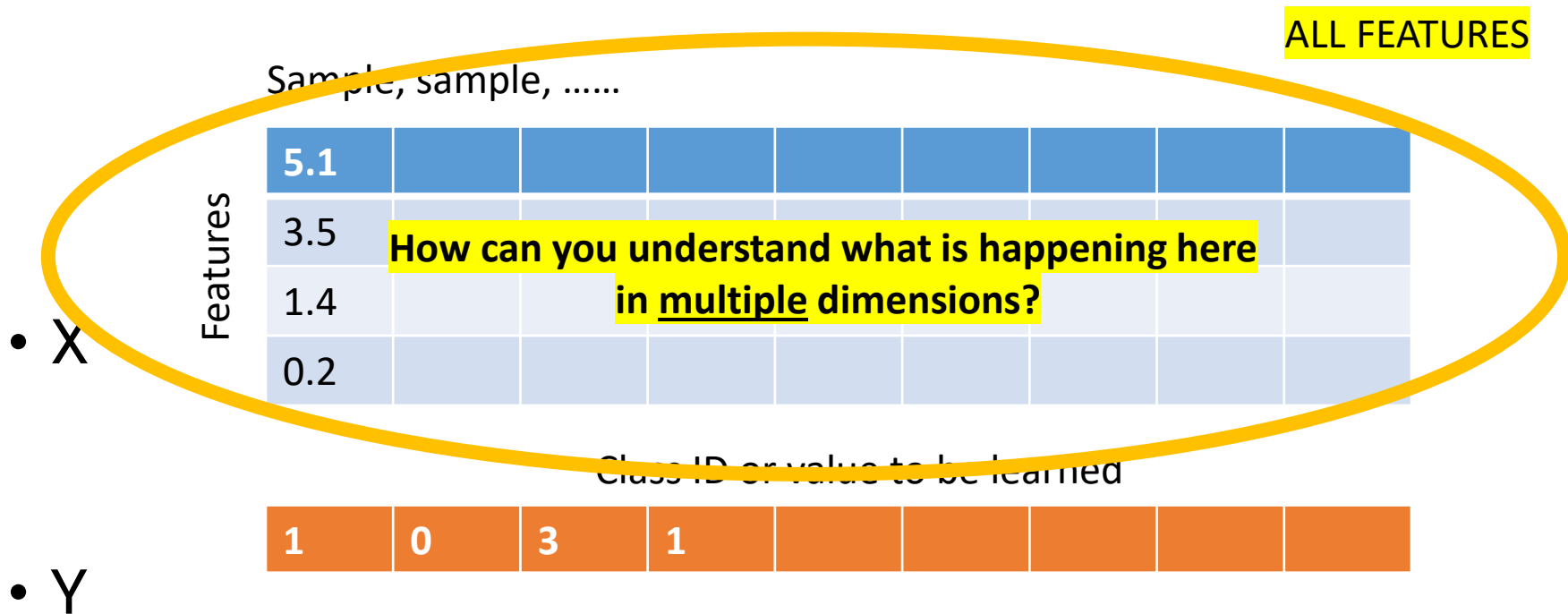


Histograms

$$Y = \text{FUNC}(X)$$

Exploratory Data Analysis / Feature engineering

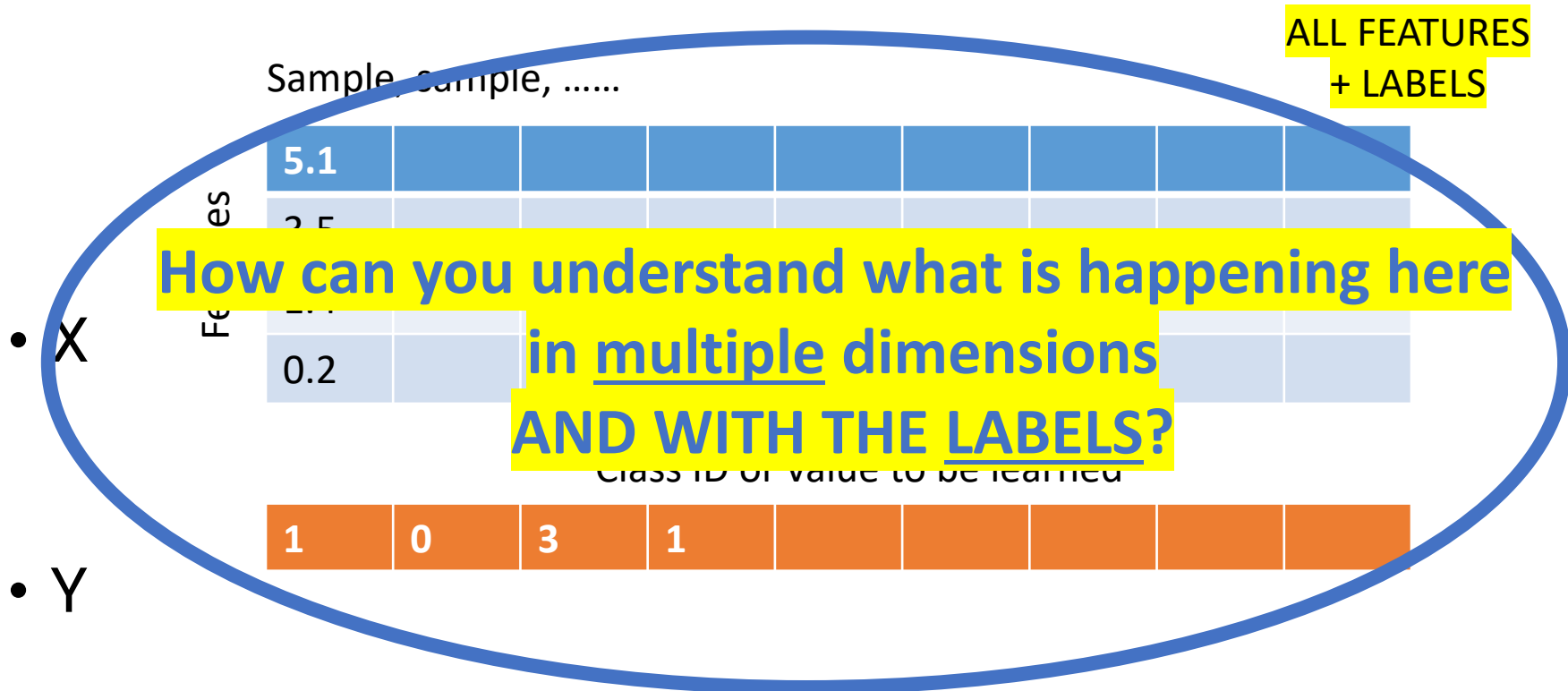
- Exploring or optimizing all features at the same time



$$Y = \text{FUNC}(X)$$

Exploratory Data Analysis / Feature engineering

- Exploring or optimizing all features and labels at the same time



What about images?

Example: Visualizing CIFAR-10



[< Back to Alex Krizhevsky's home page](#)

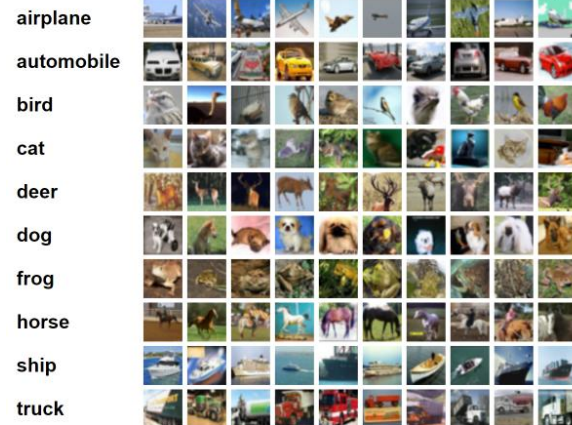
The CIFAR-10 and CIFAR-100 are labeled subsets of the [80 million tiny images](#) dataset. They were collected by Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton.

The CIFAR-10 dataset

The CIFAR-10 dataset consists of 60000 32x32 colour images in 10 classes, with 6000 images per class. There are 50000 training images and 10000 test images.

The dataset is divided into five training batches and one test batch, each with 10000 images. The test batch contains exactly 1000 randomly-selected images from each class. The training batches contain the remaining images in random order, but some training batches may contain more images from one class than another. Between them, the training batches contain exactly 5000 images from each class.

Here are the classes in the dataset, as well as 10 random images from each:



The classes are completely mutually exclusive. There is no overlap between automobiles and trucks. "Automobile" includes sedans, SUVs, things of that sort. "Truck" includes only big trucks. Neither includes pickup trucks.

Download

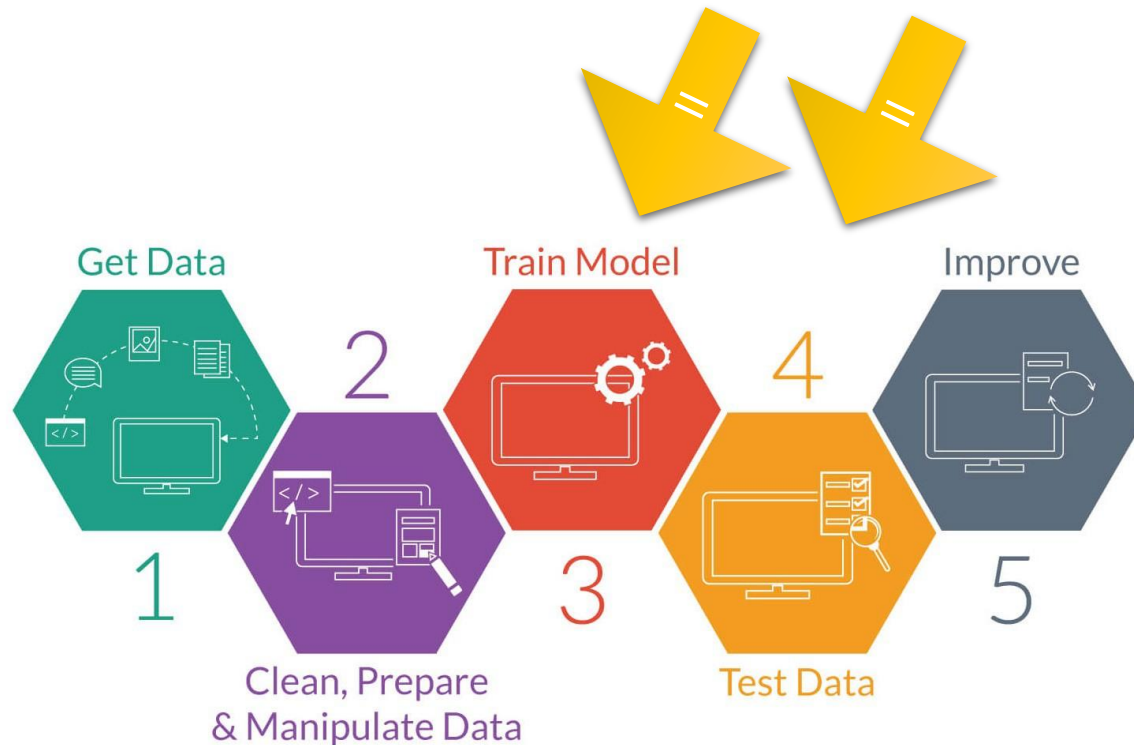
If you're going to use this dataset, please cite the tech report at the bottom of this page.

Facets:

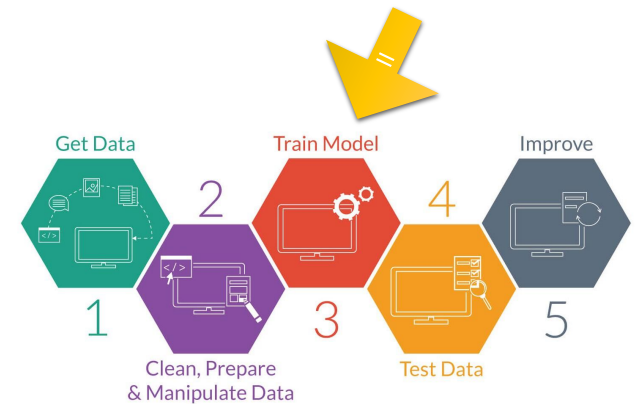
Open Source Visualization Tool for Machine Learning Training Data



Step 3-4 of the ML workflow



Why EDA after ML?

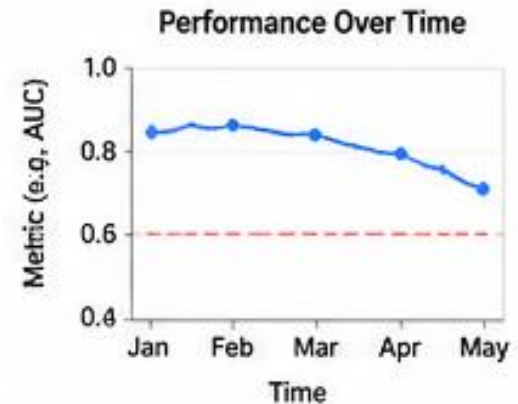


- Error analysis and interpretability
- Model performance evaluation
- Comparison of models and hyperparameters
- Confidence, calibration, and uncertainty analysis
- Deployment dashboards and drift monitoring

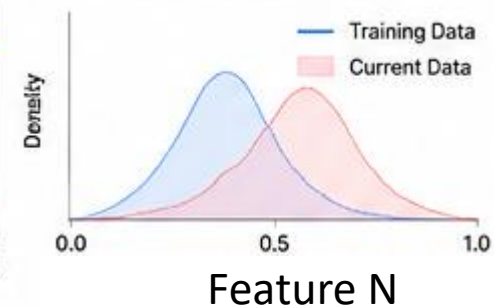
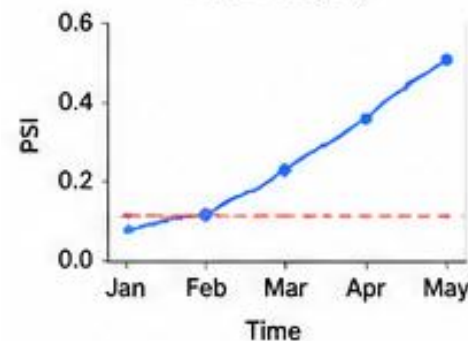


Deployment dashboards and drift monitoring

- Monitor data and model performance in production
- Detect data drift and concept drift
- Trigger alerts and take action early



Data Drift (PSI)



PSI, Population Stability Index

A numerical index used to compare the training data distribution with the current production data distribution.

Alerts and Actions

- ▲ High data drift detected
- ▲ Performance below threshold
- ✓ Model retraining scheduled

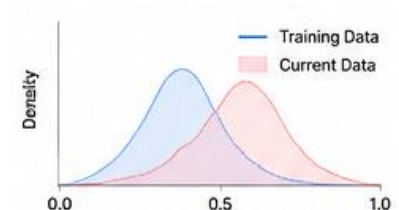
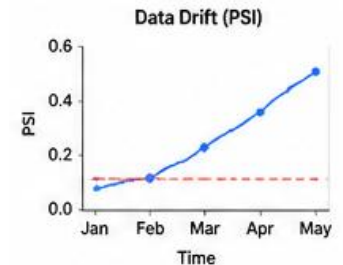
Last update: 2 min ago

Detecting Drift after deployment

The EU AI Act mandates strict, continuous performance monitoring for "high-risk" AI systems and "General-Purpose AI (GPAI)"

AI Act perspective

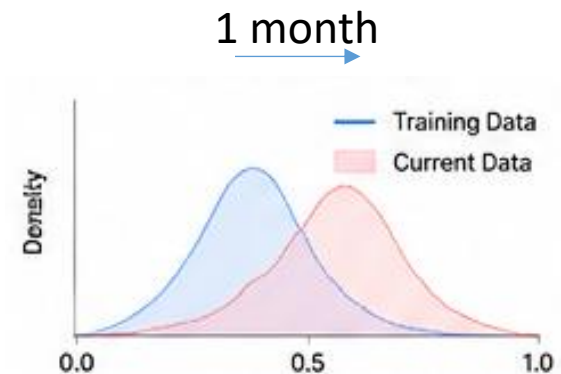
- Post-deployment monitoring is required for high-risk AI systems.
- That can be done analyzing typically
 - The **Input of the model**
 - E.g., the Population Stability Index
 - The **Output of the model**
 - E.g., the Prediction Distribution Drift



Population Stability Index (PSI)

- PSI can help detect **input data** drift, which may lead to bias, accuracy degradation, or the need for recalibration and retraining monitoring whether the deployed model is still receiving familiar input data.

$$PSI = \sum_i (p_i - q_i) \ln \left(\frac{p_i}{q_i} \right)$$



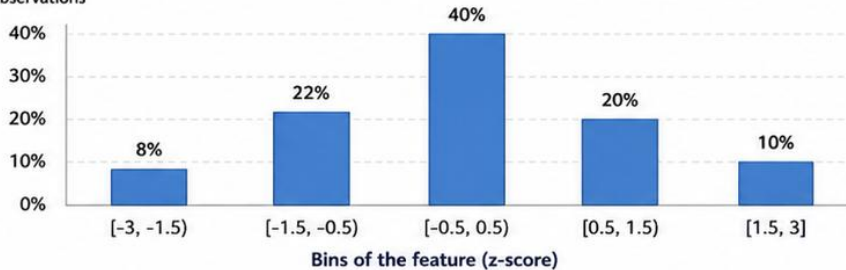
where p_i is the percentage of samples in bin i in the reference data, and q_i is the percentage of samples in the same bin in the current production data.

Population Stability Index (PSI)

REFERENCE DISTRIBUTION (TRAINING DATA)

Feature: transaction amount (normalised)

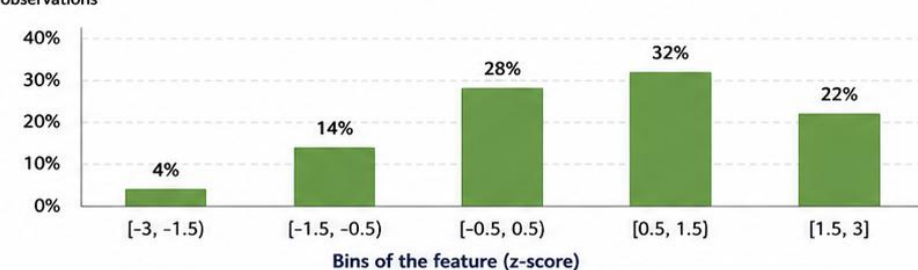
Percent of observations



CURRENT DISTRIBUTION (PRODUCTION DATA)

Same feature, new data in production

Percent of observations



PSI = 0.2576

Interpretation (common rule of thumb)

- PSI < 0.1 No significant drift
- 0.1 ≤ PSI < 0.25 Moderate drift
- PSI ≥ 0.25 Significant drift

$$PSI = \sum_i (p_i - q_i) \ln \left(\frac{p_i}{q_i} \right)$$

Bin (z-score)	Reference % p_i	Current % q_i	Difference $p_i - q_i$	$\ln \left(\frac{p_i}{q_i} \right)$	PSI contribution $(p_i - q_i) \ln(p_i / q_i)$
[-3, -1.5]	0.08	0.04	0.04	$\ln(0.08 / 0.04) = 0.6931$	$0.04 \times 0.6931 = 0.0277$
[-1.5, -0.5]	0.22	0.14	0.08	$\ln(0.22 / 0.14) = 0.4511$	$0.08 \times 0.4511 = 0.0361$
[-0.5, 0.5]	0.40	0.28	0.12	$\ln(0.40 / 0.28) = 0.3567$	$0.12 \times 0.3567 = 0.0428$
[0.5, 1.5]	0.20	0.32	-0.12	$\ln(0.20 / 0.32) = -0.4700$	$-0.12 \times -0.4700 = 0.0564$
[1.5, 3]	0.10	0.22	-0.12	$\ln(0.10 / 0.22) = -0.7885$	$-0.12 \times -0.7885 = 0.0946$
TOTAL	1.00	1.00			PSI = 0.2576

Here the PSI method is applied to an **input** feature

Prediction Distribution Drift

Prediction drift is not the same as feature drift.

- **Feature drift concerns the input data distribution**
- **Prediction drift concerns the output distribution of the model**
- Prediction drift is a compact post-deployment signal that the behavior of the model may be changing, even before full ground-truth labels are available.

Prediction Distribution Drift

Assume a typical classification system. We monitor the distribution of the output (before the decision w.r.t) to all possible input X in an interval of time (using PSI)

Here the PSI method is applied to the **output** value

Monitoring the output of a classifier: $P(Y = 1 | X)$

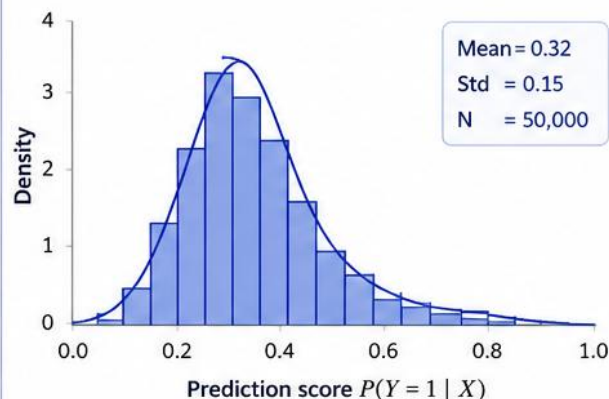
A classifier outputs a probability score $s = P(Y = 1 | X)$ for each input X .
Prediction drift occurs when the distribution of these scores changes over time.

Why it matters

- May indicate changes in data, concept drift, or model degradation
- Allows early detection before performance drops

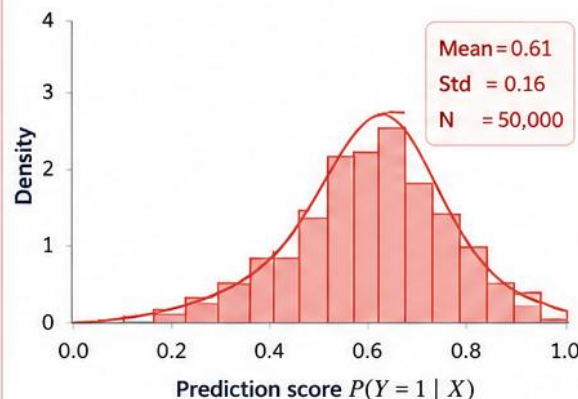
1) REFERENCE PERIOD (training/validation)

Distribution of predicted scores $P(Y = 1 | X)$



2) CURRENT PERIOD (production)

Distribution of predicted scores $P(Y = 1 | X)$



3) COMPARE DISTRIBUTIONS USING PSI

Bins on prediction score

Bin range	Reference % (p_i)	Current % (q_i)	$(p_i - q_i) \ln\left(\frac{p_i}{q_i}\right)$
[0.0, 0.2)	25.0%	8.2%	0.2032
[0.2, 0.4)	45.0%	18.5%	0.3236
[0.4, 0.6)	20.0%	30.3%	0.0926
[0.6, 0.8)	8.0%	30.6%	0.3185
[0.8, 1.0]	2.0%	12.4%	0.1229
TOTAL PSI			1.0608

PSI = 1.06 → Significant prediction drift

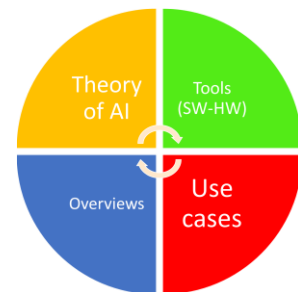
(Rule of thumb: PSI < 0.1 no drift, 0.1–0.25 moderate, > 0.25 significant)



THEORY

Saliency maps

Checking what the model is looking at!

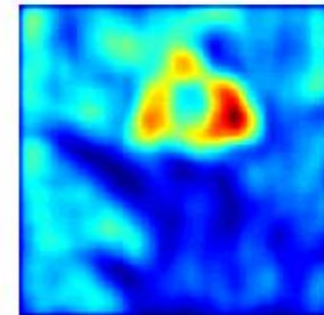


Saliency maps

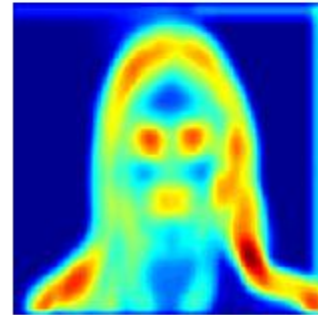
- Highlight input regions that influence the prediction
- Help detect spurious correlations or unwanted biases
- Support error analysis and model interpretability
- A diagnostic tool, not definitive explanations



Butterfly Saliency Map



Person Saliency Map



Saliency maps: example

(in the next lessons)....Take the gradient of a class prediction neuron with respect to the input pixels.

SmoothGrad technique often significantly denoises the sensitivity mask.

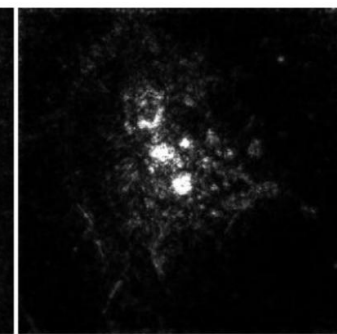
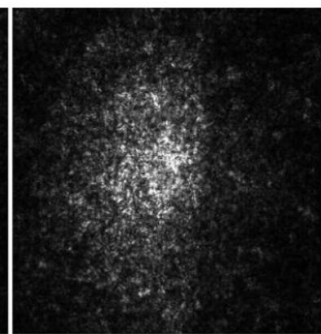
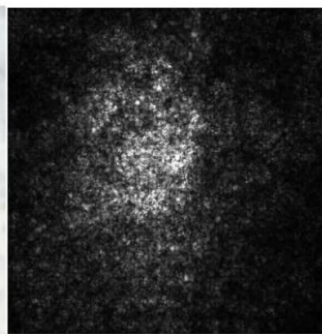
This technique adds pixel-wise Gaussian noise to many copies of the image, and simply averages the resulting gradients.

Image

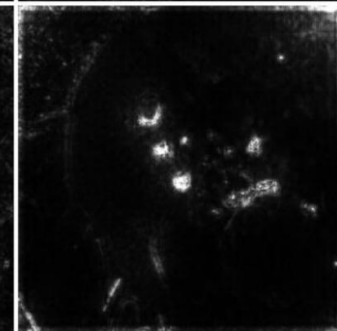
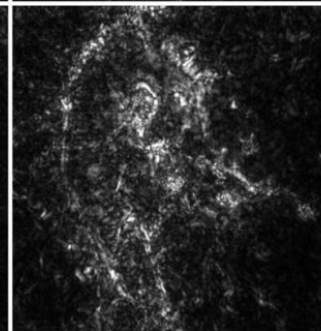
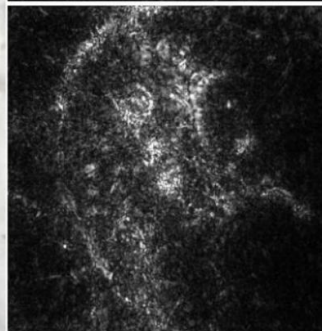
Gradient

Integrated

Guided Backprop



Gradient



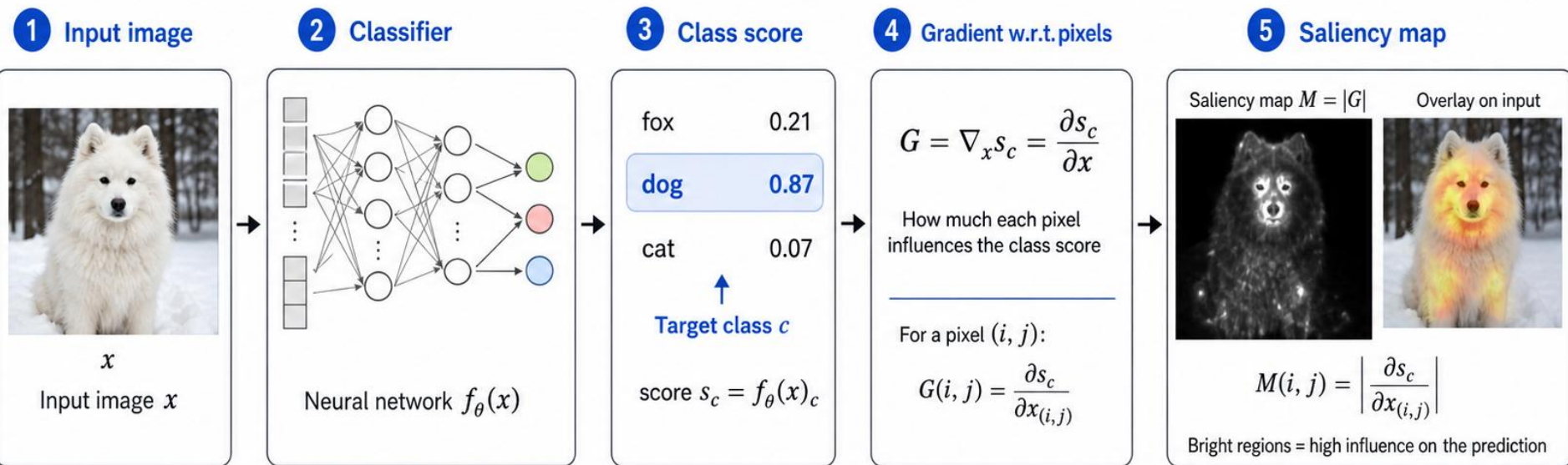
Gradient $\times$ Image

Determine which input features (pixels, for images) were important for the prediction

The saliency map is built using gradients of the output over the input. This highlights the areas of the images which were relevant for the classification

How the saliency map is computed

The gradient of the class score with respect to the input pixels measures how much the output score changes when each input pixel is slightly changed.



1. Choose a target class score

$$\frac{\partial}{\partial x}$$

2. Differentiate the score with respect to the input pixels



3. Visualize the magnitude of the gradient as a saliency map



Takeaway:

Saliency maps highlight image regions that most affect the prediction.

How to use the saliency maps

Input image



NN₁

Dog = 0,95

**What is the
best NN?**

Input image

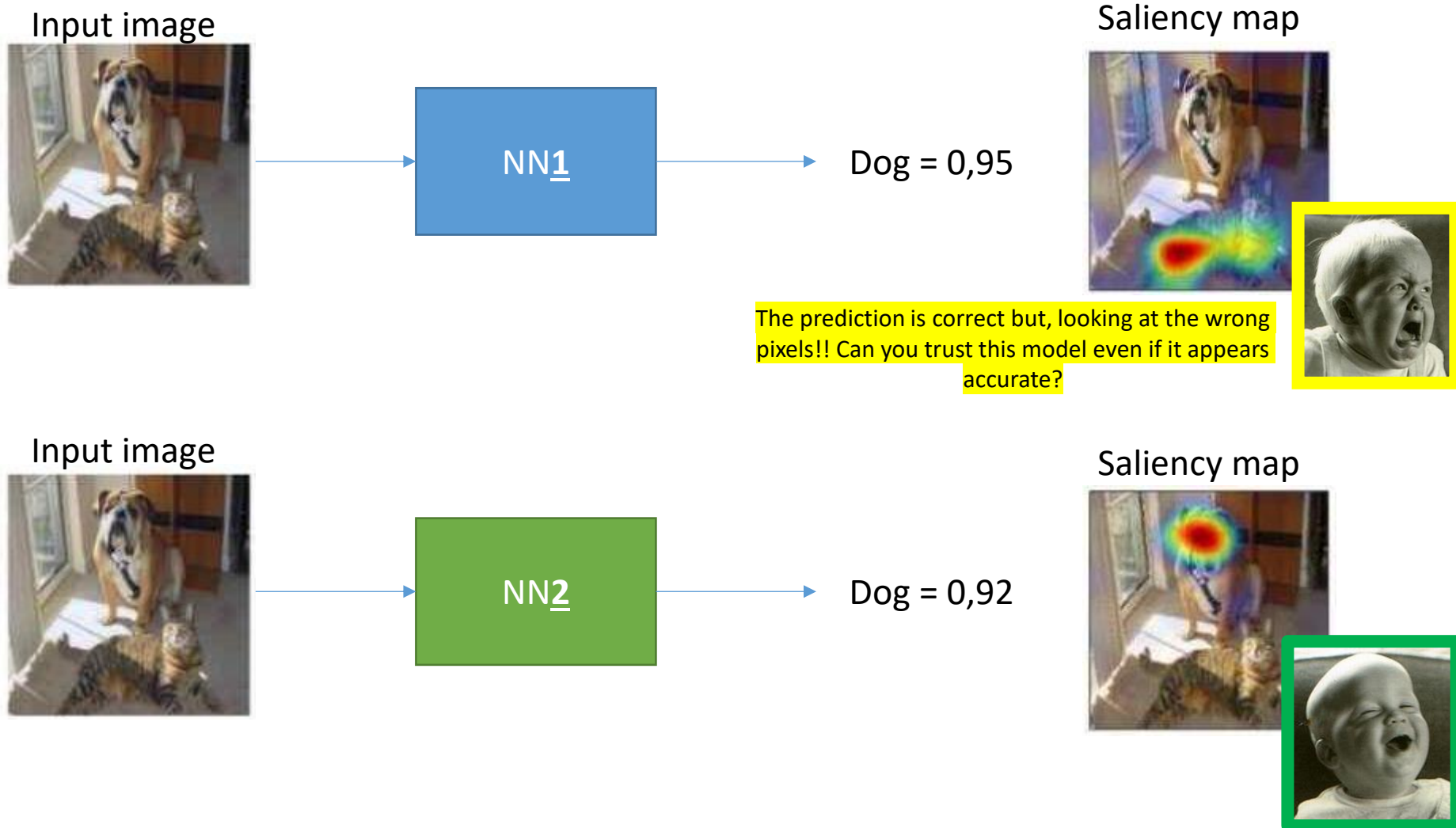


NN₂

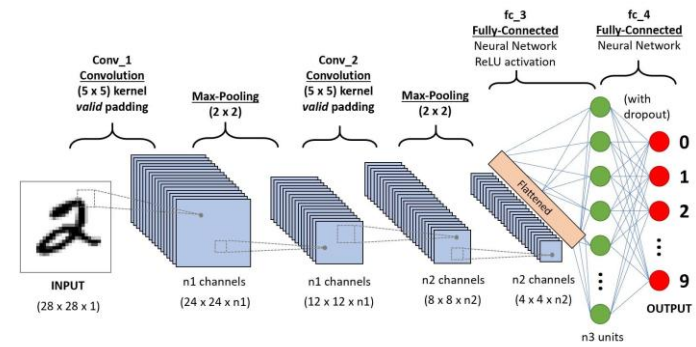
Dog = 0,92

... sure?

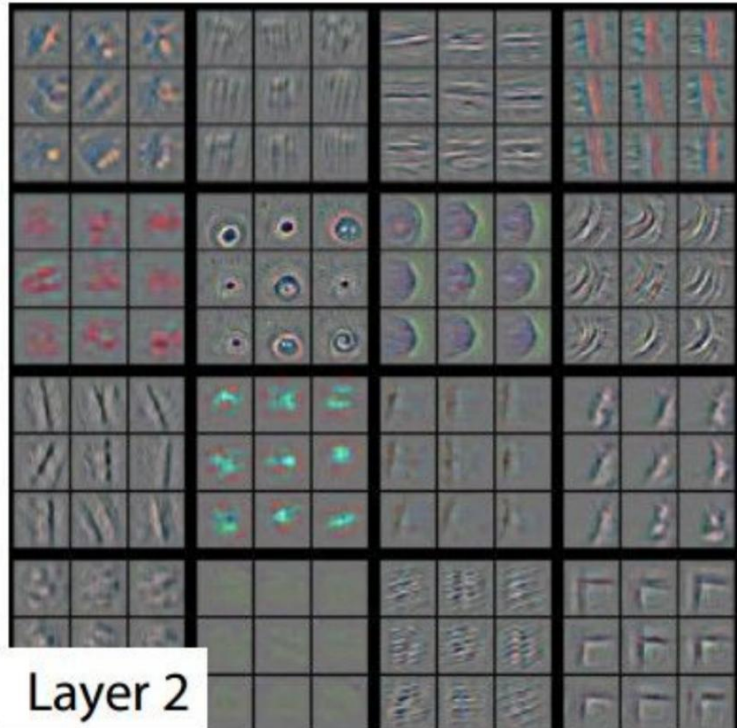
How to use the saliency maps



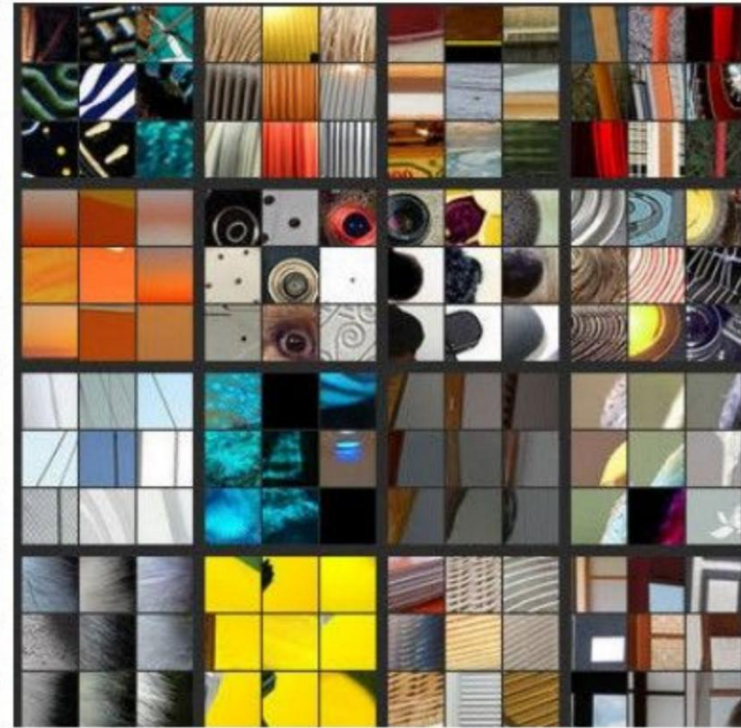
Visualizing Kernels in CNNs(What patterns are they detecting?)



Layer 1



Layer 2



Gray squares: kernels

Colorful squares: maximal examples from an image data set producing the greatest output of the kernel

Build your ML model visualization...

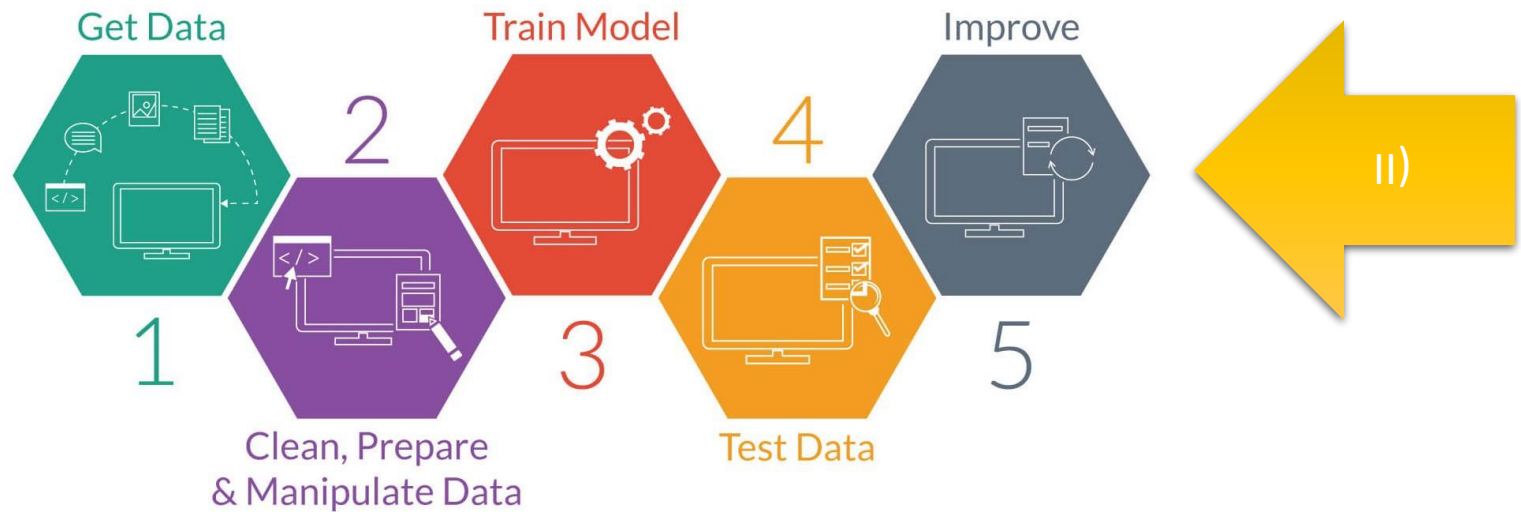
Maybe too dense?



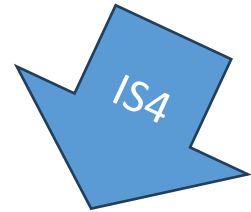
DQNViz:
A Visual
Analytics
Approach to
Understand
Deep Q-
Networks

(DQN) is one type of deep reinforcement learning model

Step 5 of the ML workflow



BI / monitoring dashboard tools con AI assistant,



ML experiment visualization

TensorBoard, Weights & Biases, MLflow, Neptune, ClearML

Training curves, loss, accuracy, hyperparameters, model comparison

BI / AI-powered dashboards

Tableau, Power BI, Zoho Analytics, Domo, ThoughtSpot

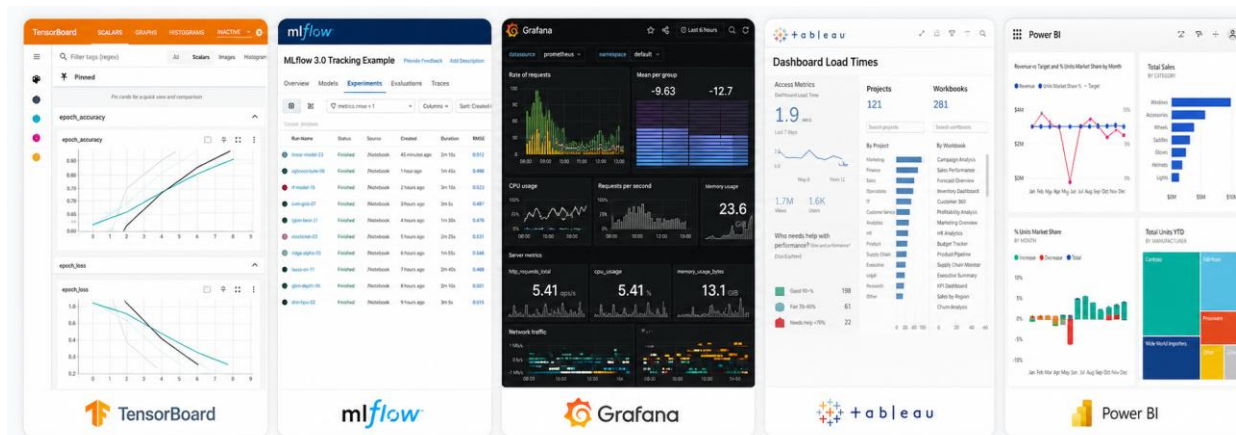
Business metrics, natural language queries, KPI monitoring, reports

Observability / operational monitoring

Grafana, Datadog, Prometheus, Evidently AI, WhyLabs

Production systems, telemetry, drift, alerts, reliability

(Observability means monitoring a deployed system through measurable signals in order to understand its behavior, detect failures, and support corrective actions)

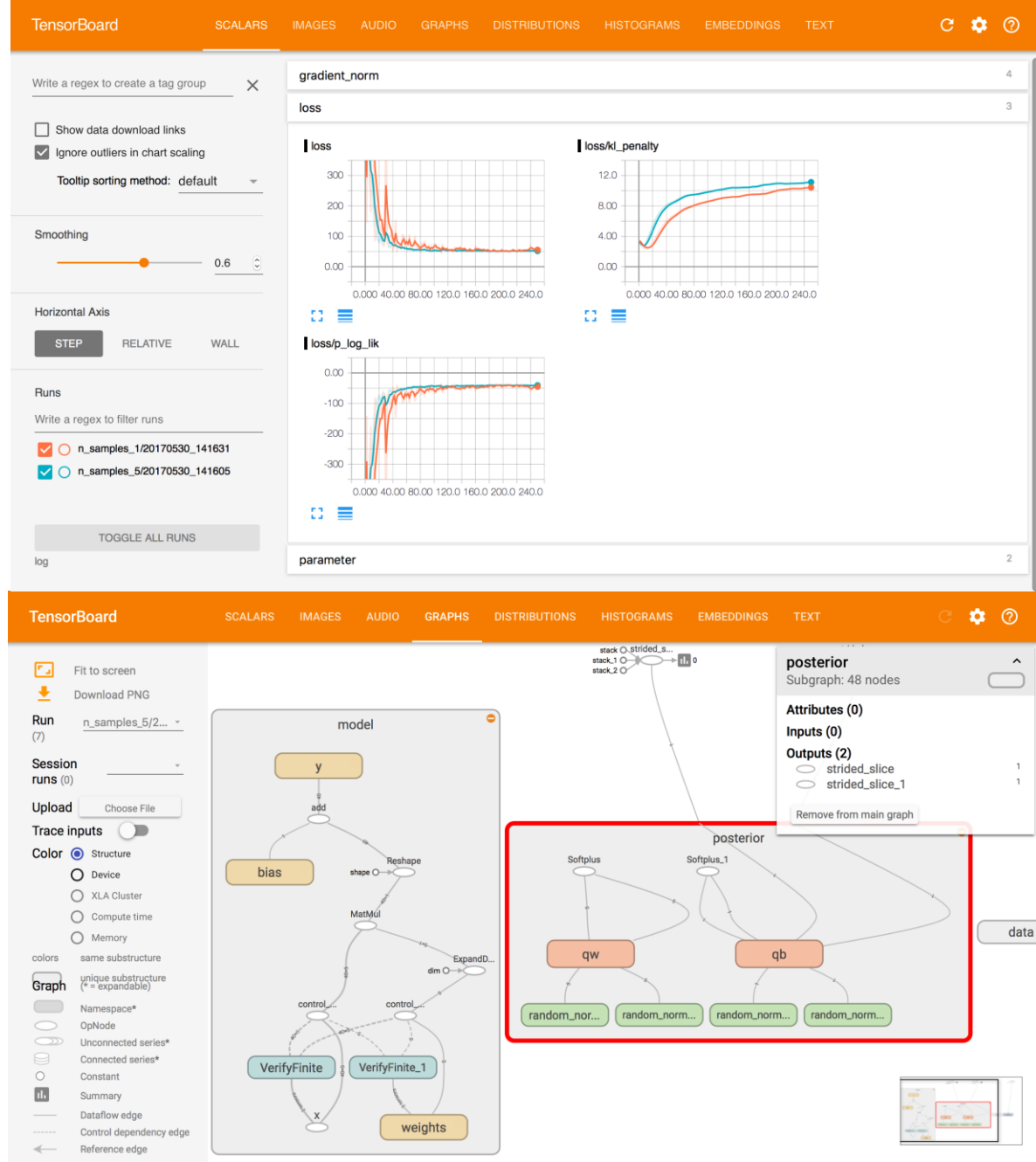


Monitoring dashboards Standard visualization tools

TensorBoard (Google)

TensorBoard provides a suite of visualization tools to make it easier to understand, debug, and optimize your models

You can use it “to visualize your TensorFlow graph, plot quantitative metrics about the execution of your graph, and show additional data like images that pass through it”



Monitoring dashboards Standard visualization tools

Visdom (Meta AI)

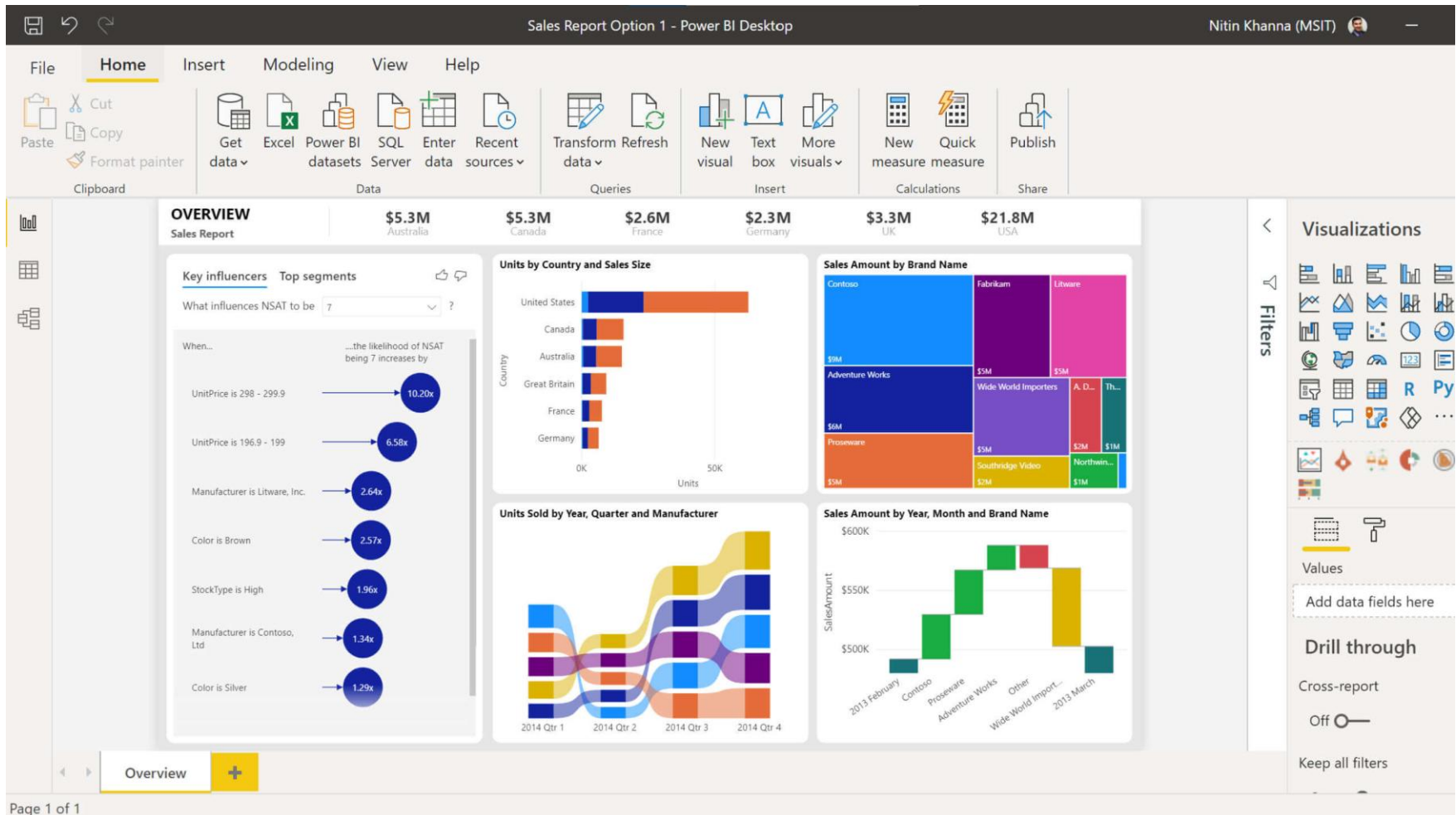
Visdom is a visualization tool that generates rich visualizations of live data to help researchers and developers stay on top of their scientific experiments that are run on remote servers.

Visualizations in Visdom can be viewed in browsers and easily shared with others.



Example: MS Power Bi

“Connect to and visualize any data using the unified, scalable platform for self-service and enterprise business intelligence (BI) that’s easy to use and helps you gain deeper data insight.”



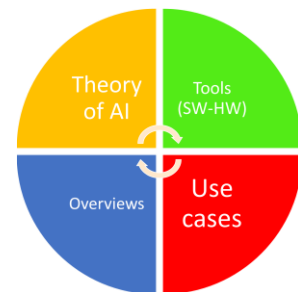


THEORY

Data visualization

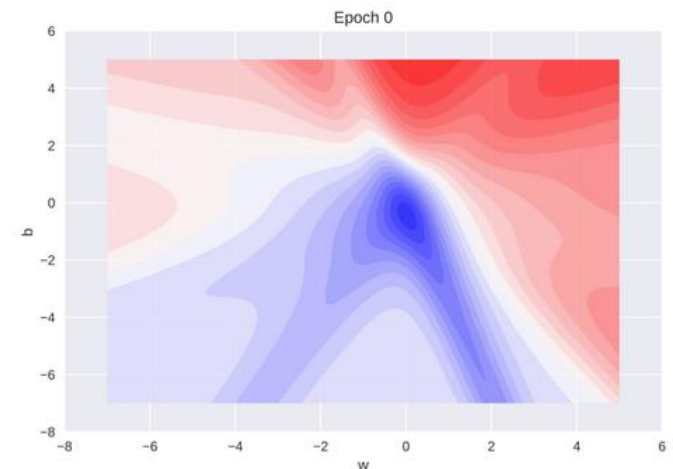
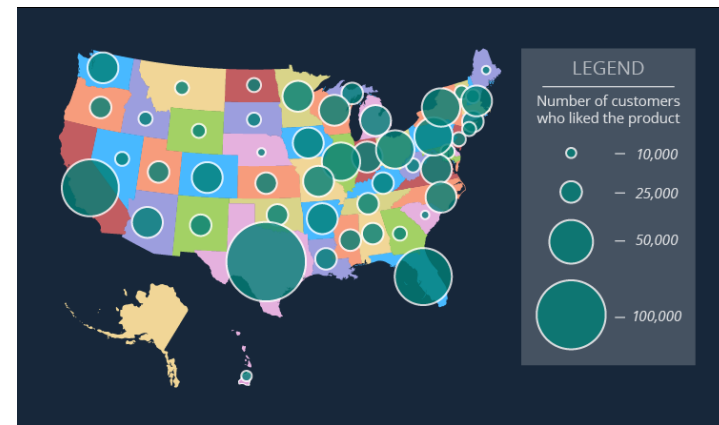


General concepts



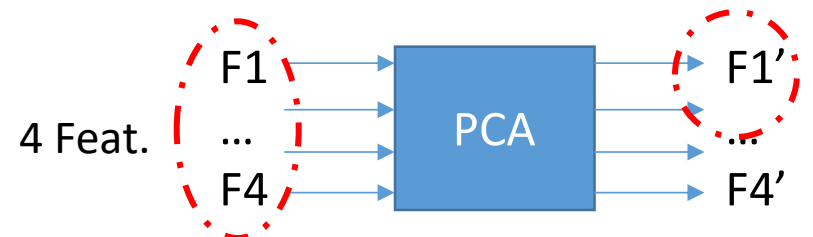
Introduction

- What is Data Visualization?
- How does Data Visualization Work?
 - Image Theory
 - Definition of an image representation
 - Data Visualization and its use today
- What are the benefits of Data Visualization?
- Examples of Data Visualization



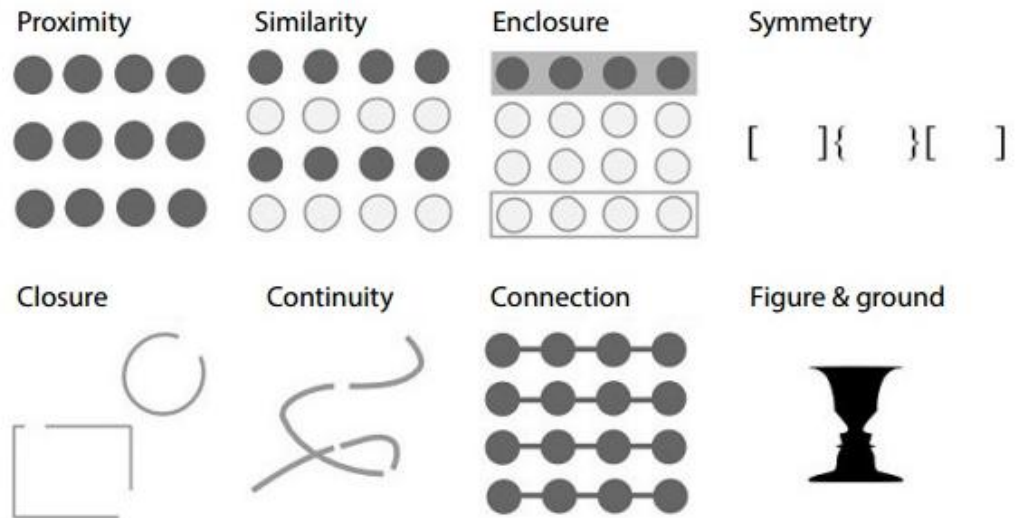
Why (almost) 2D and 3D plots?

- We live in the *Space-Time*
 - a mathematical model fusing the three dimensions of space and the one dimension of time into a single four-dimensional continuum....
- Ok, but our perception is limited...
 - Only 2D and (rarely) 3D plots can be easily evaluated and understood
- That is also why it is so important to manage dimensionality reduction!
- Let's try to add some information beyond just 2-3 axes!



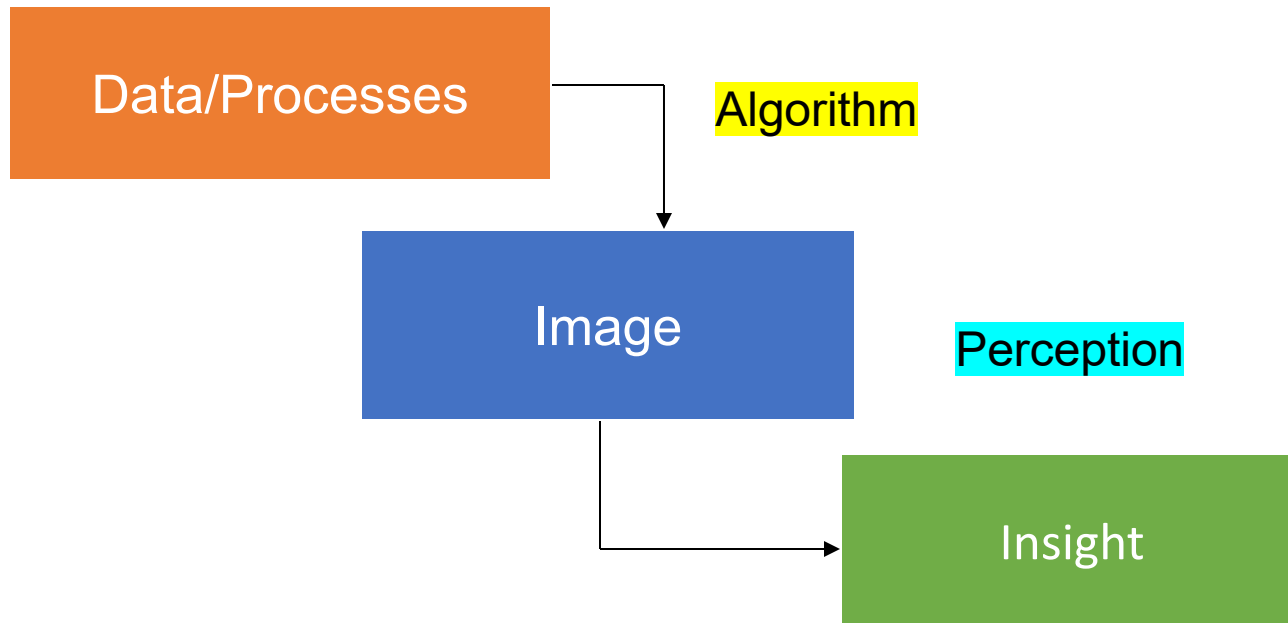
What is Data Visualization?

- Data visualization is the process of converting raw data into easily understood pictures of information that enable fast and effective decisions.
- Early in the 20th-century, gestalt psychologists observed that when elements were gathered into a figure, the figure took on a perceptual salience that exceeded the sum of its parts;



Data -> Easily Understood Pictures

- Jacques Bertin who wrote the classic works of graphical visualization “Semiology of Graphics” states that the “transformation from numbers to insight requires two stages.”



Bertin's 7 Visual Variables

- 7 Visual Variables
 - position
 - form
 - orientation
 - color
 - texture
 - value
 - size
- combined with a visual semantics for linking data attributes to visual elements

Visualization: encoding data using visual cues

Length



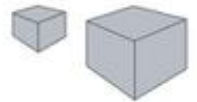
Slope



Color hue



Volume



Angle



Length (aligned)



Area



Color intensity



Not all the
visualization
features are
of the same
kind...

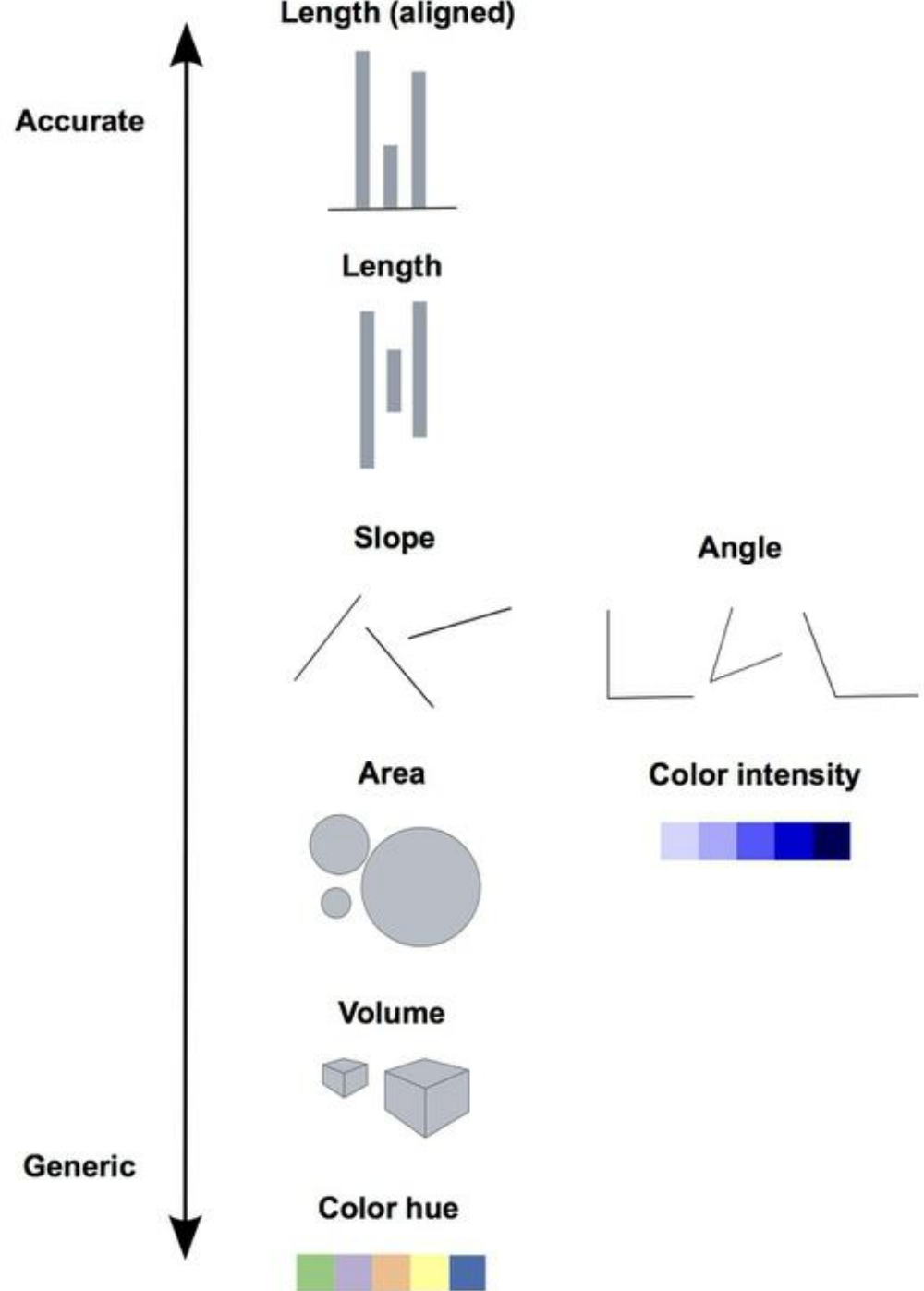
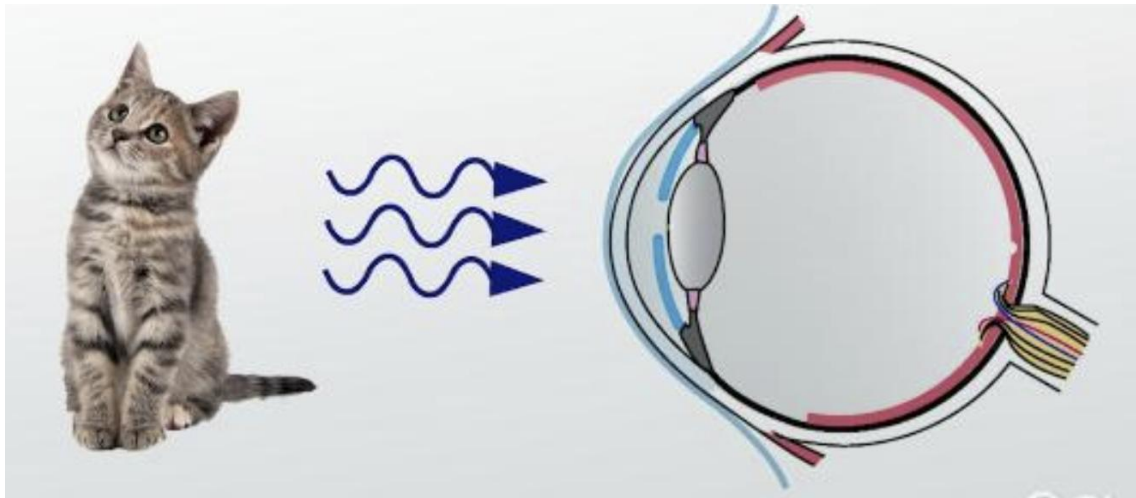
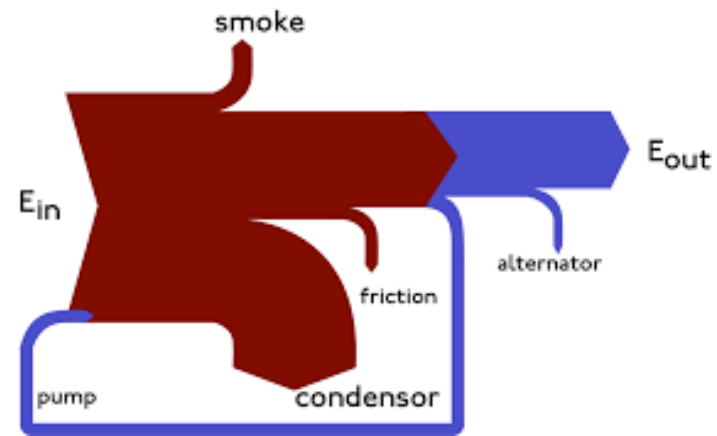


Image Theory

- Visual Processing occurs in 3 steps.
 1. Formation of the retinal image
 2. Decomposition of the retinal image information into an array of specialized representations
 3. Reassembly of the information into object perception.



Image

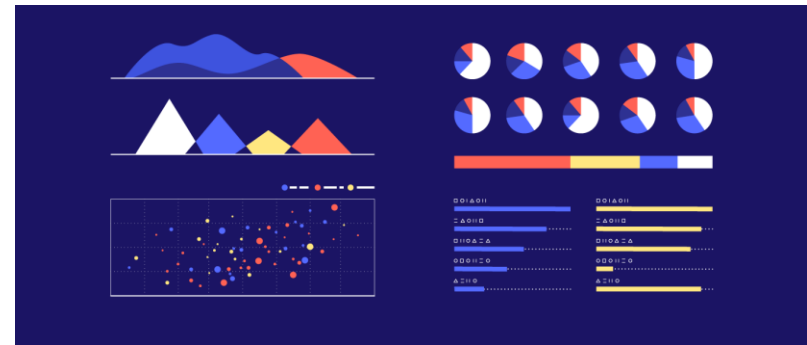


Where are you losing energy?

- Bertin's key concept is the *image*, from which the theory derives its name.
- Simplifying.... an *image* is the fundamental perceptual unit of a visualization.
 - An ideal visualization contains only a single image in order to optimize "efficiency" the speed with which observer can extract the information

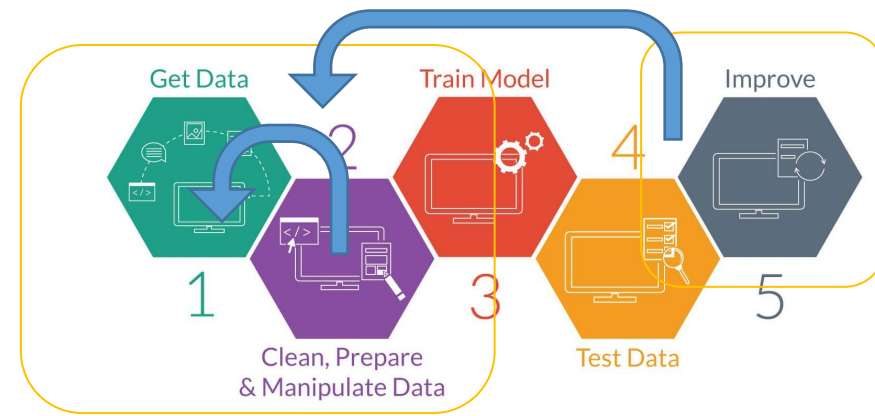
A fusion of methods for your applications

- Information visualization is emerging as an important **fusion** of
 - graphics,
 - scientific visualization,
 - database management,
 - human-computer interaction.
- In your applications use **Data Visualization to convey complex results as understandable images**, improving your HMI

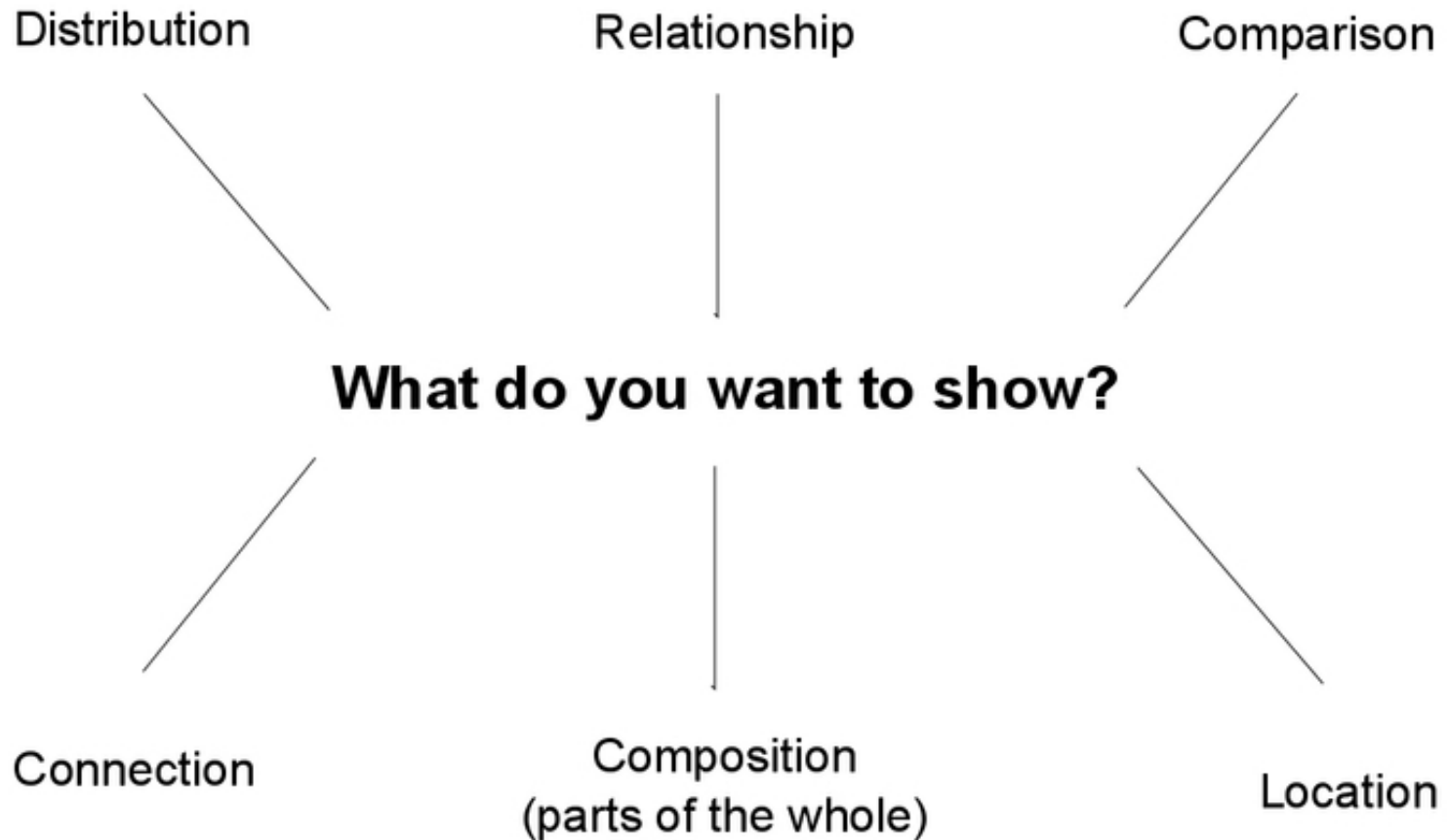


Important for users and the designers of intelligent systems

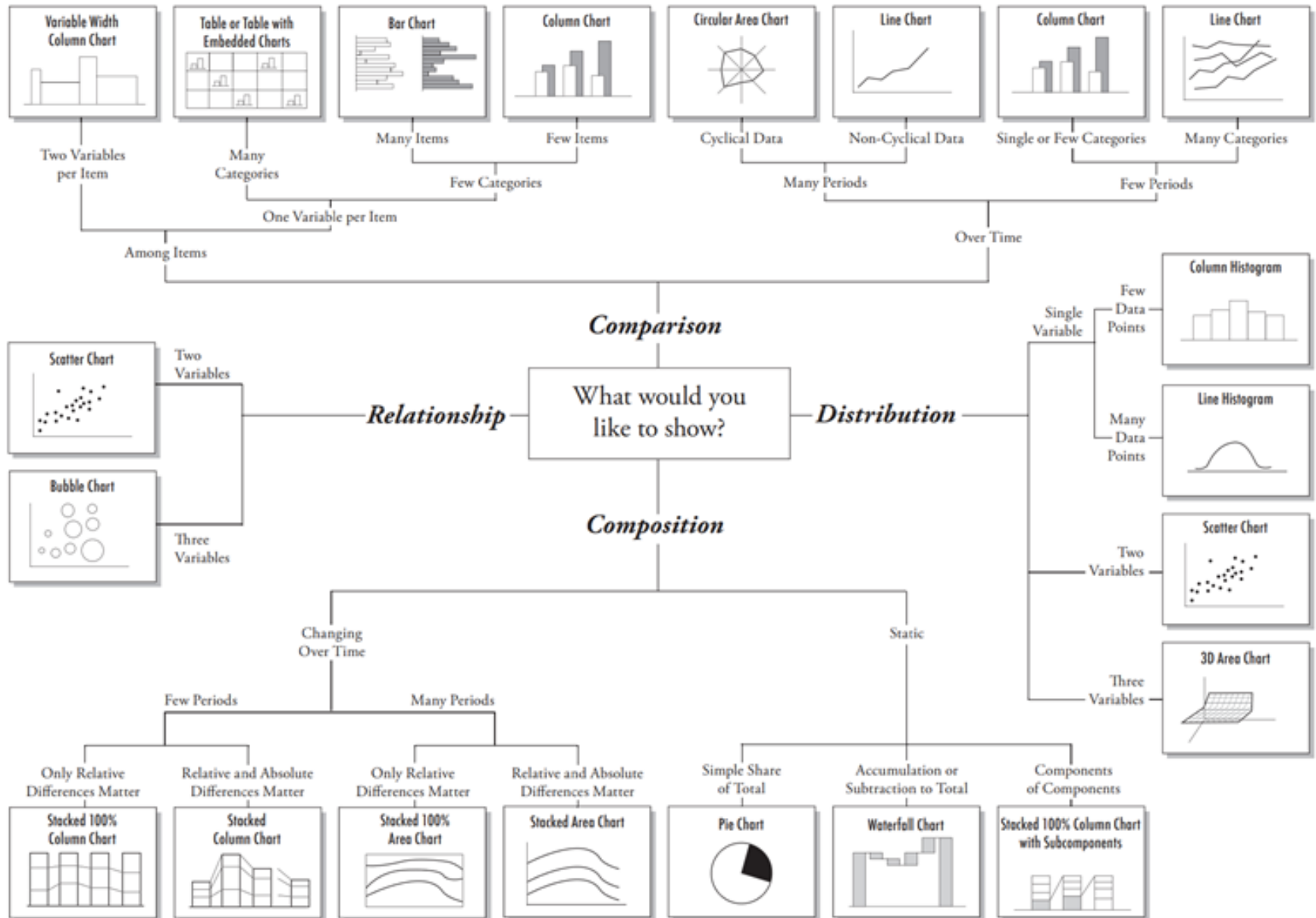
- Data visualization is used in software applications to provide an **intuitive graphical interface**.
- It is applied to many areas to enable users to glean useful information from their data for faster, more **informed decision making**.
- Users but also you, the designer!



What chart?



What chart? (cont..)

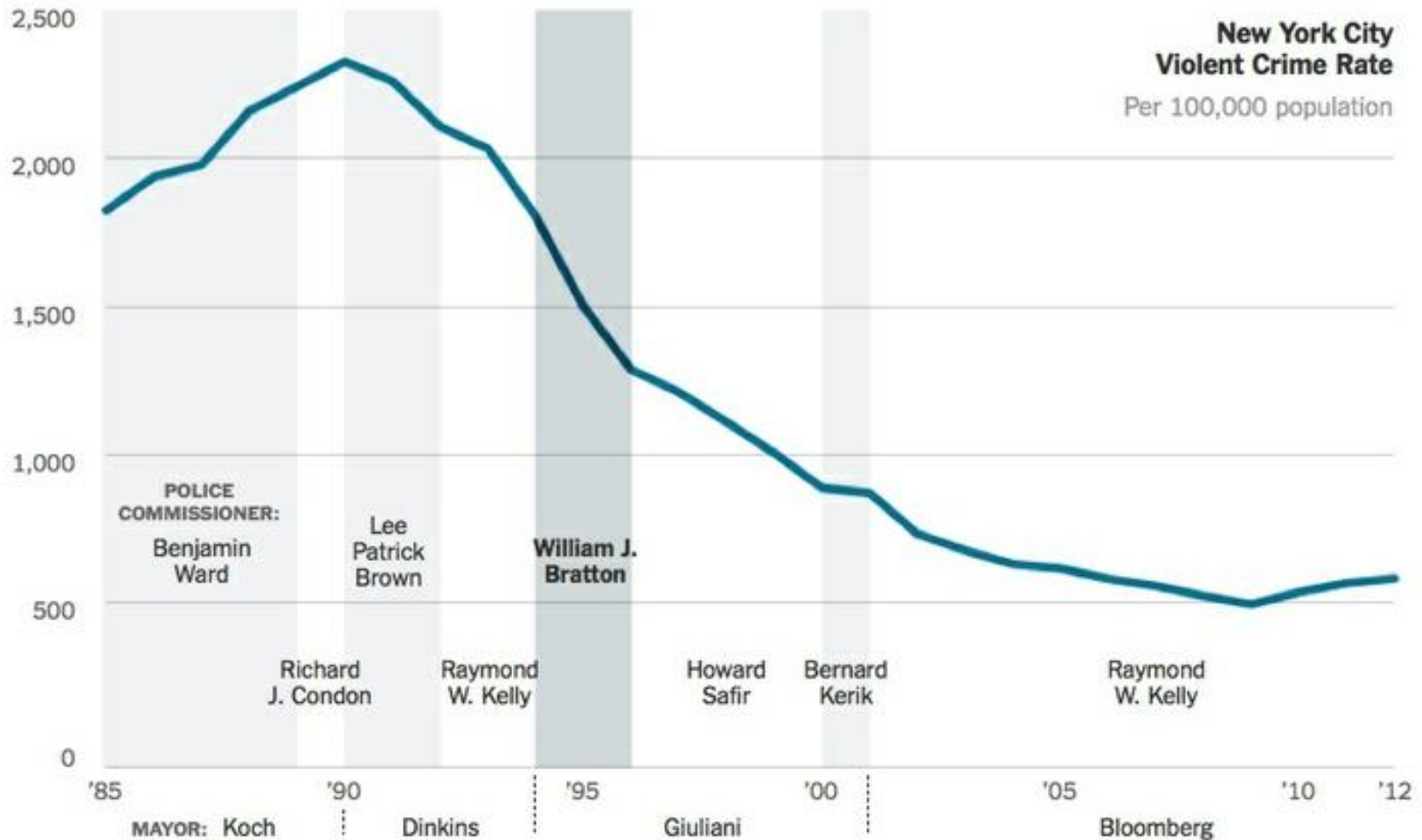


What are the benefits of Data Visualization?

- Data visualization allows users to see several different perspectives of the data.
- Data visualization makes it possible to interpret vast amounts of data
- Data visualization offers the ability to detect exceptions in the data.
- Data visualization allows the user to analyze visual patterns in the data.
- Visualization lets analysts explore trends by navigating data and recognizing patterns directly.

A picture is worth a thousand words ...

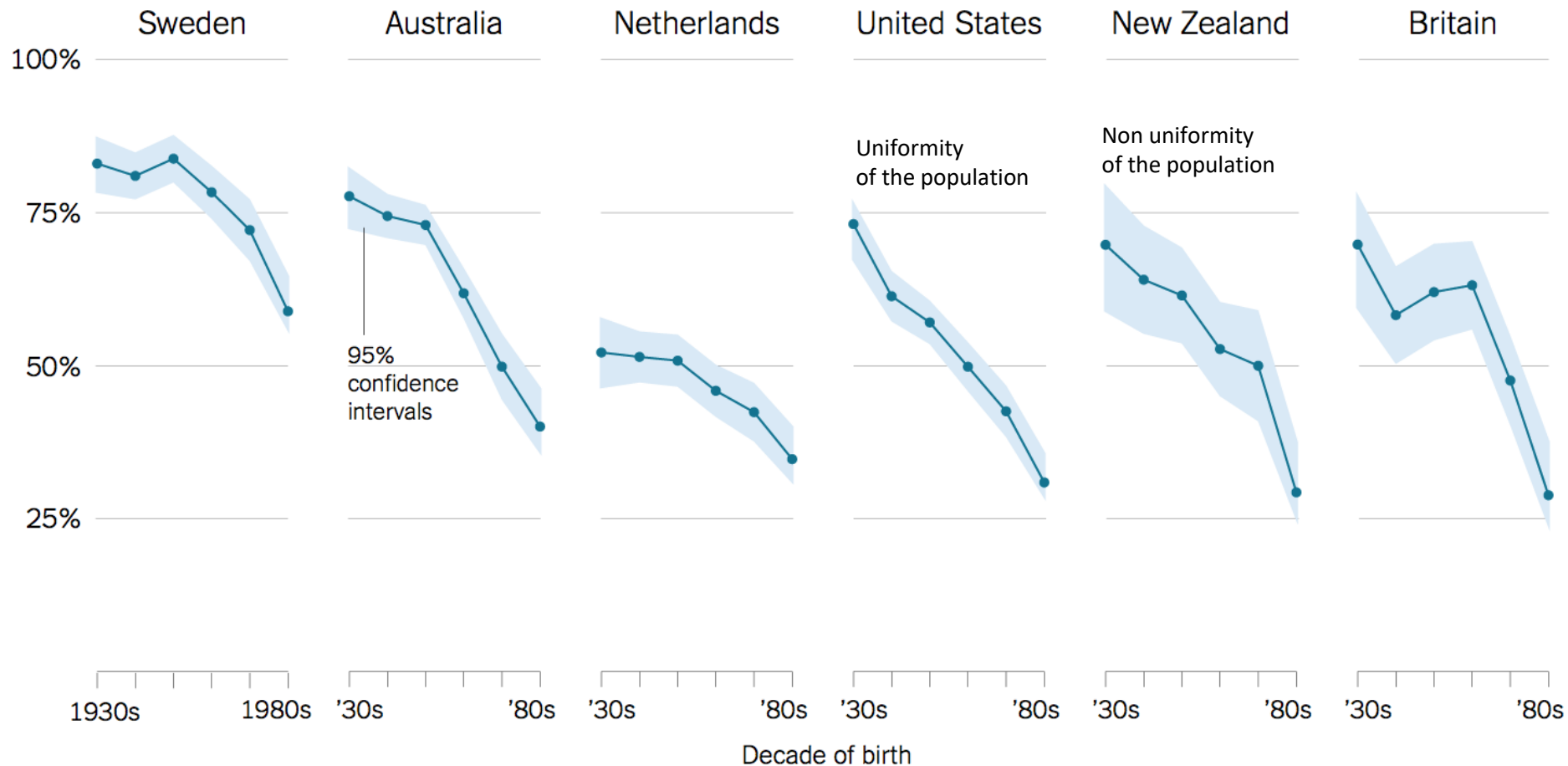
English language idiom



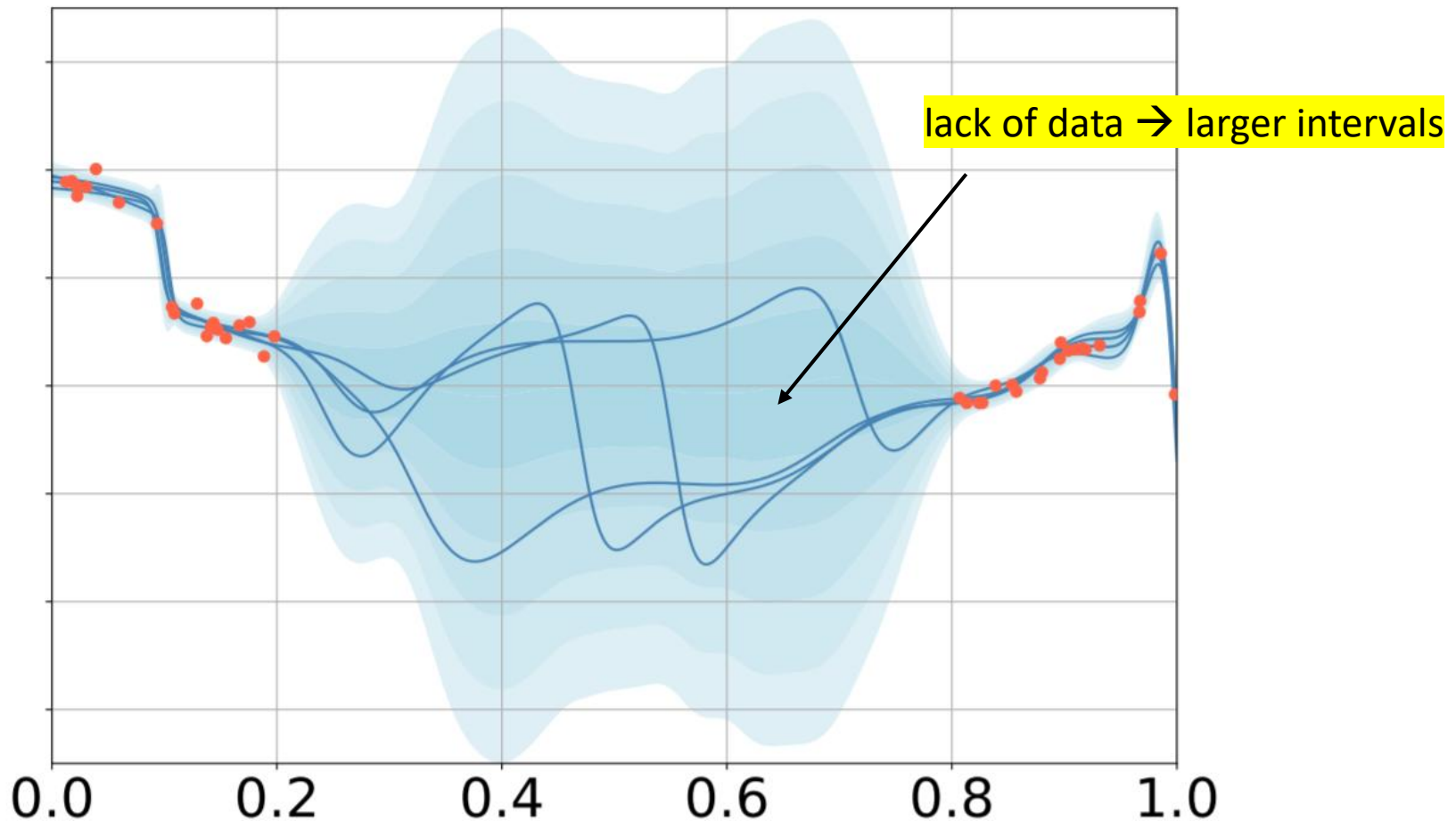
Confidence...

Fabio Scotti - Università degli Studi di Milano

Percentage of people who say it is “essential” to live in a democracy

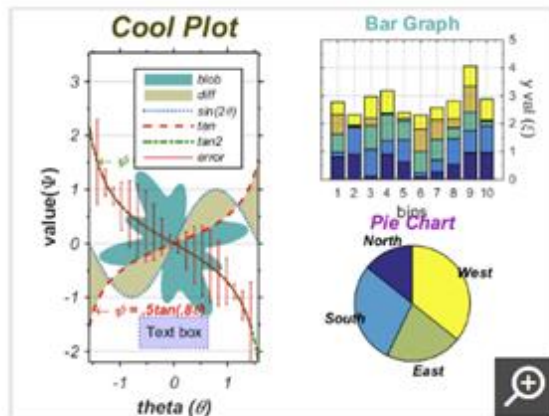


Showing the confidence intervals of trained NN

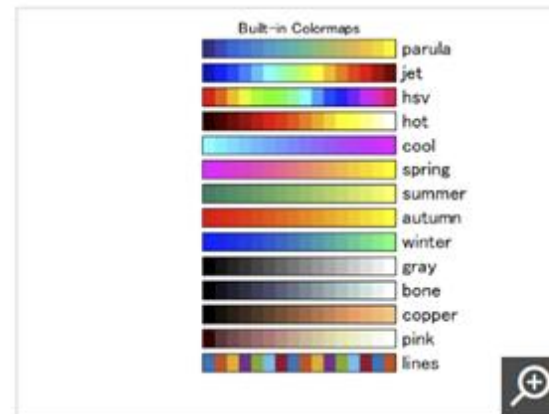


Annotations, colormaps bounds

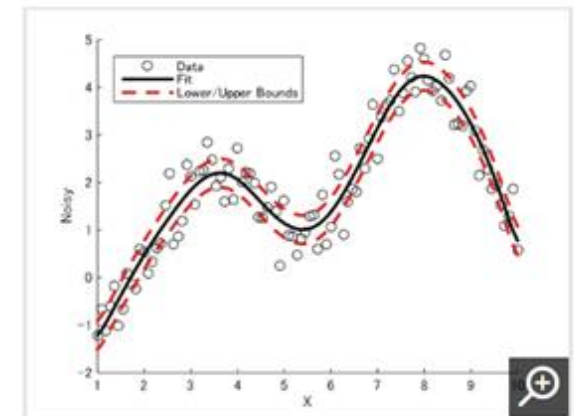
Area Bar Pie Charts with Annotations



Colormap Chart

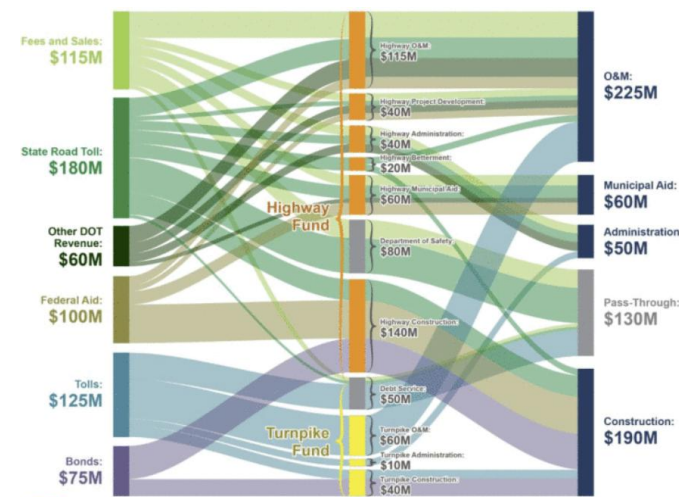


Curve with Lower and Upper Bounds



Benefits (2)

- Data visualization can help translate data patterns into insights, making it a highly effective decision-making tool.
 - reduce the time and difficulty it takes to move from data to decision.
- Data visualization equips users with the ability to see influences (outputs VS inputs) that would otherwise be difficult to find.
- With all the data available, it is difficult to find the nuances that can make a difference.

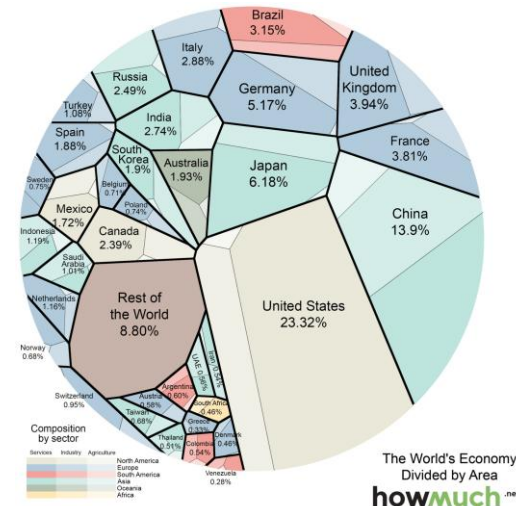


Conclusions: Graphical displays should...



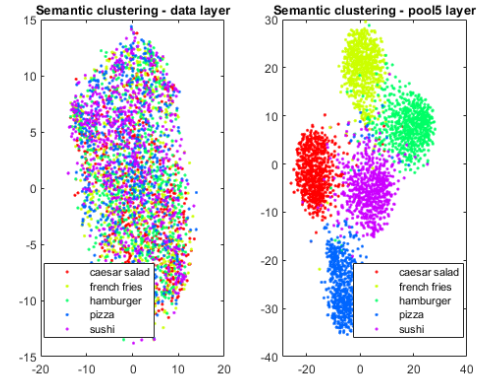
Too much graphic design...

- Show the data
- Induce the viewer to think about the substance rather than about methodology, graphic design, the technology of graphic production, or something else
- Avoid distorting what the data have to say
- Present many numbers in a small space
- Make large data sets coherent
 - Support data harmonization and comparison.



Conclusions: Graphical displays should (2)...

- Encourage the eye to compare different pieces of data
- Reveal the data at several levels of detail, from a broad overview to the fine structure
- Serve a reasonably clear purpose:
 - description,
 - exploration,
 - tabulation,
 - or decoration
- Be closely integrated with the statistical and verbal descriptions of a data set

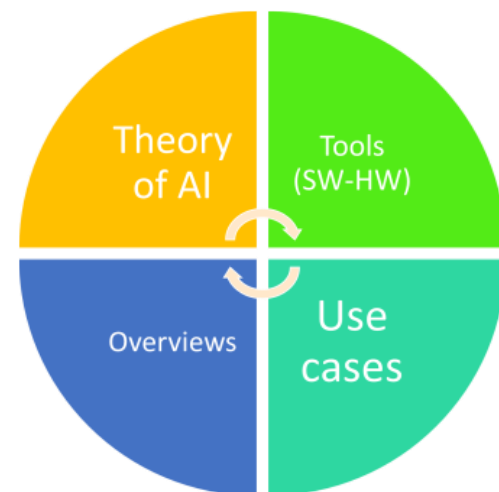




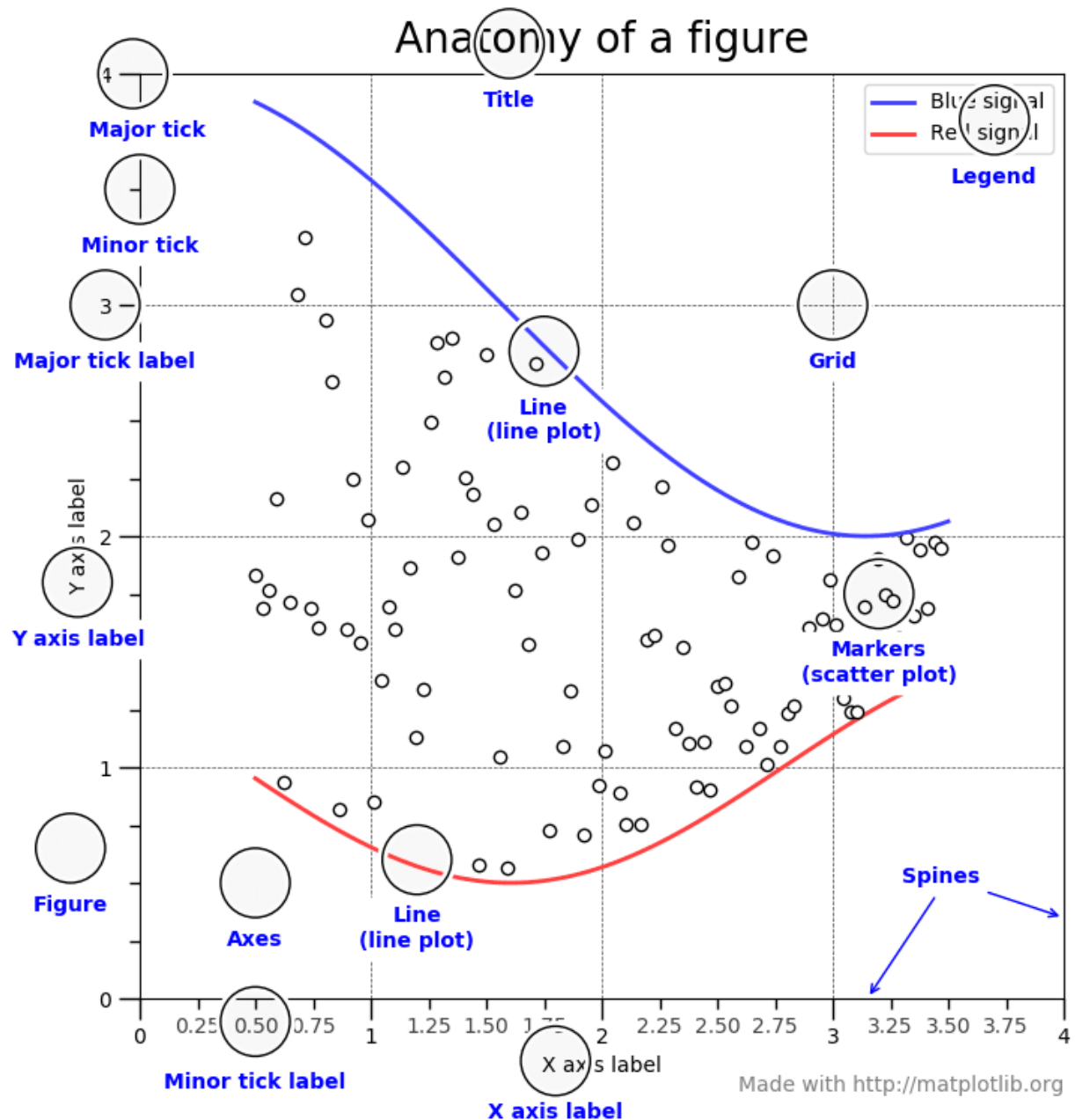
Toolboxes

Data visualization

How to check/plot/ look at the data



The structure of a very basic (but complete) plot

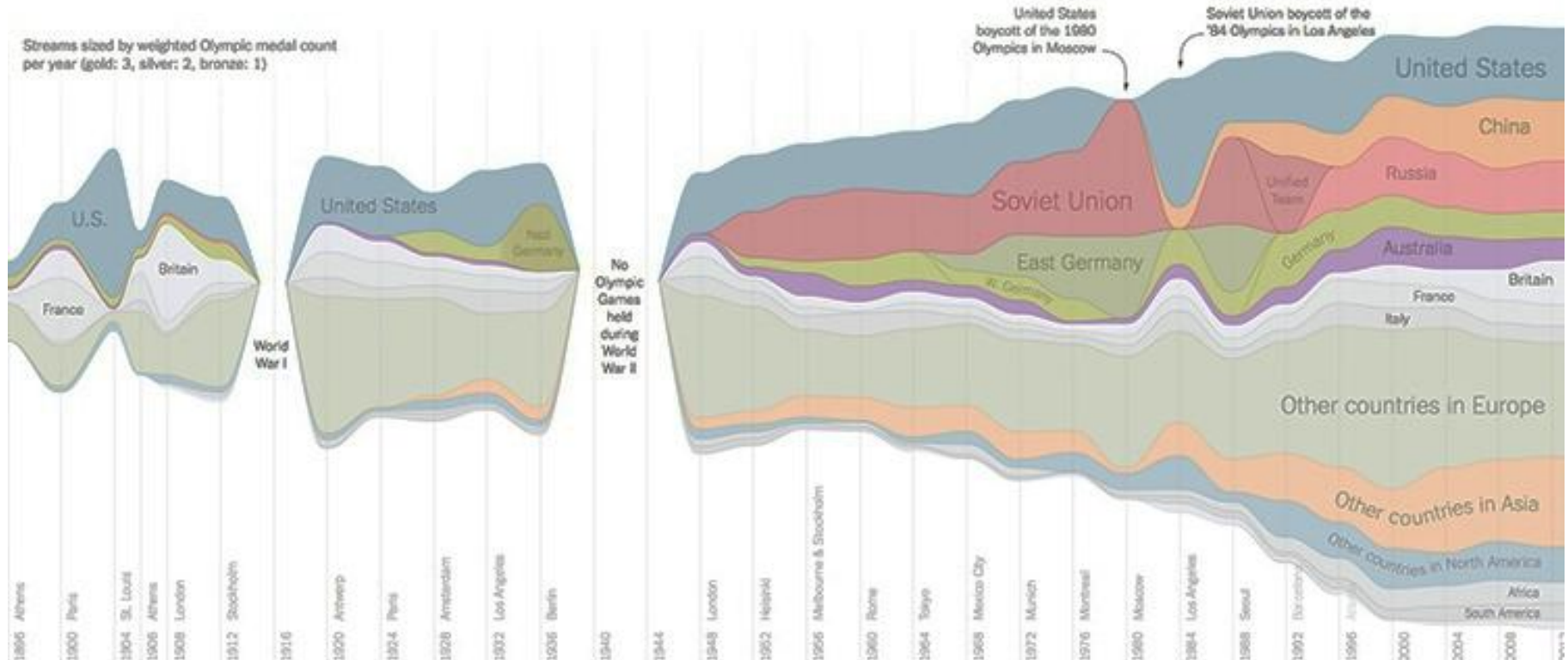


Distributions in time

No data, winners, events...

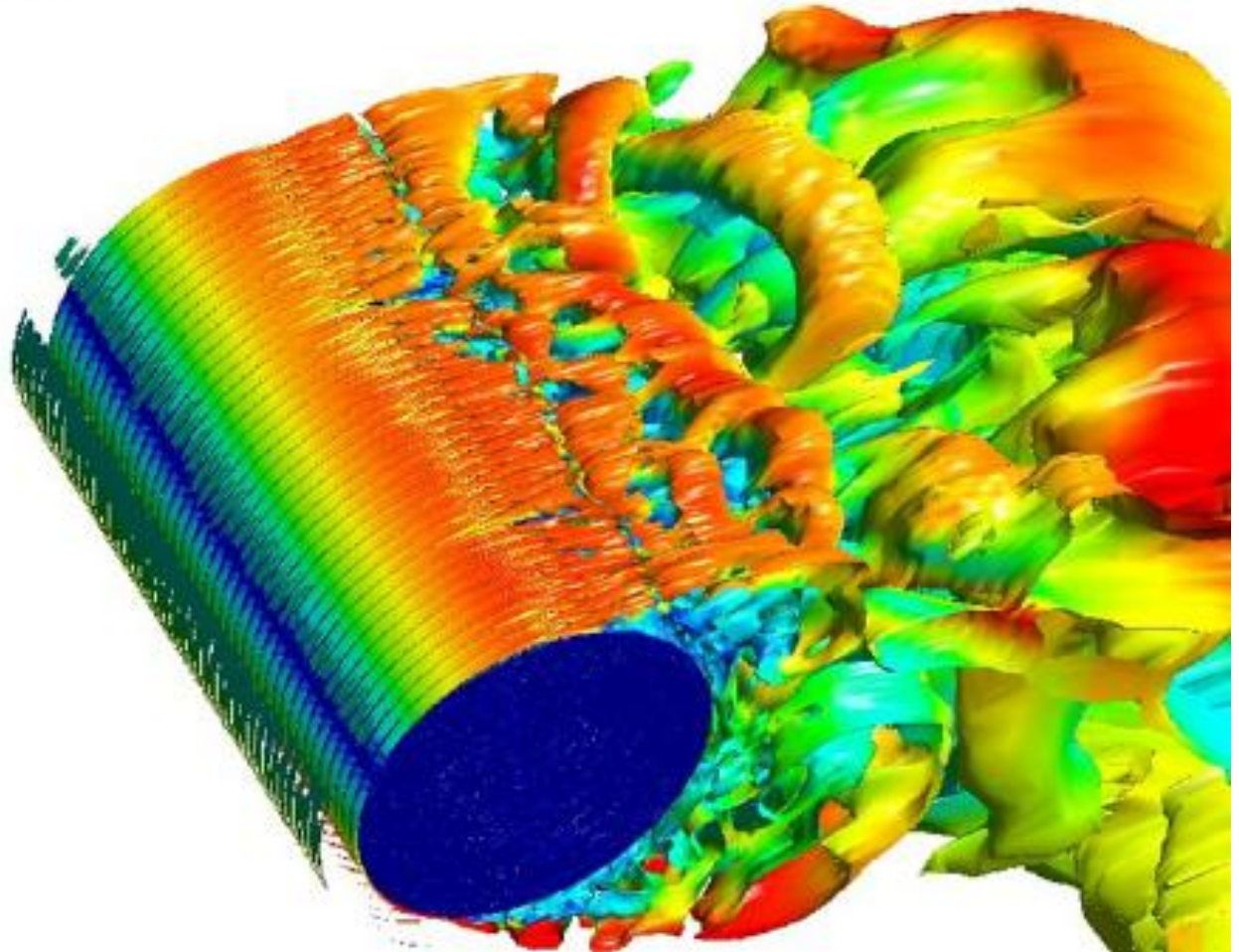
A Visual History of Which Countries Have Dominated the Summer Olympics

By GREGOR AISCH and LARRY BUCHANAN AUG. 8, 2008



Complex phenomena...

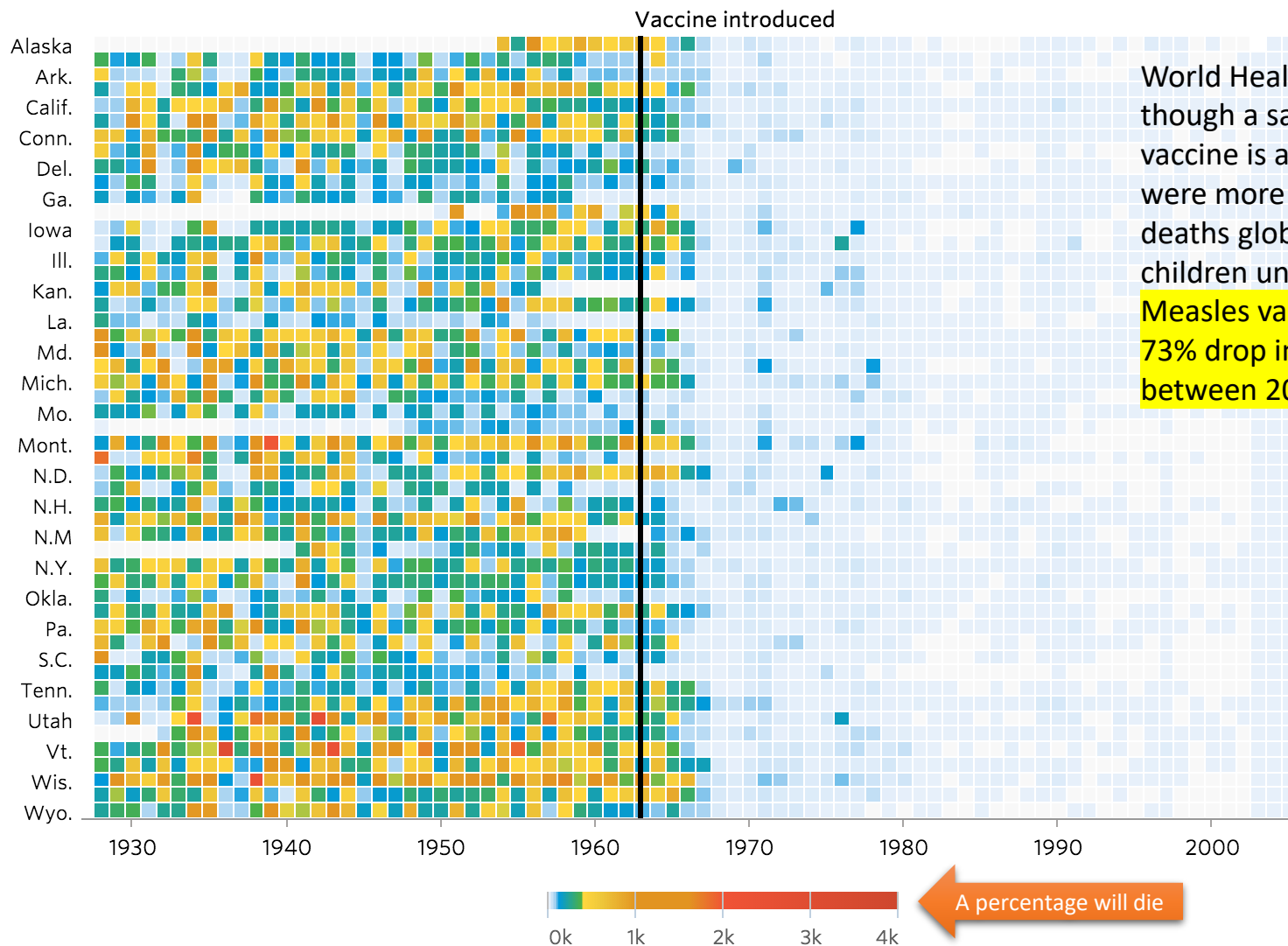
Velocity Magnitude



The heat maps below show number of cases per 100,000 people.

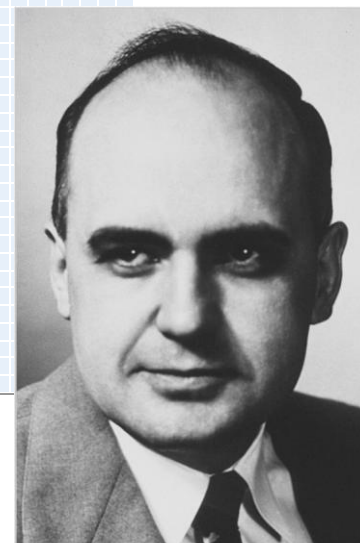
Measles (Morbilli)

Vax or no Vax?



World Health Organization: Even though a safe and cost-effective vaccine is available, in 2018, there were more than 140 000 measles deaths globally, mostly among children under the age of five.

Measles vaccination resulted in a 73% drop in measles deaths between 2000 and 2018 worldwide

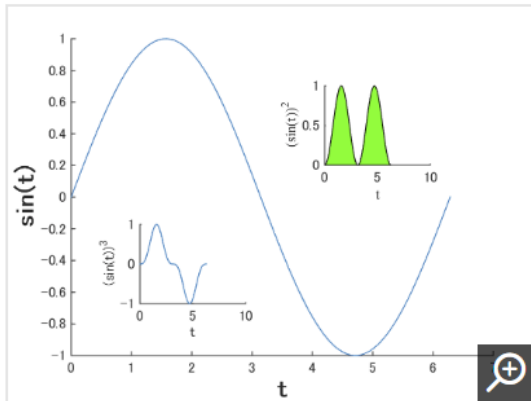


Maurice Hilleman's measles vaccine is estimated to prevent 1 million deaths per year.^[117]

Note: CDC data from 2003-2012 comes from its Summary of Notifiable Diseases, which publishes yearly rather than weekly and counts confirmed cases as opposed to provisional ones.

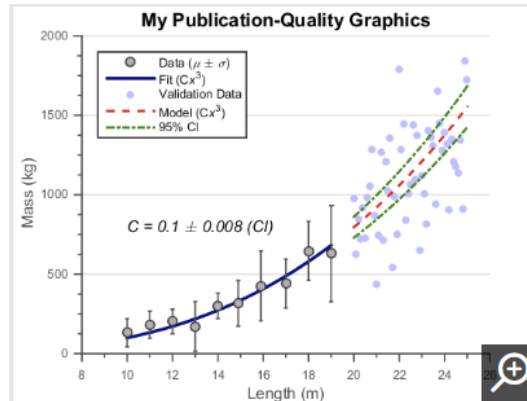
Plot in plot, 3D line vectors

Plot in Plot



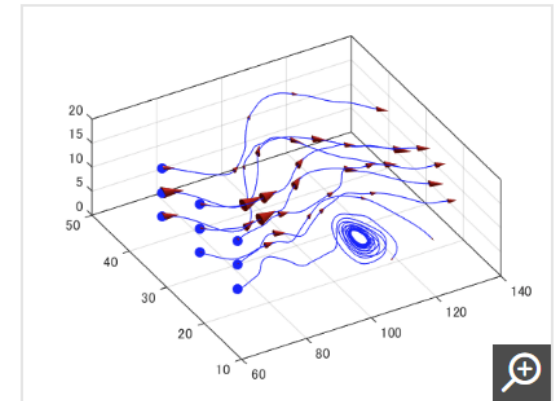
[View source](#)

Publication Quality Graphics



[View source](#)

Streamline

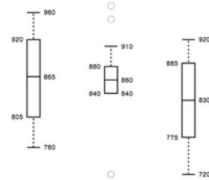


[View source](#)

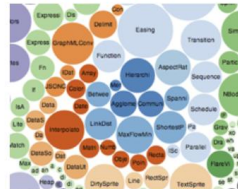
Online tools for interactive data visualization

Example: observablehq.com

Box Plots



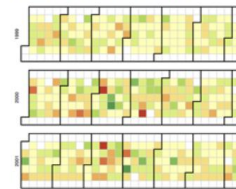
Bubble Chart



Bullet Charts



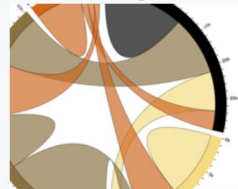
Calendar View



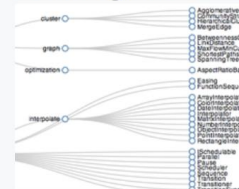
Non-contiguous Cartogram



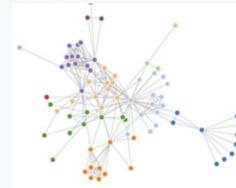
Chord Diagram



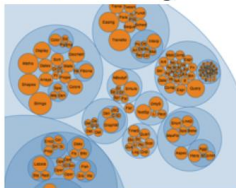
Dendrogram



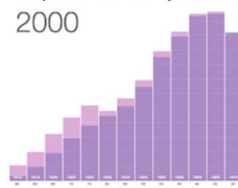
Force-Directed Graph



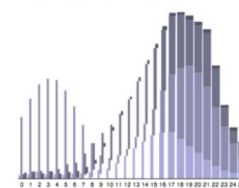
Circle Packing



Population Pyramid 2000



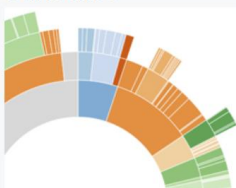
Stacked Bars



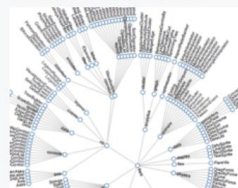
Streamgraph



Sunburst



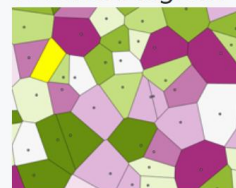
Node-Link Tree



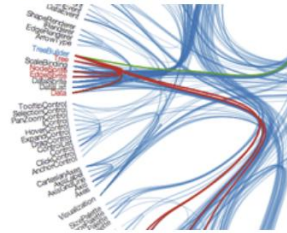
Treemap



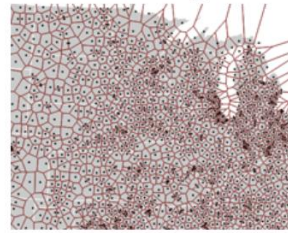
Voronoi Diagram



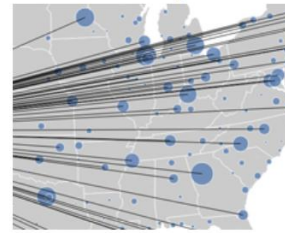
Hierarchical Edge Bundling



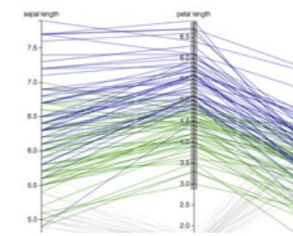
Voronoi Diagram



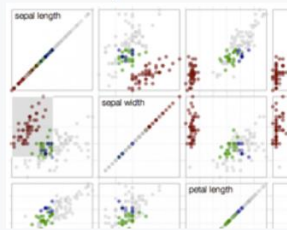
Bubble Map



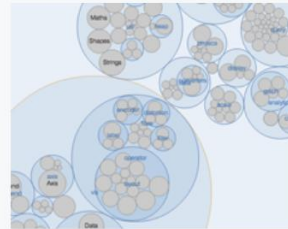
Parallel Coordinates



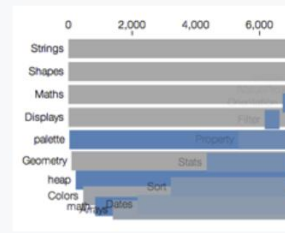
Scatterplot Matrix



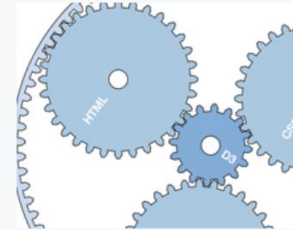
Zoomable Pack Layout



Hierarchical Bars



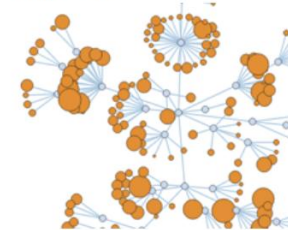
Epicyclical Gears



Collision Detection



Collapsible Force Layout



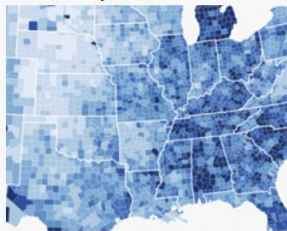
Force-Directed States



Versor Dragging



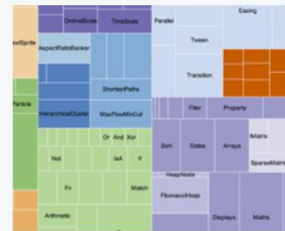
Choropleth



Collapsible Tree Layout



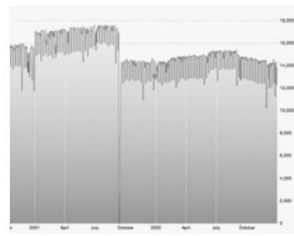
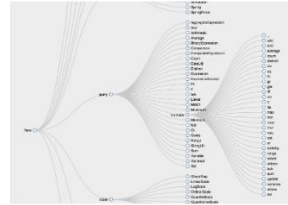
Zoomable Treemap



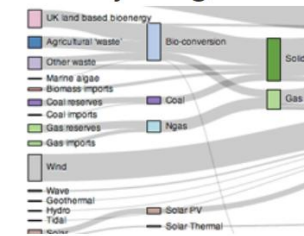
Zoomable Icicle



Zoomable Area Chart

Drag and Drop
Collapsible Tree
LayoutRadial Cluster
Layout

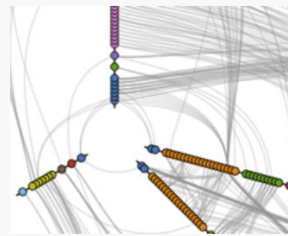
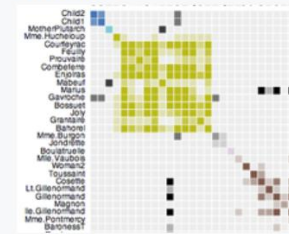
Sankey Diagram



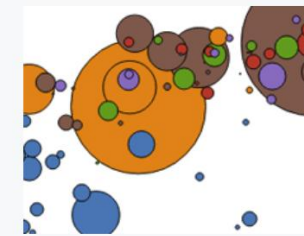
Fisheye Distortion



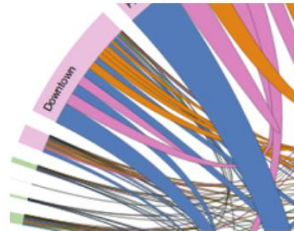
Hive Plot

Co-occurrence
Matrix

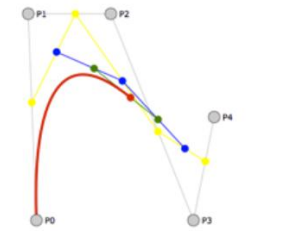
Motion Chart



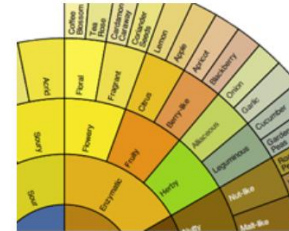
Chord Diagram



Animated Bézier's



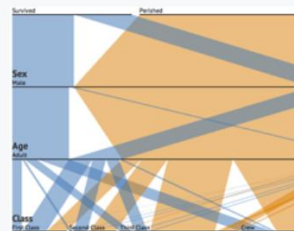
Zoomable Sunburst



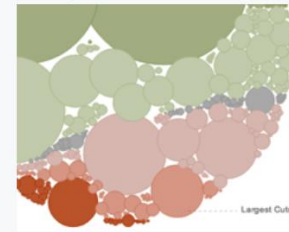
Collatz Graph



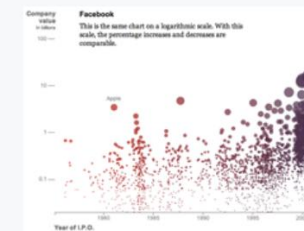
Parallel Sets



Word Cloud

Obama's Budget
Proposal

Facebook IPO



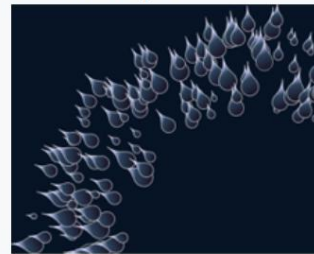
Voronoi Picking



Zoomable Map



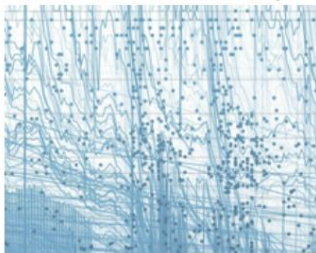
Raindrops



Color



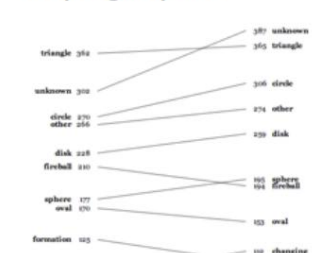
Hacker News Popularity



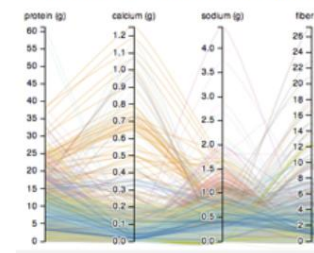
Life Expectancy



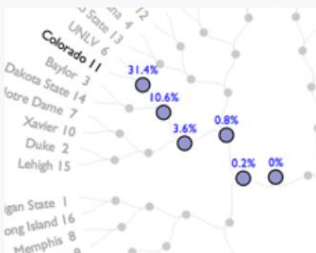
Slopegraphs



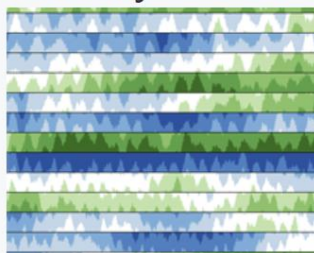
Parallel Coordinates



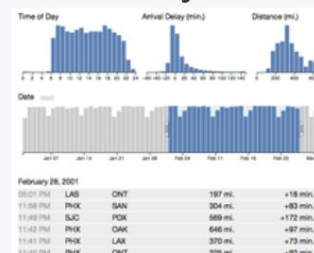
NCAA Predictions



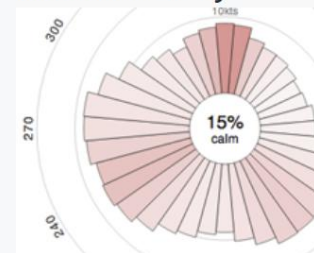
Cubism.js



Crossfilter.js



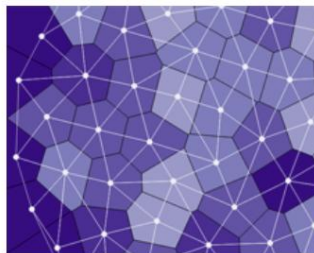
Wind History



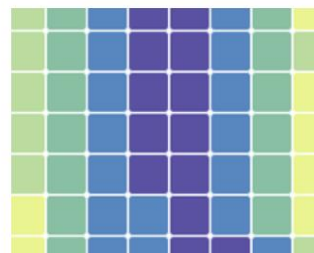
Cubic Hamiltonian Graphs



Force-Directed Voronoi



Trulia Trends



Trulia Trends



Fancy plots...

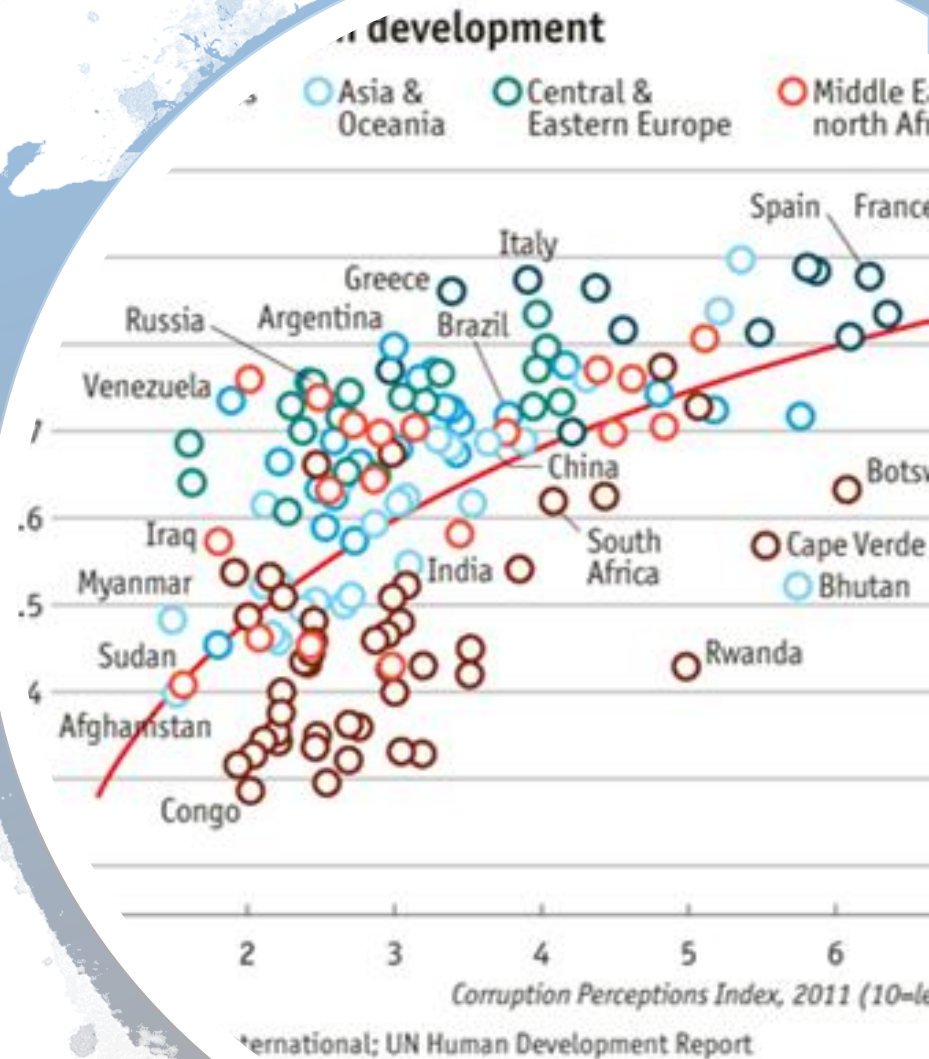
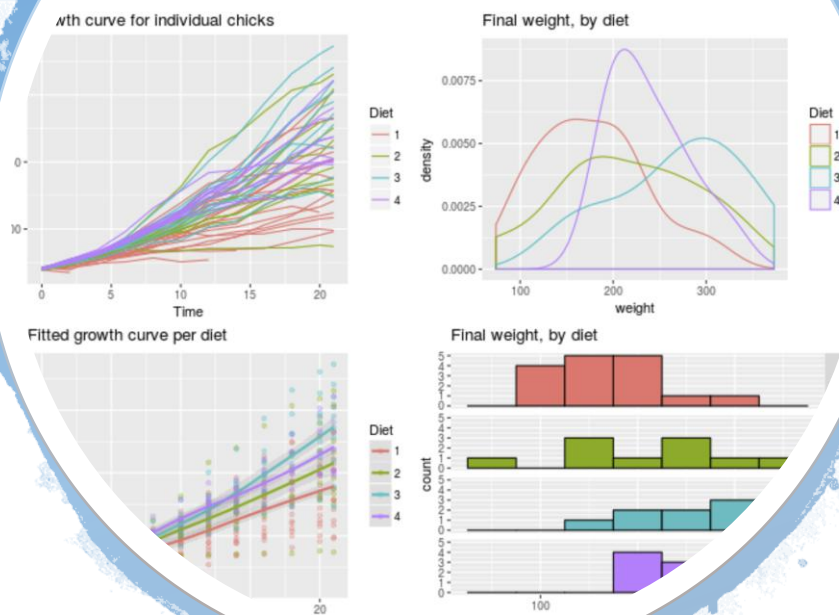
- The graphic power of the plots and charts is useless if....
 - data are not representative
 - data are corrupted
 - data are not properly preprocessed
 - data are FAKE!!
(e.g., too much or bad augmentation)

R's ggplot2 library

- R is a programming language and free software environment for statistical computing and graphics supported by the R Foundation for Statistical Computing.



R's ggplot2 library

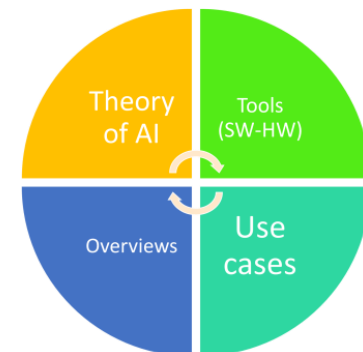
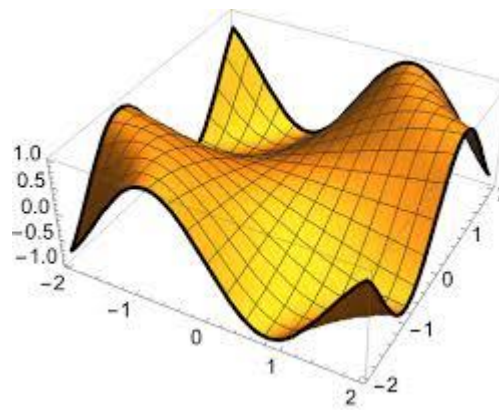




Toolboxes

3D Data visualization

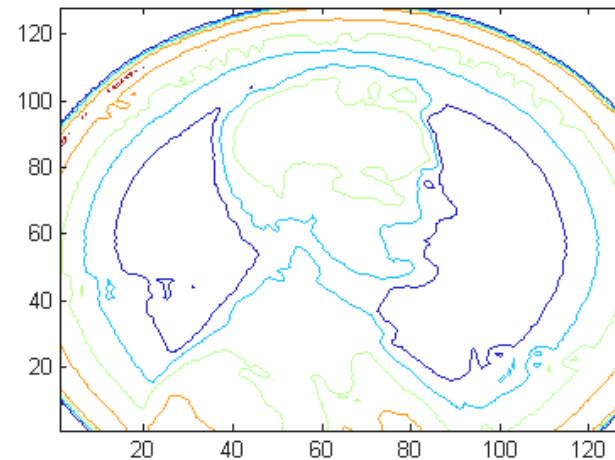
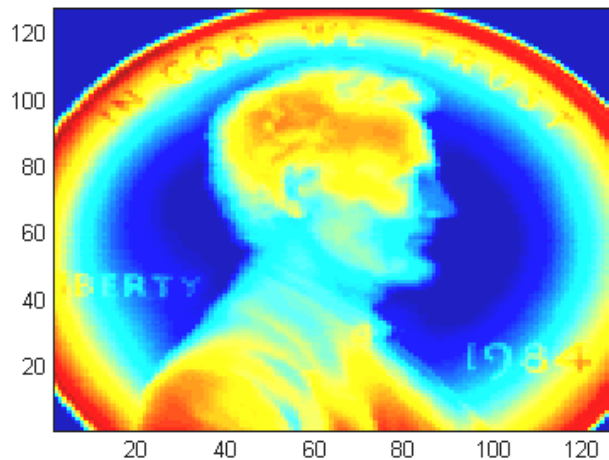
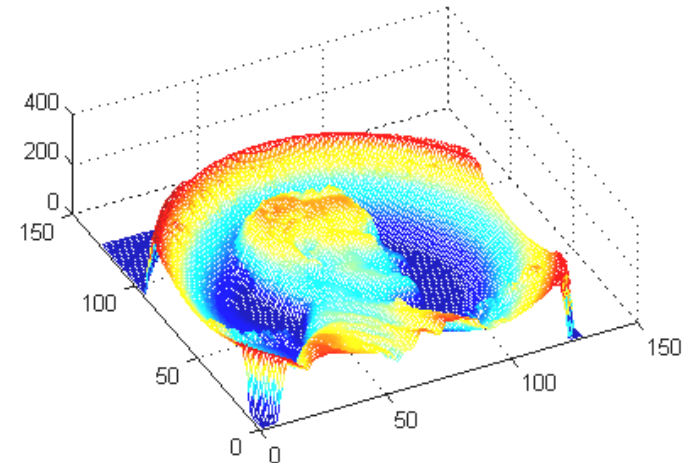
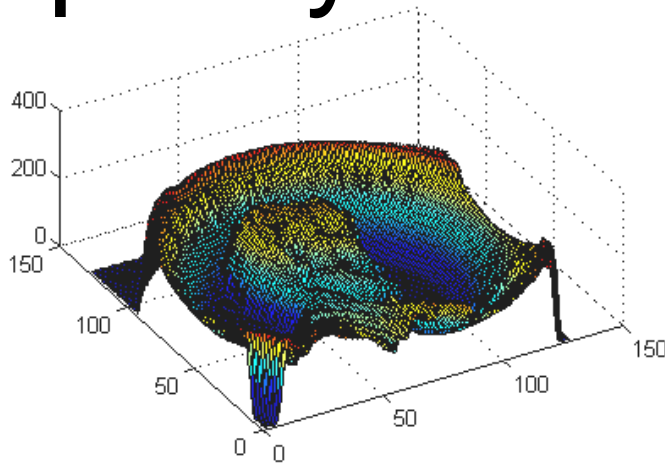
The best we can do in 3D



The scanned 3D penny



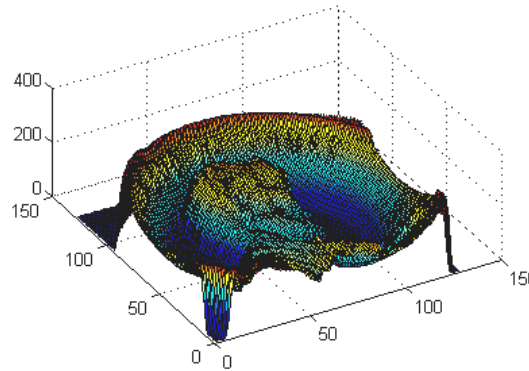
3D imaging devices



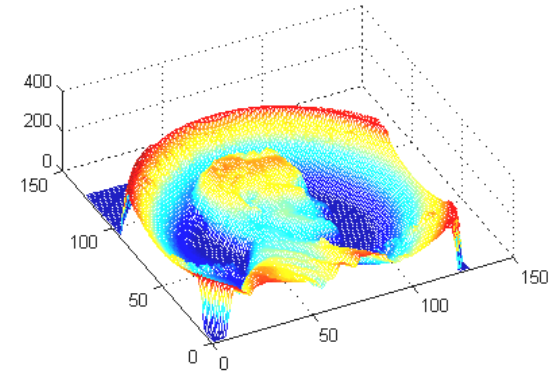
Some very simple examples with 2D data (matlab)

4 different plots of a penny

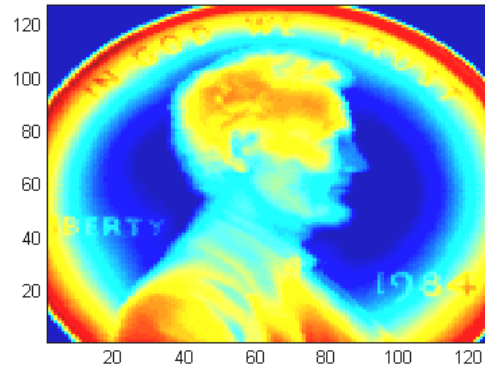
```
load penny ;  
P = flipud(P);  
subplot(2,2,1)  
surf(P);
```



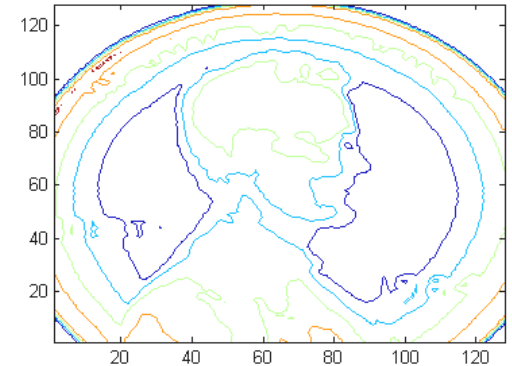
```
subplot(2,2,2)  
mesh(P);
```



```
subplot(2,2,3)  
pcolor(P);  
shading flat
```

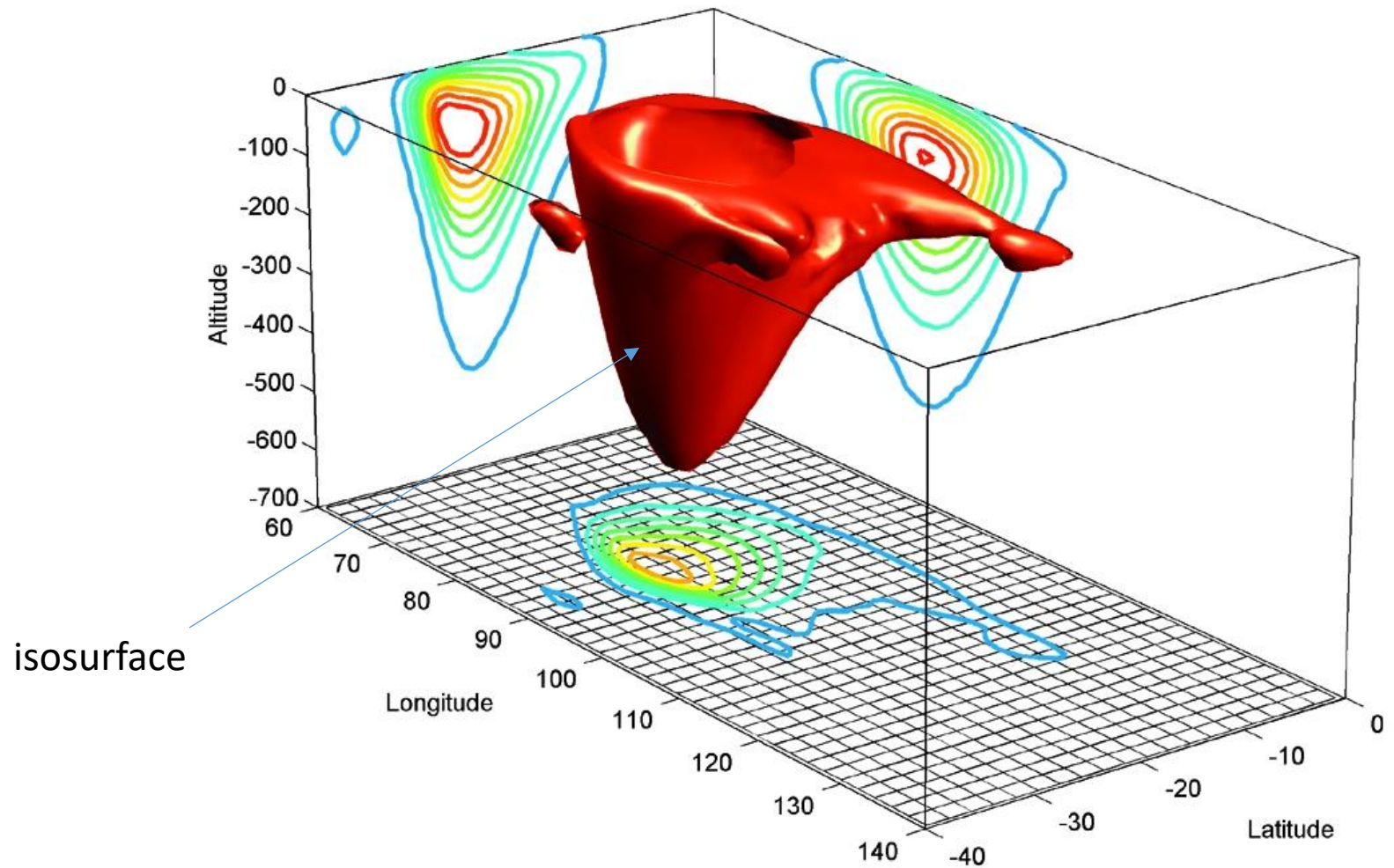


```
subplot(2,2,4)  
contour(P);
```

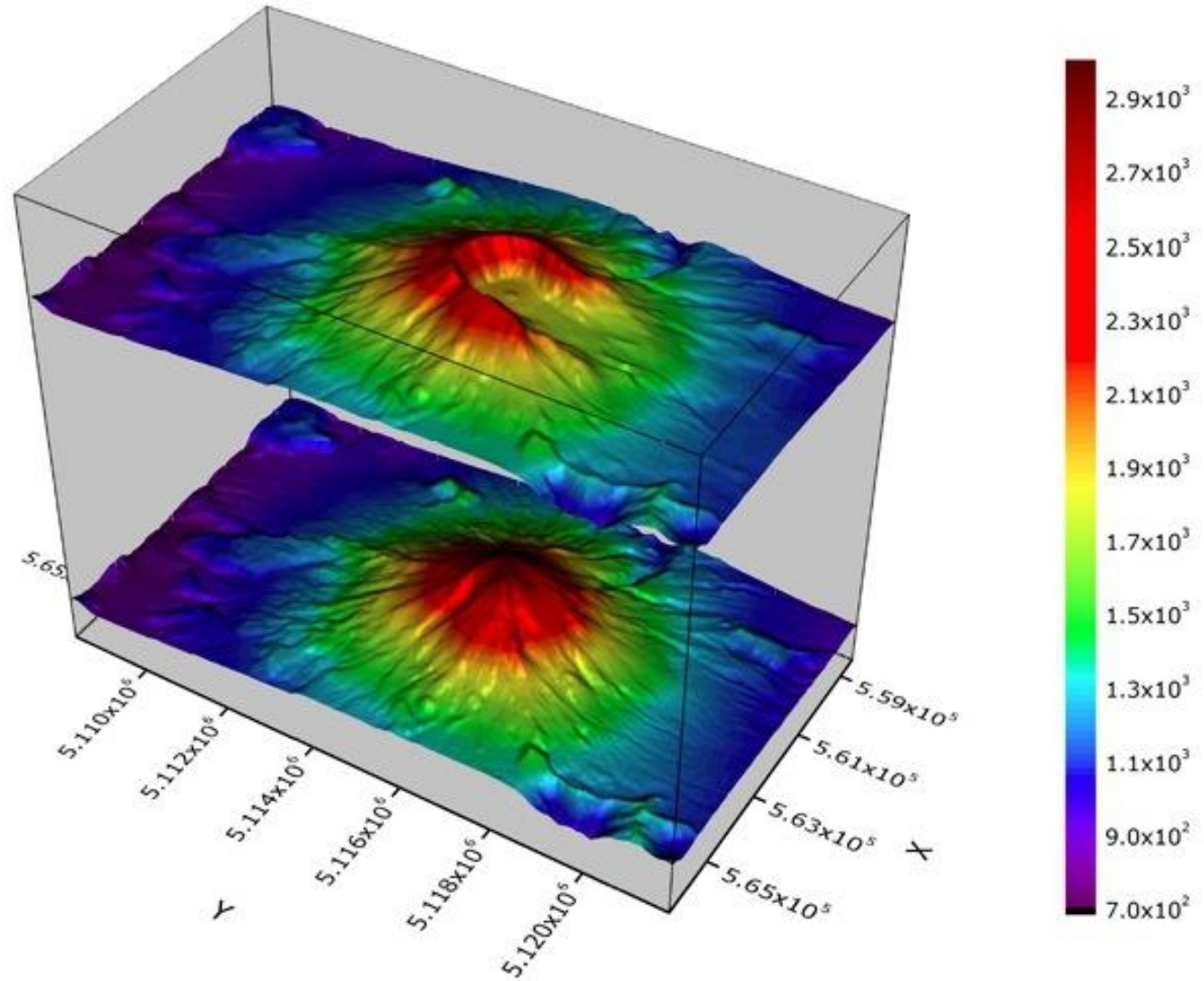


3D Surfaces

Intelligent systems decisions
can be represented
as surfaces in internal input spaces



3D data in time

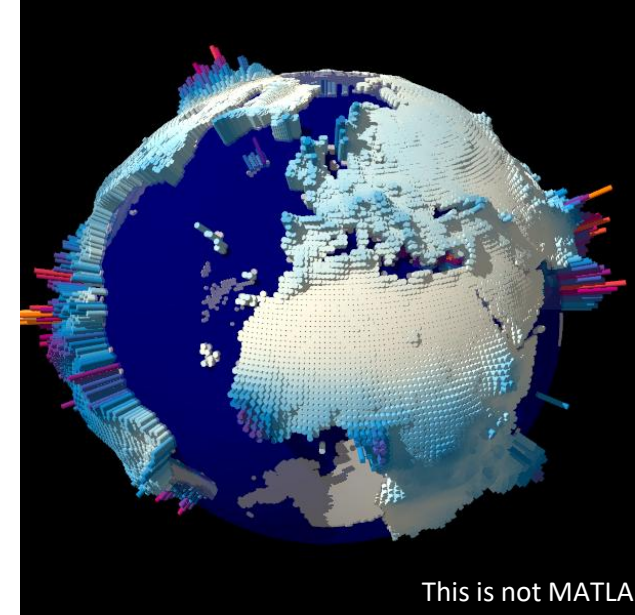
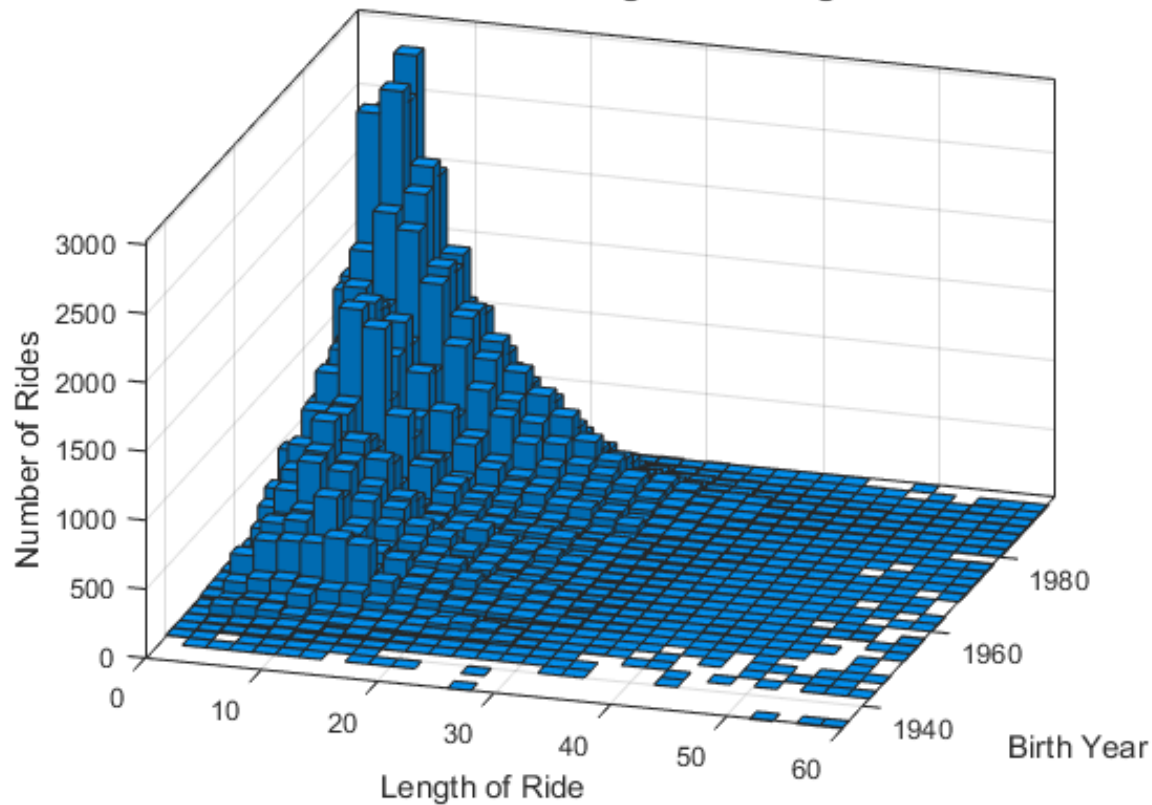


Representations of
Mount St. Helens
in Washington State,
before and after its 1980 eruption

3D Barchart

(Data from Boston's bike sharing program)

Ride counts based on ride length and the age of the rider



This is not MATLAB

```
% Create bivariate histogram plot using  
the histogram2 function  
histogram2(rideData.Duration,  
rideData.birth_date, 'BinWidth', [2 2])  
xlabel('Length of Ride')  
ylabel('Birth Year')  
zlabel('Number of Rides')  
title('Ride counts based on ride length  
and the age of the rider')  
% Adjust view  
view(17,30)
```

Find the maximum of a 2D function

Learning means OPTIMIZATION
that means find a minimum or a
maximum in a function

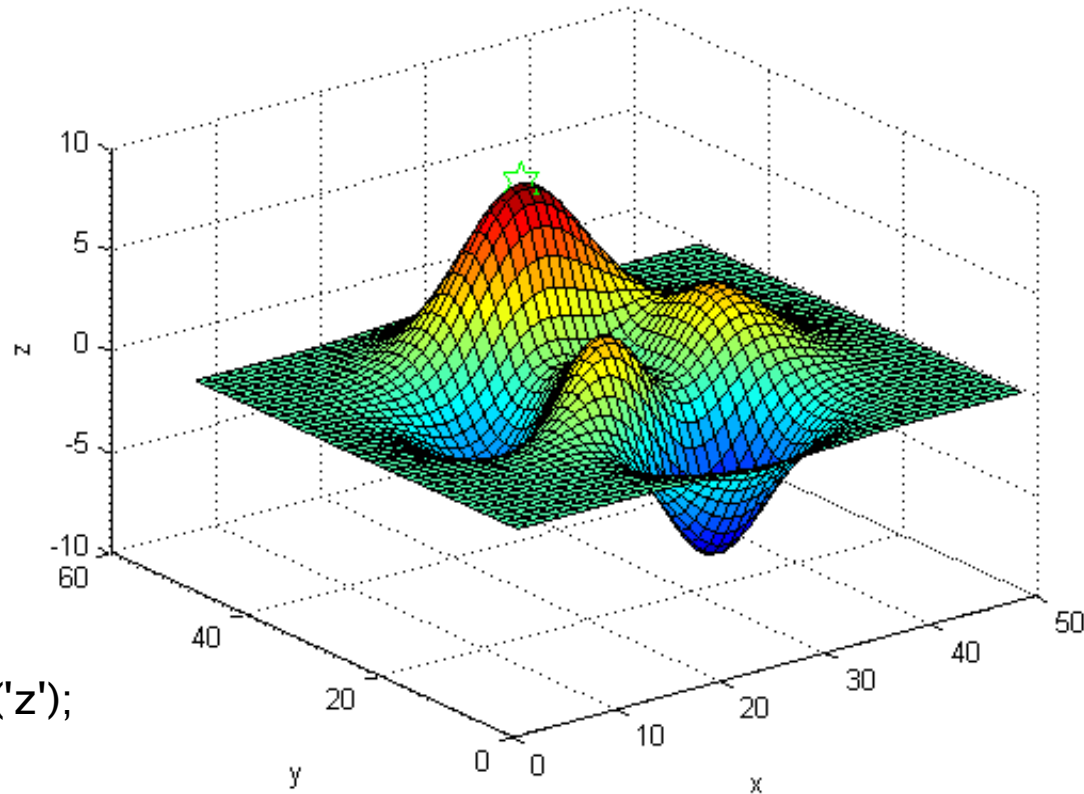
Homework:
consider the function «peaks»,
plot and locate the maximum value

```
[Z] = peaks;  
% it creates a smooth curved surface
```

```
surf(Z); xlabel('x'); ylabel('y'); zlabel('z');
```

```
Zmax = max(max(Z))  
[ymax, xmax] = find(Z == Zmax)
```

```
%  
hold on;  
plot3( xmax, ymax, Zmax, 'hg', 'Markersize', 15 );  
hold off;
```



Vectors in 3D

Showing a complex
property of a single point
such as vectors

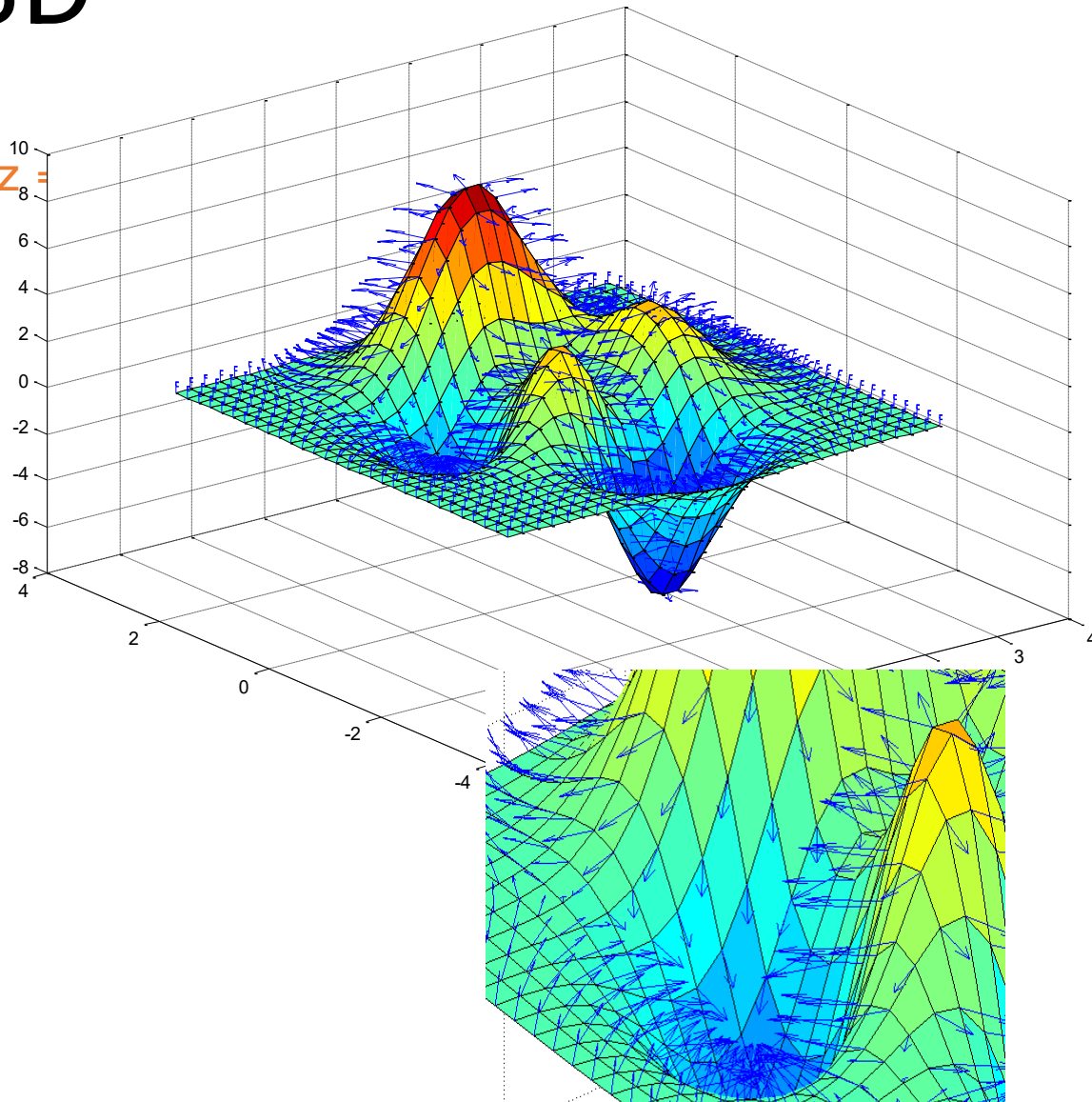
Plot the orthogonal vectors of $z =$

```
[x,y] = meshgrid(-2:2:2,-1:15:1);  
z = x .* exp(-x.^2 - y.^2);
```

```
[u,v,w] = surfnorm(x,y,z);
```

```
quiver3(x,y,z,u,v,w);
```

```
hold on,  
surf(x,y,z),  
hold off
```



Showing a property in 3D

Map with a proper palette the gradient of $z = x \cdot \exp(-x.^2 - y.^2)$

```
[x,y] = meshgrid([-2:1:2]);
```

```
Z = x.*exp(-x.^2-y.^2);
```

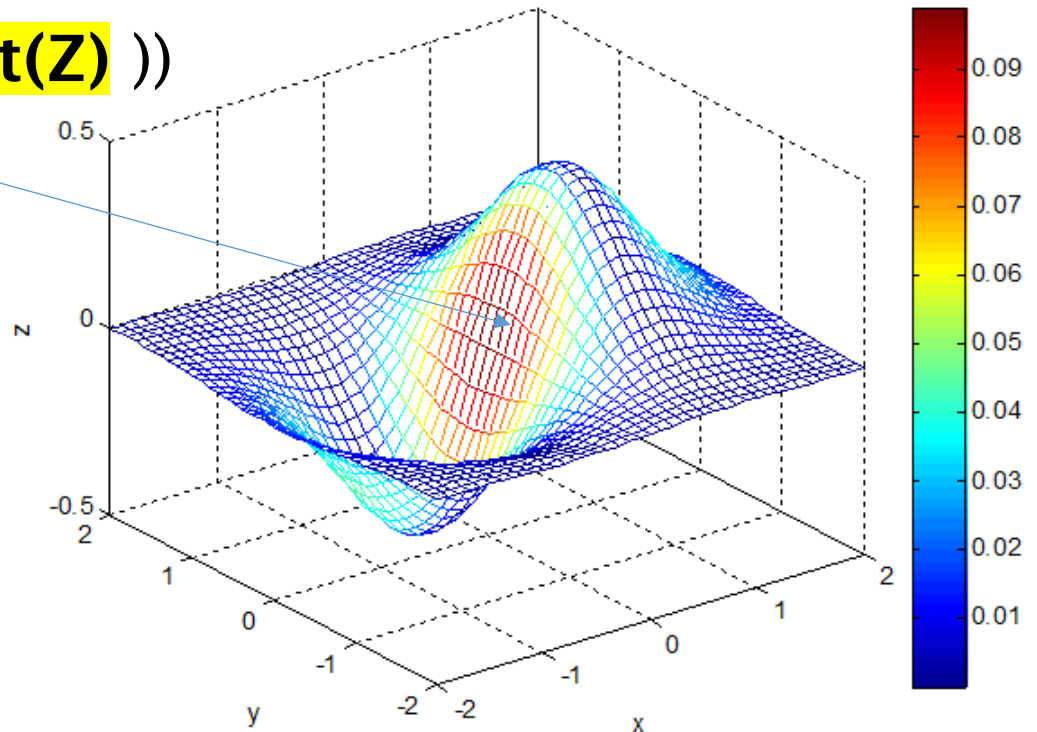
```
mesh(x,y,Z,abs(gradient(Z)) )
```

```
xlabel('x'); ylabel('y'); zlabel('z');
```

```
title(' .... ');
```

```
colorbar; colormap jet ;
```

The color is the slope gradient



Inspect decision surfaces adding lights

`ezsurf('funzione simbolica, dominio)`

es:

```
ezsurf('sin(sqrt(x^2+y^2))/sqrt(x^2+y^2)',[-6*pi,6*pi])
```

```
view(0,75)
```

```
shading interp
```

```
lightangle(-45,30)
```

```
set(gcf,'Renderer','zbuffer')
```

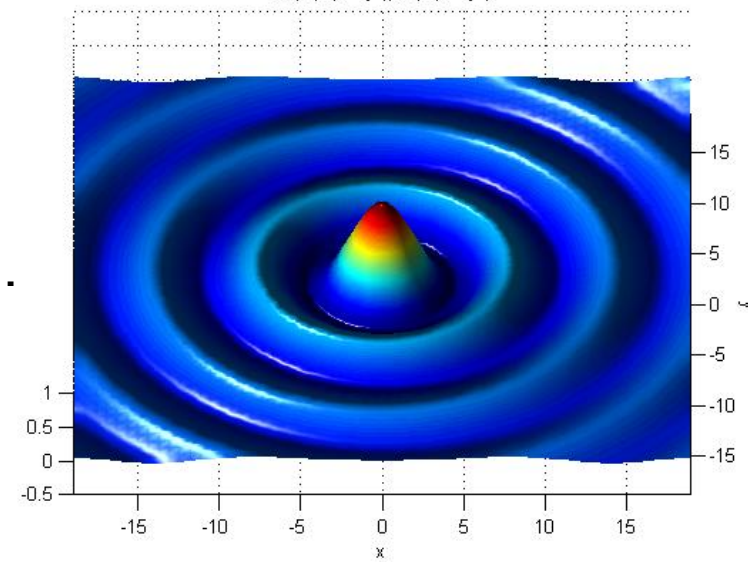
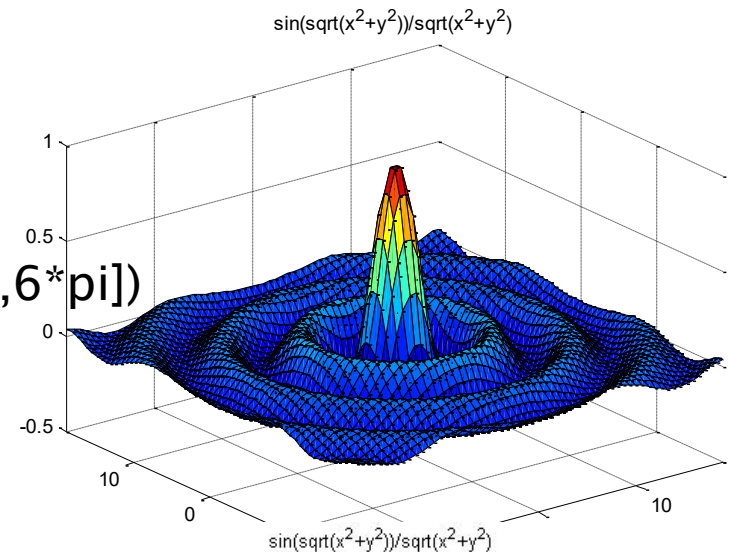
```
set(findobj(gca,'type','surface'),...
```

```
'FaceLighting','phong',...
```

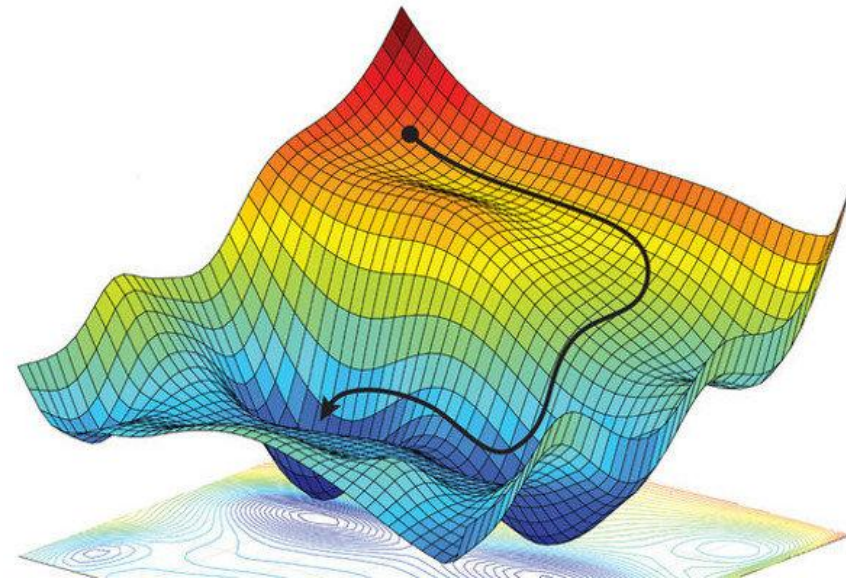
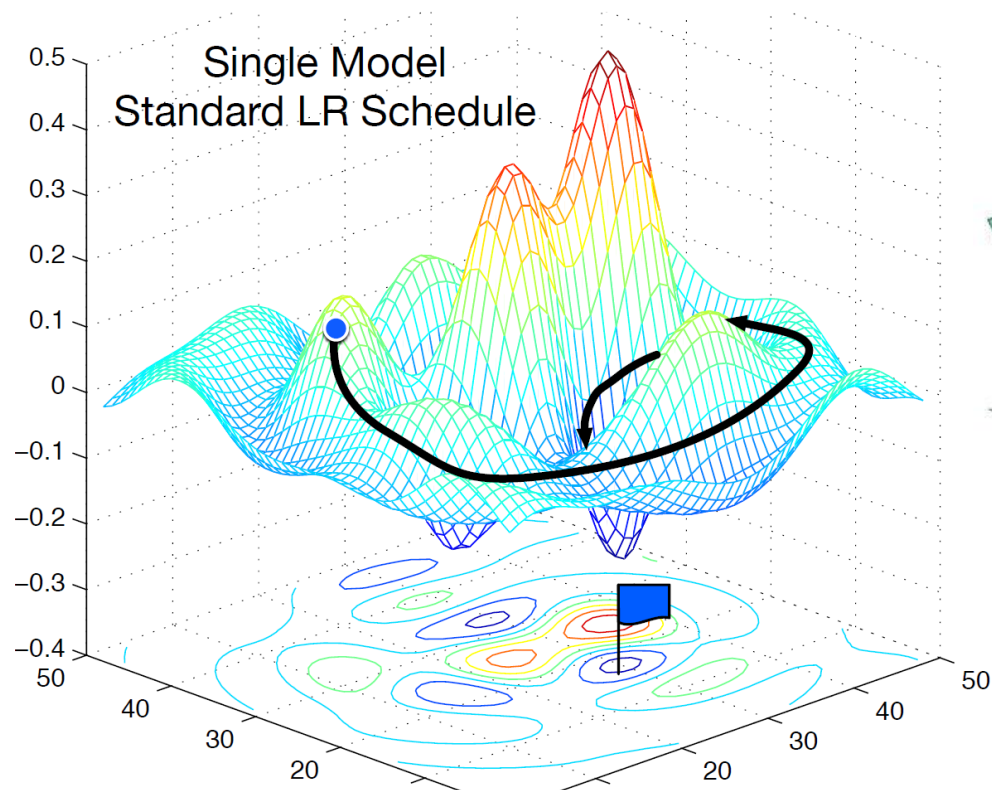
```
'AmbientStrength',.3,'DiffuseStrength',.8,...
```

```
'SpecularStrength',.9,'SpecularExponent',25,...
```

```
'BackFaceLighting','unlit')
```

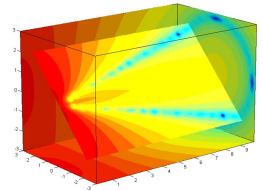


Showing the optimization process



Example of
stochastic gradient descent
in non-convex optimization

3D data



Speed vectors in a fluid jet in water (with no boundaries)

% Generate the volume data with the command

```
[x,y,z,v] = flow;
```

% Determine the range of the volume by finding the minimum and maximum of the coordinate data.

```
xmin = min(x(:)); ymin = min(y(:)); zmin = min(z(:));
```

```
xmax = max(x(:)); ymax = max(y(:)); zmax = max(z(:));
```

```
hslice = surf(linspace(xmin,xmax,100), linspace(ymin,ymax,100), zeros(100));
```

```
rotate(hslice,[-1,0,0],-45)
```

```
xd = get(hslice,'XData'); yd = get(hslice,'YData'); zd = get(hslice,'ZData'); delete(hslice)
```

```
h = slice(x,y,z,v,xd,yd,zd); set(h,'FaceColor','interp', 'EdgeColor','none', 'DiffuseStrength',.8)
```

hold on

```
hx = slice(x,y,z,v,xmax,[],[]); set(hx,'FaceColor','interp','EdgeColor','none')
```

```
hy = slice(x,y,z,v,[],ymax,[]); set(hy,'FaceColor','interp','EdgeColor','none')
```

```
hz = slice(x,y,z,v,[],[],zmin); set(hz,'FaceColor','interp','EdgeColor','none')
```

```
daspect([1,1,1])
```

```
axis tight
```

```
box on
```

```
view(-38.5,16)
```

```
camzoom(1.4)
```

```
camproj perspective
```

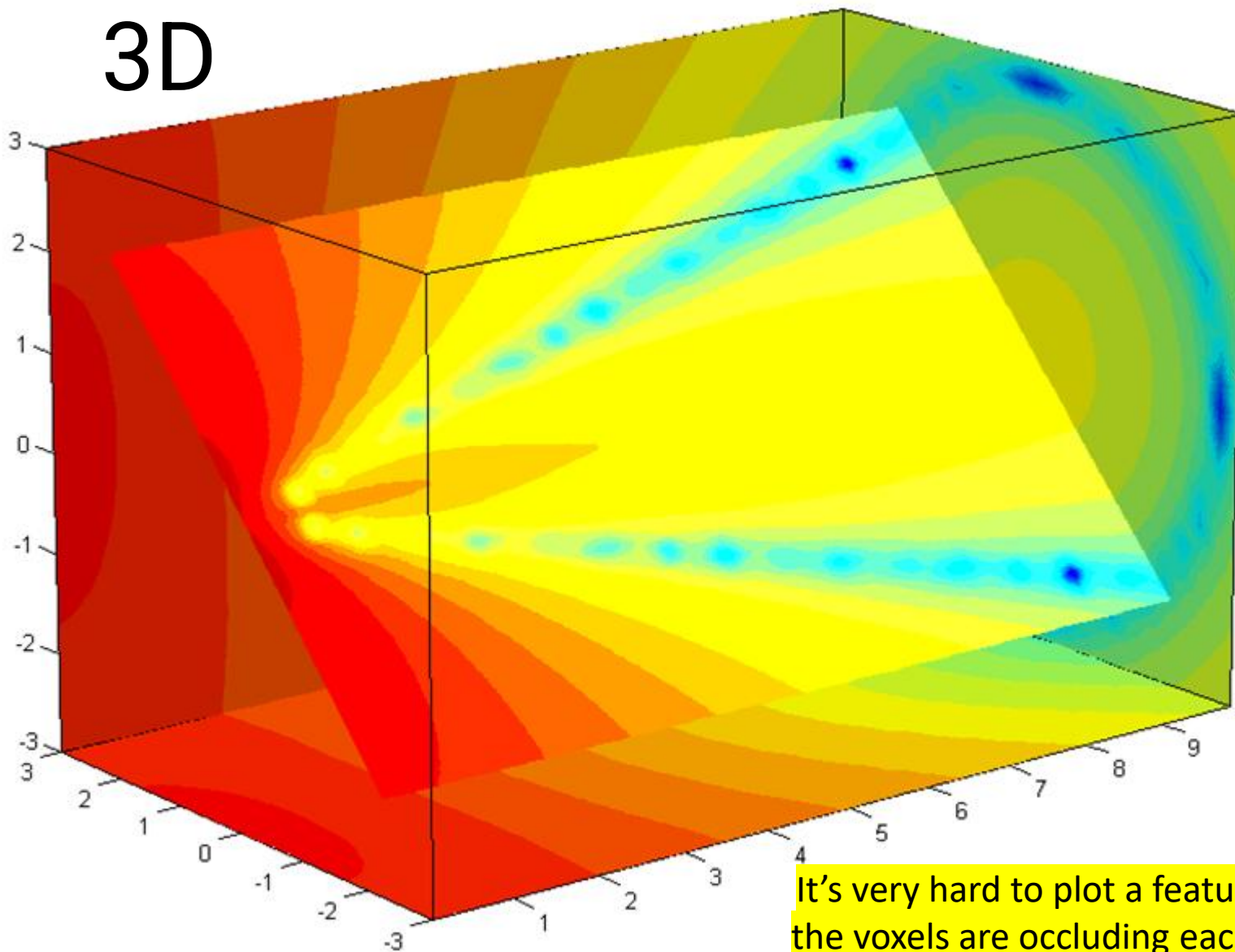
```
lightangle(-45,45)
```

```
colormap (jet(24))
```

```
set(gcf,'Renderer','zbuffer');
```

```
title('Velocity profile in a fluid jet in an infinite container');
```

3D



It's very hard to plot a feature on 3 axes, since the voxels are occluding each other. Preferring projections and slices can be a better idea

3D and surfaces

```
clear; close all; load mri; D = squeeze(D); image_num = 8;
```

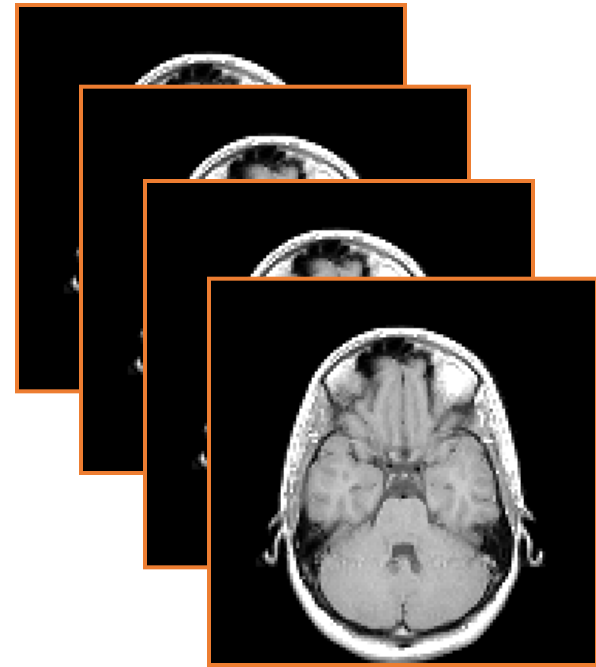
```
subplot(1,2,1); %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
phandles = contourslice(D,[],[],[1,12,19,27],8);  
view(3); axis tight  
set(phandles,'LineWidth',2)  
axis square
```

```
subplot(1,2,2) %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
Ds = smooth3(D);  
hiso = patch(isosurface(Ds,5),...  
    'FaceColor',[1,.75,.65],...  
    'EdgeColor','none');
```

```
hcap = patch(isocaps(D,5),...  
    'FaceColor','interp',...  
    'EdgeColor','none');  
colormap(map)
```

```
view(45,30)  
axis tight  
daspect([1,1,.4])
```

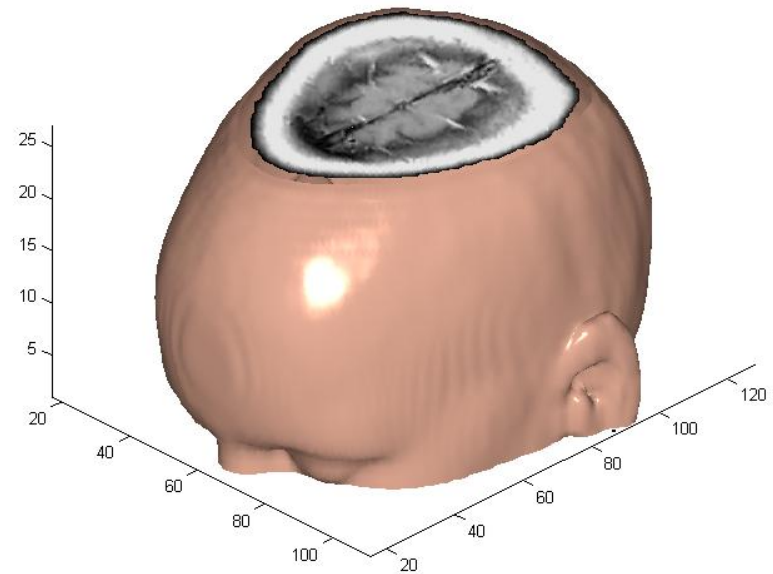
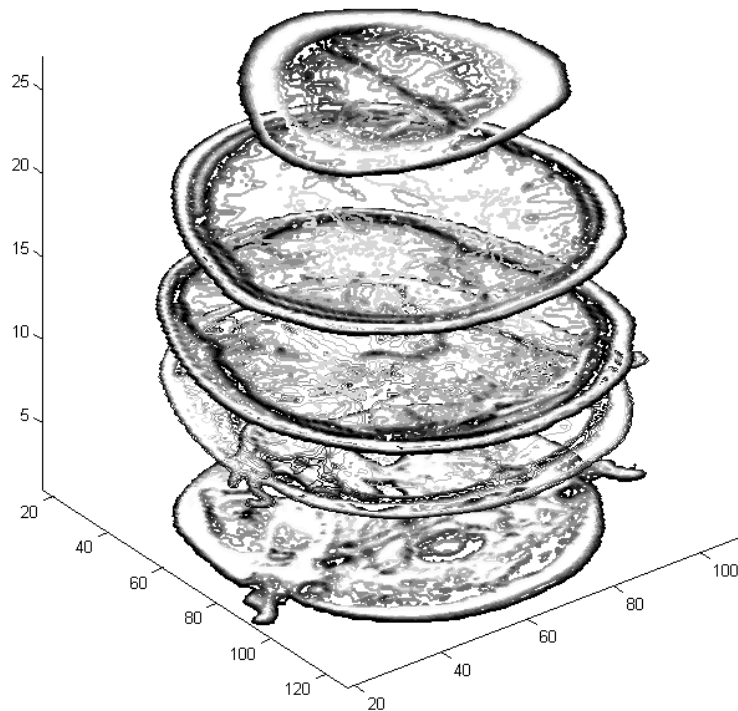
```
lightangle(45,30);  
set(gcf,'Renderer','zbuffer'); lighting phong  
isonormals(Ds,hiso)  
set(hcap,'AmbientStrength',.6)  
set(hiso,'SpecularColorReflectance',0,'SpecularExponent',50)
```



Starting from a 3D matrix ***D***
of MRI [128x128x27]

3D and surfaces

According to the goal of your visualization, your choice of the proper format can greatly differ



Now, just some funny slides before closing the lesson

NOT IN THE EXAM!

- Can I compare the different 007s?
- Are you creative?
Can I compare different lifestyles?

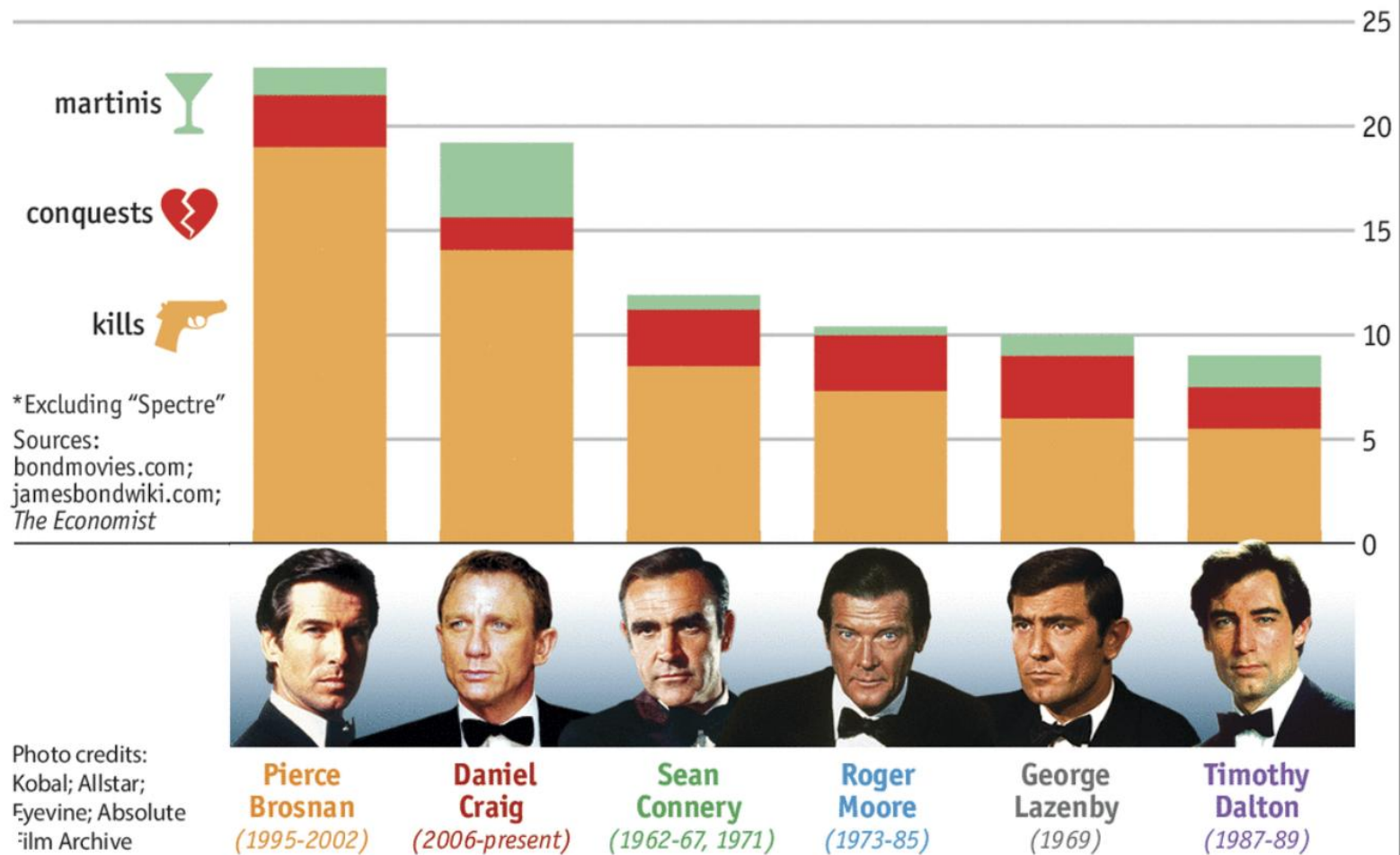
Excessive use of data visualization (humor not in the exam)

Even for very peculiar «applications» the choice of a proper input space permits comparisons and to get insights

Booze, bonks and bodies

Average per James Bond film* of:

007



Are you creative?

THE DAILY ROUTINES OF FAMOUS CREATIVE PEOPLE

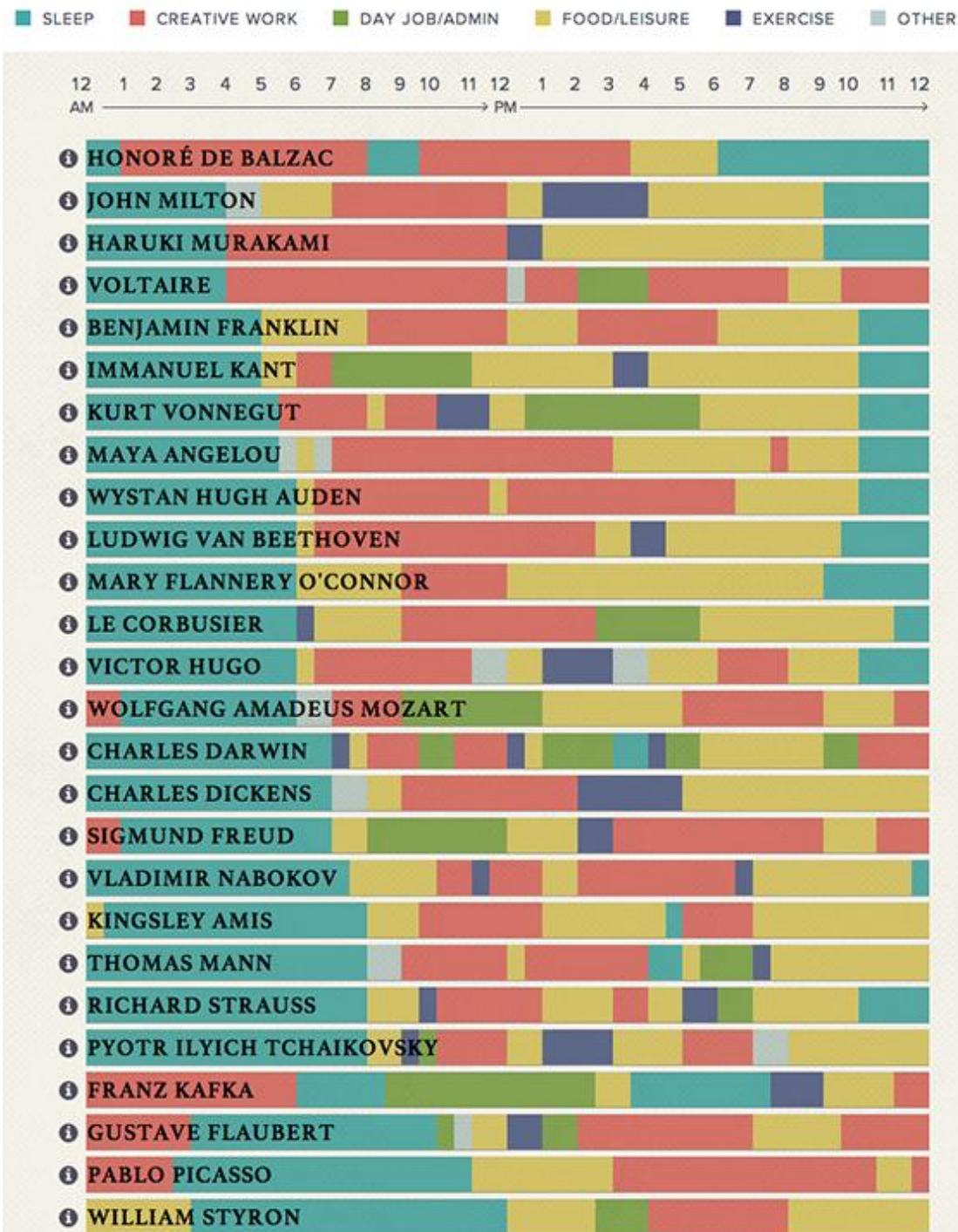
Turns out great minds don't think alike. Discover how some of the world's most original artists, writers and musicians structured their day.

Alignment is critical here,
Try to think this plot without it...

How to find the one
who is the most similar to you?

- 1) Decide the input space
- 2) Decide the metrics (distance)
- 3) Loop to find the closest creative!

This way you are finding the
«**First Nearest Neighbor**»
which is a typical ML method



Main points

- How using visualizations in ML
 - Inputs and image datasets
 - Dashboards
 - Saliency maps
 - Kernels
- Good visualization rules
 - Avoid funny plot and choose wisely your visual features
 - One concept at time in 1 powerful image

